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Abstract

The **Learnova** project is a pioneering initiative to revolutionize school management by offering a robust, role-based educational platform that streamlines operations and fosters collaboration among five key roles: Admin, Manager, Teacher, Parent, and Student. Educational institutions often face challenges such as fragmented communication, cumbersome administrative tasks, and limited access to centralized resources, which hinder academic and operational efficiency. Learnova addresses these challenges with an intuitive platform that provides tailored functionalities for each role, ensuring a cohesive and engaging environment. Key features include:

- **Role-Based Dashboards:** Personalized interfaces for managing tasks specific to each role, enhancing user experience and productivity.
- **Course and Schedule Management:** Tools for creating, tracking, and exporting schedules, including weekly classes, and exams, and virtual classroom integration.
- **AI-Powered Chatbot:** An NLP-driven assistant supporting English and Arabic, aiding users with navigation and queries.
- **Notification System:** Real-time alerts, reminders, and messages with customizable recipient options, improving communication.
- **Performance Analytics:** AI and ML-driven insights into student performance, offering predictive scores and improvement plans.
- **Resource Marketplace:** A platform for sharing and accessing educational materials, fostering community engagement.
- **Motivation System:** A gamified scoring system with badges (Green, Golden, Diamond) to encourage user participation.
- **Multi language and Accessible Design:** Support for Arabic and English, with light/dark modes and a responsive website.
- **Attendance Tracking:** Real-time monitoring and reporting of student attendance across classes and semesters.
- **Grade Management:** Comprehensive tools for uploading, editing, and viewing grades, supporting both online and offline assessments.
- **Activity and Contest Management:** Features for creating school-wide events and contests, with participant tracking and prize allocation.

By integrating these tools into a single, accessible system, Learnova empowers educational communities to overcome operational challenges, enhance communication, and focus on academic excellence. We aspire for this project to set a new benchmark for school management solutions, providing sustainable and impactful support for future generations.

Acknowledgment

Thanks to God, the most Gracious, the Most Merciful, who granted us the patience and endurance to complete this work.

We express our deep gratitude to Menoufia University, Faculty of Computers and Information, for providing the resources, devices, and capabilities that enabled us to develop the **Learnova - Smart School System** project in an excellent manner.

Special appreciation goes to our professors and teaching assistants who supported us throughout our learning journey, helping us acquire new skills, guiding us on the right path, and patiently addressing our numerous questions with care and dedication.

We owe a profound debt of thanks to our supervisor, **Dr. Mahmoud Hussein**, whose expertise taught us the essential knowledge for this project and whose wisdom allowed us to grow independently. His continuous support from the outset to the completion of the project, instilling courage and confidence to take bold steps without fear of initial setbacks or frustrations, has been invaluable. We are immensely grateful for his guidance as we progressed toward success. Additionally, we extend our gratitude to **Eng. Abdullah**, who provided significant technical assistance and support during the development process.

This project is also dedicated to everyone who contributed to its development, including the staff and students of **Salah Khattab Secondary School for Languages** in Meet Khaqan. We held meetings with **Mohamed**, the school's technology coordinator, to understand the specific needs of the school. Based on these discussions, we designed the features of our system to align with their requirements, ensuring that **Learnova** addresses their operational challenges effectively. Furthermore, we benefited from studying previously developed educational websites such as **Classera** and **Google Classroom**, which inspired our design and functionality.

Without the collective support of all these individuals and resources, the success of the **Learnova** would not have been possible.

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Chapter 1

Introduction

1.1 - Problem Definition

- **Fragmented communication among educational stakeholders:** Ineffective coordination between Admin, Manager, Teacher, Parent, and Student roles hinders smooth school operations.
- **Cumbersome administrative tasks:** Manual processes for scheduling, attendance tracking, and grade management reduce efficiency and increase workload.
- **Limited access to centralized resources:** Lack of a unified platform restricts access to educational materials, schedules, and performance data.
- **Inconsistent student performance monitoring:** Absence of AI-driven insights makes it challenging to track and improve student progress.
- **Poor engagement and motivation:** Lack of gamified systems or recognition mechanisms reduces student and staff participation.
- **Inefficient event and contest management:** Manual organization of school activities leads to delays and poor participant tracking.
- **Inadequate support for virtual learning:** Limited integration of virtual classrooms and attendance tracking affects remote education quality.
- **Difficulty in resource sharing:** Lack of a marketplace for educational materials limits collaboration and resource exchange among users.
- **Manual grade and attendance reporting:** Time-consuming processes for generating reports hinder timely feedback and decision-making.
- **Lack of real-time notifications:** Absence of automated alerts for schedules, exams, or updates disrupts communication and preparedness.
- **Limited parental oversight:** Difficulty for parents to monitor and understand their child's academic progress, activities, and overall school experience.

1.2 - Goals

- **Enhance communication efficiency:** Improve coordination among Admin, Manager, Teacher, Parent, and Student roles for seamless school operations.
- **Streamline administrative processes:** Automate scheduling, attendance tracking, and grade management to boost efficiency and reduce workload.
- **Provide centralized resource access:** Develop a unified platform for easy access to educational materials, schedules, and performance data.
- **Improve student performance tracking:** Utilize AI-driven insights to monitor and enhance student progress effectively.
- **Boost engagement and motivation:** Implement gamified systems and recognition mechanisms to increase participation among students and staff.
- **Optimize event and contest management:** Automate the organization of school activities for timely execution and accurate participant tracking.
- **Strengthen virtual learning support:** Integrate virtual classrooms and attendance tracking to enhance remote education quality.
- **Facilitate resource sharing:** Create a marketplace for educational materials to promote collaboration and resource exchange among users.
- **Automate reporting:** Enable quick generation of grade and attendance reports for timely feedback and decision-making.
- **Enable real-time notifications:** Provide automated alerts for schedules, exams, and updates to improve communication and preparedness.
- **Enhance parental oversight:** Equip parents with tools to monitor and understand their child's academic progress, activities, and overall school experience.

1.3 - Solutions

Traditional Solutions

Traditional approaches to school management often rely on manual coordination and paper-based systems, placing significant pressure on administrators, teachers, and parents. These methods, such as handwritten schedules or face-to-face meetings, can lead to miscommunication and inefficiencies. Additionally, the lack of centralized resource sharing results in duplicated efforts and limited access to educational materials, while manual reporting increases workload and delays feedback. Traditional practices may also fail to engage students or track performance effectively, hindering overall educational outcomes.

Technological Solution

To address these challenges and enhance school management, we developed the **Learnova - Smart School System**, a comprehensive technological solution. This platform integrates all educational stakeholders—Admin, Manager, Teacher, Parent, and Student—into a unified system to ensure smooth operations and effective communication. It automates administrative tasks like scheduling, attendance tracking, and grade management, reducing workload and improving efficiency. The system provides centralized access to resources, including educational materials and performance analytics, powered by AI to monitor and enhance student progress. Gamified features and recognition systems boost engagement, while automated event management and virtual classroom integration optimize activity organization and remote learning. A resource marketplace facilitates collaboration, and real-time notifications keep all stakeholders informed. Additionally, parents gain tools to oversee their child's academic journey, ensuring comprehensive support for educational success.

Chapter 2

Previous Work

Introduction

The **Learnova - Smart School System** was developed to address the shortcomings of existing educational platforms and provide a comprehensive solution for modern school management. By analyzing systems like **Classera** and **Google Classroom**, we identified significant limitations in user engagement, communication, and functionality. This chapter briefly reviews these platforms, highlights their drawbacks, and compares them with **Learnova**, demonstrating how our system offers superior features, including support for five distinct user roles and enhanced student-teacher interaction, to meet the needs of educational institutions.

2.1 - Classera

Classera is an e-learning platform used in regions like the Middle East, offering tools for virtual classrooms and resource sharing. Key features include:

- Virtual classrooms for assignments and discussions.
- Basic gamification through points and rewards.
- Resource marketplace for educational materials.
- Support for multiple languages, including Arabic.

Limitations:

- Limited rules supporting only student and teacher engagement.
- Complex navigation and functionality, requiring extensive training.
- Limited gamification, failing to sustain student engagement.
- Weak performance analytics, lacking predictive or personalized insights.
- Poor customization for diverse educational needs.
- Inadequate communication tools, limiting student-teacher interaction.
- Scalability challenges for smaller schools with limited technical support.

2.2 - Google Classroom

Google Classroom is a free platform integrated with Google Workspace, used for classroom management. Key features include:

- Assignment creation and distribution.
- Integration with Google Docs and Drive.
- Basic communication via announcements and emails.
- Accessible interface across devices.

Limitations:

- No AI-driven analytics for student performance tracking.
- Absence of gamification or motivation systems.
- Limited customization for user roles or workflows.
- Basic resource sharing without a dedicated marketplace.
- Weak support for multilingual interfaces, especially Arabic.
- Minimal tools for managing school-wide activities or contests.
- Ineffective communication features, hindering student-teacher collaboration.

Chapter 3

Theoretical Consideration

System Requirement Specification

1-Functional Requirements :

The system supports five roles (Admin, Manager, Teacher, Student, Parent) with tailored functionalities:

Onboarding and Authentication

- **Onboarding Page:** Displays introductory content about Learnova with a "Login" button redirecting to role selection.
- **Role Selection:** Allows users to choose their role (Admin, Manager, Teacher, Student, Parent) and redirects to a role-specific login page.
- **Login:** Validates credentials using JWT authentication (Express.js backend, MongoDB). First-time users receive an email verification link.

Admin Role

- **Dashboard:** Displays system metrics (e.g., user counts, schedules) and an AI-powered chatbot (Arabic/English).
- **Notification System:** Sends customizable notifications (Reminder, Alert, Message) to specific users or groups, stored in MongoDB.
- **Members Management:** CRUD operations for users (add, edit, delete) with bulk upload via Excel. Displays role-specific data (e.g., teacher's assigned classes).
- **Term, Course, Grade, and Schedule Management:** Manages academic terms, courses, grades, and schedules with CRUD APIs, ensuring no duplicates. Schedules are exportable as CSV.

Manager Role

- **Dashboard:** Shows metrics, grades/absence statistics, and top-ranked users with badges.
- **School Hubs:** Manages school-wide events with fields for title, dates, location, and prizes.

- **Absence and Grade Management:** Displays attendance reports and allows grade uploads via Excel/CSV/PDF.
- **Virtual Classrooms:** Creates and manages virtual rooms with links stored in MongoDB.

Teacher Role

- **Dashboard:** Displays assigned courses, notifications, and motivation scores (Green, Golden, Diamond badges).
- **Courses:** Manages content (videos, PDFs, assignments, exams) via APIs. Tracks student interactions.
- **Absence and Grades:** Records attendance and uploads grades (manual for essays, automatic for MCQs).
- **Activities and Library:** Manages contests and uploads materials to a general/subject-specific library.

Student Role

- **Dashboard:** Shows scores, attendance, progress bars, and AI-driven performance predictions.
- **Courses:** Accesses course content (videos, assignments, exams) and tracks interactions for motivation scores.
- **Absence, Grades, and Schedules:** Displays attendance records, grades, and schedules in responsive tables/charts.
- **Activities and Library:** Allows joining contests and accessing educational materials.

Parent Role

- **Dashboard:** Summarizes child's performance, attendance, grades, and activities.
- **Monitoring:** Accesses child-specific data (courses, schedules, grades) in a summarized view.

2- Non-Functional Requirements :

1. Technology Stack:

- **Frontend:** React.js with Tailwind CSS for responsive, accessible UI with light/dark mode and Arabic/English support.
- **Backend:** Express.js for RESTful APIs, handling authentication, CRUD operations, and data processing.
- **Database:** MongoDB for scalable, schema-flexible storage of users, courses, schedules, and other data.
- **API Testing:** Postman and API Dog for developing and testing APIs.
- **Authentication:** JWT with SECRET_KEY for secure user sessions.
- **AI/ML:** ML-based chatbot (supporting Arabic/English) and ML model for student performance analysis, deployed via APIs.

2. Security:

- JWT authentication ensures secure login and session management.
- HTTPS for all API communications.
- Role-based access control to limit data access per role.

3. Performance:

- TanStack Query uses an in-memory cache to store the results of server data fetching, improving performance and reducing unnecessary network requests.

4. Scalability:

- MongoDB's horizontal scaling with sharding for large datasets.
- Cloud deployment (e.g., AWS/Google Cloud) with load balancers for traffic management.

5. Reliability:

- Error handling for failed API calls with user-friendly toast messages.
- The system shall maintain high availability with minimal downtime, ensuring consistent access to services for users.

6. Maintainability:

- Modular React components and Express routes for easy updates.
- Comprehensive API documentation via Postman/API Dog.

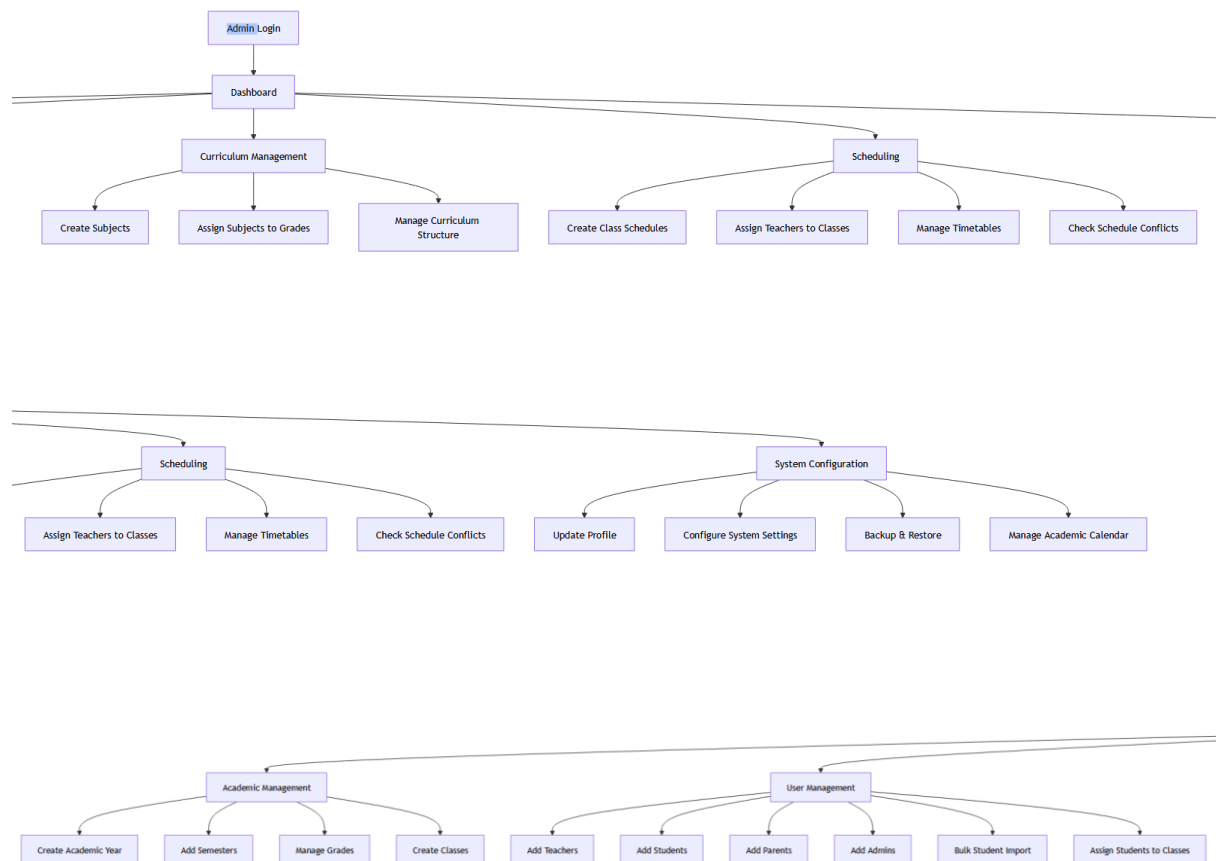
Chapter 4

System Design

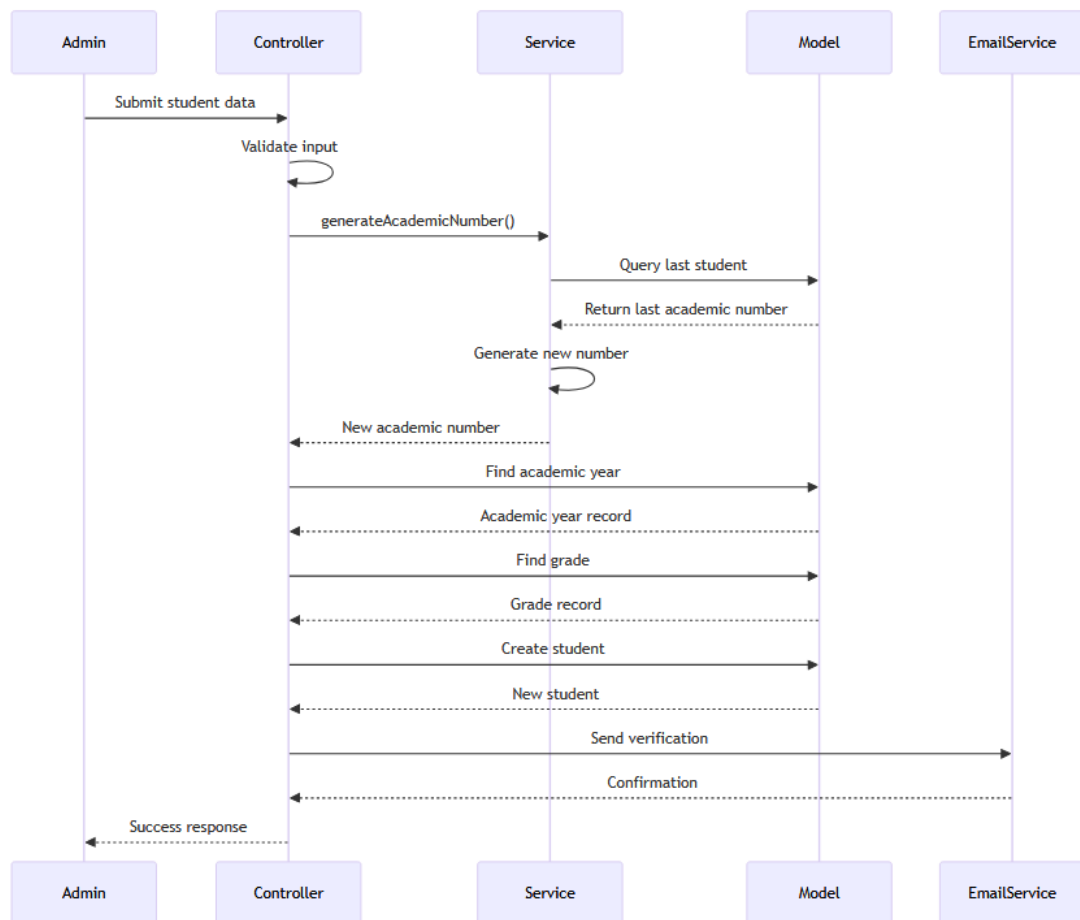
System Models

1- Admin Diagrams :

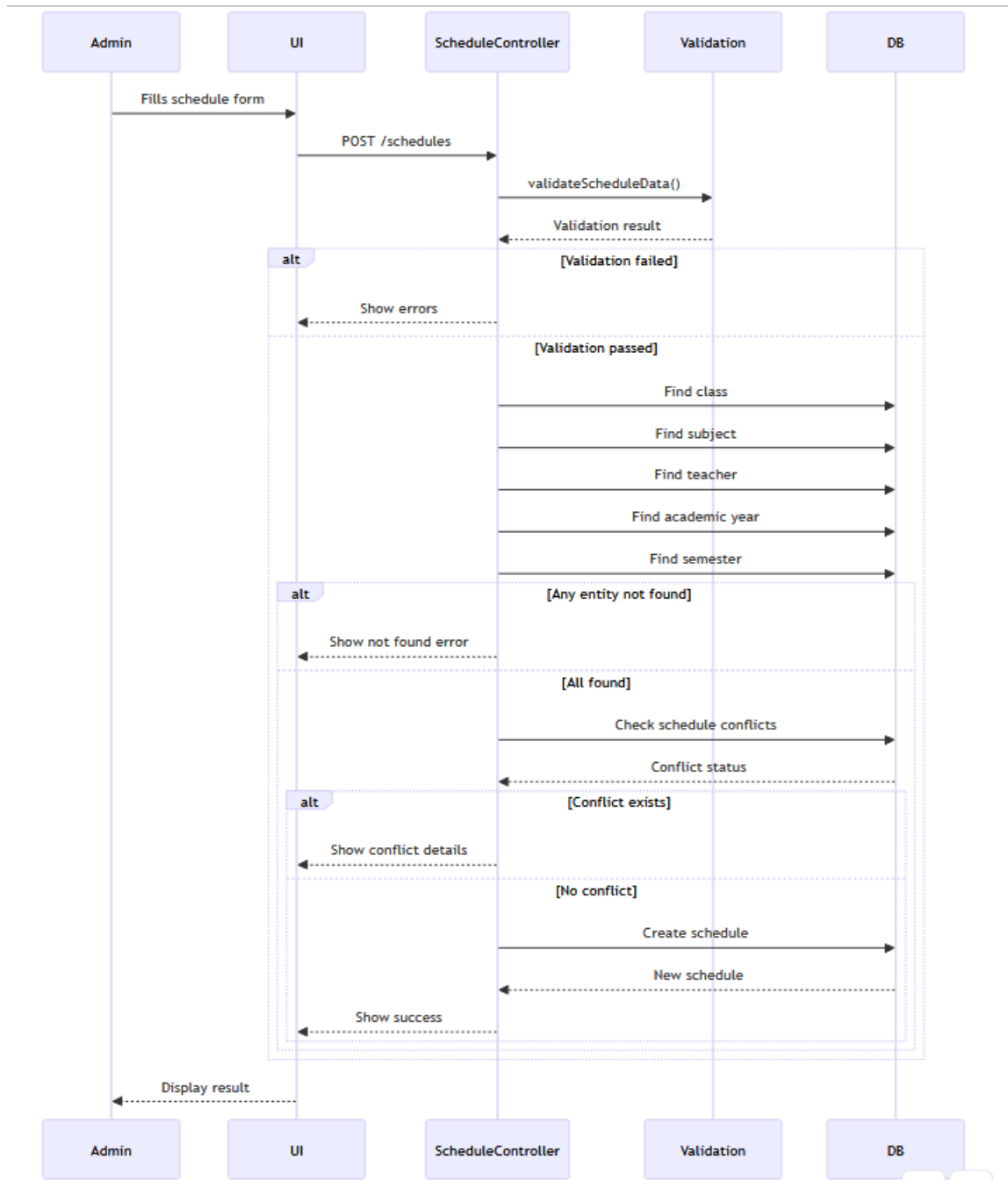
➤ Flowchart



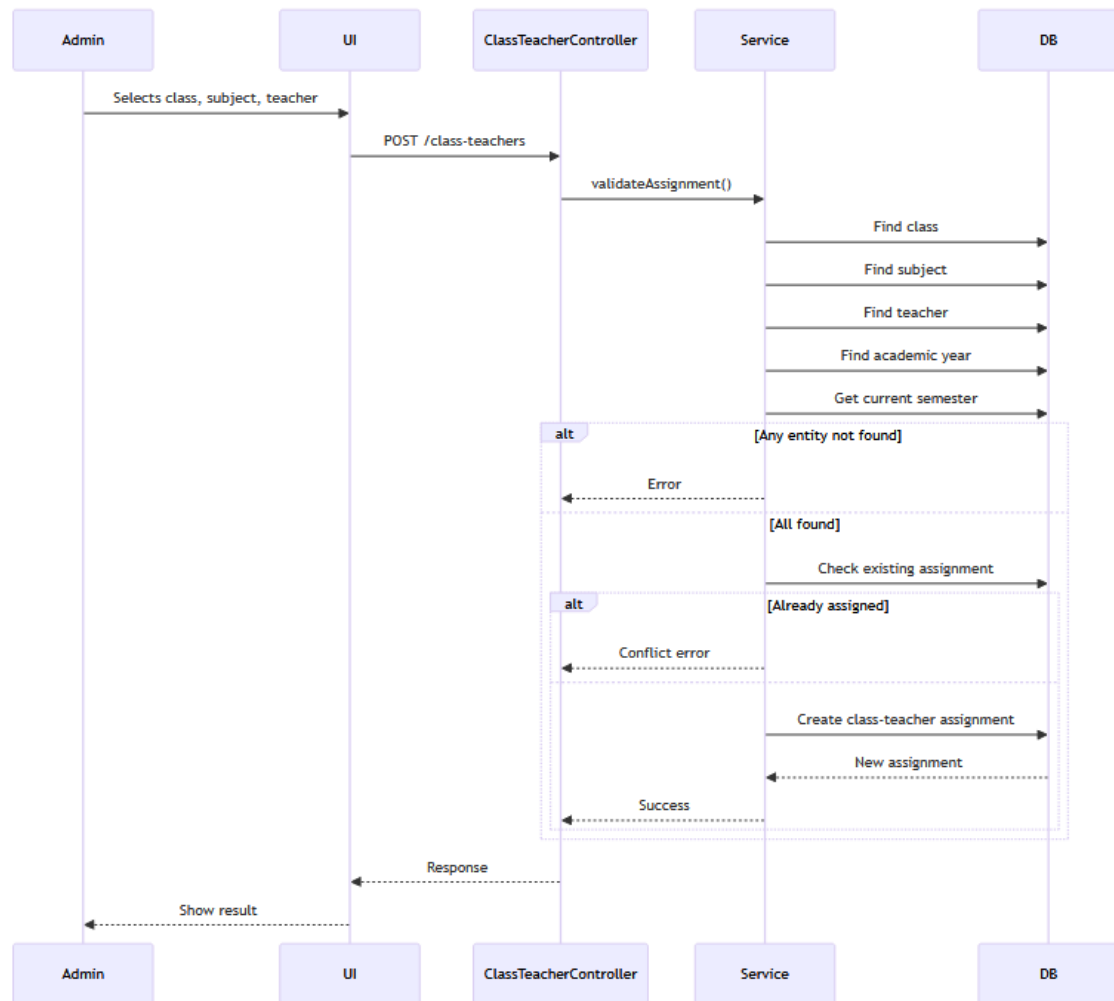
➤ Sequence Diagram : student creation



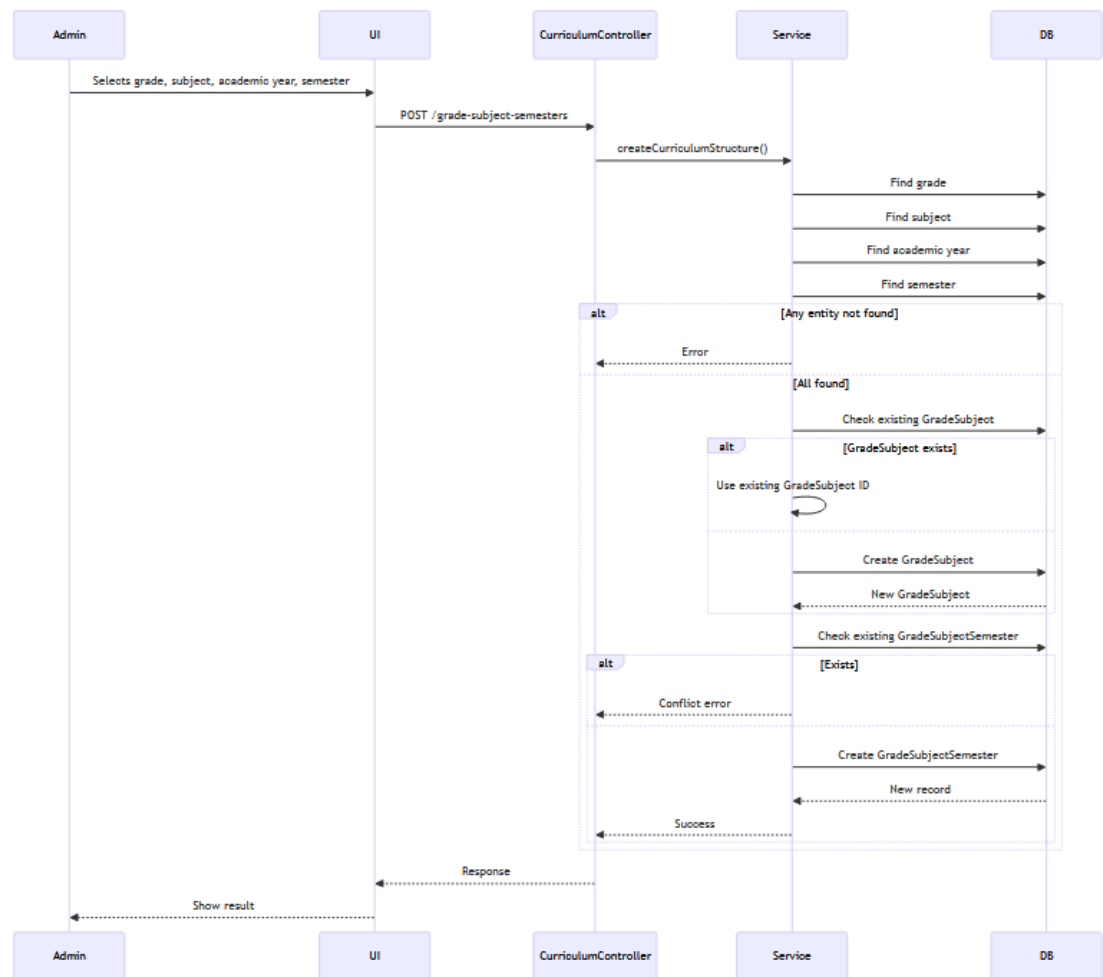
➤ Sequence Diagram : Admin creating a class schedule



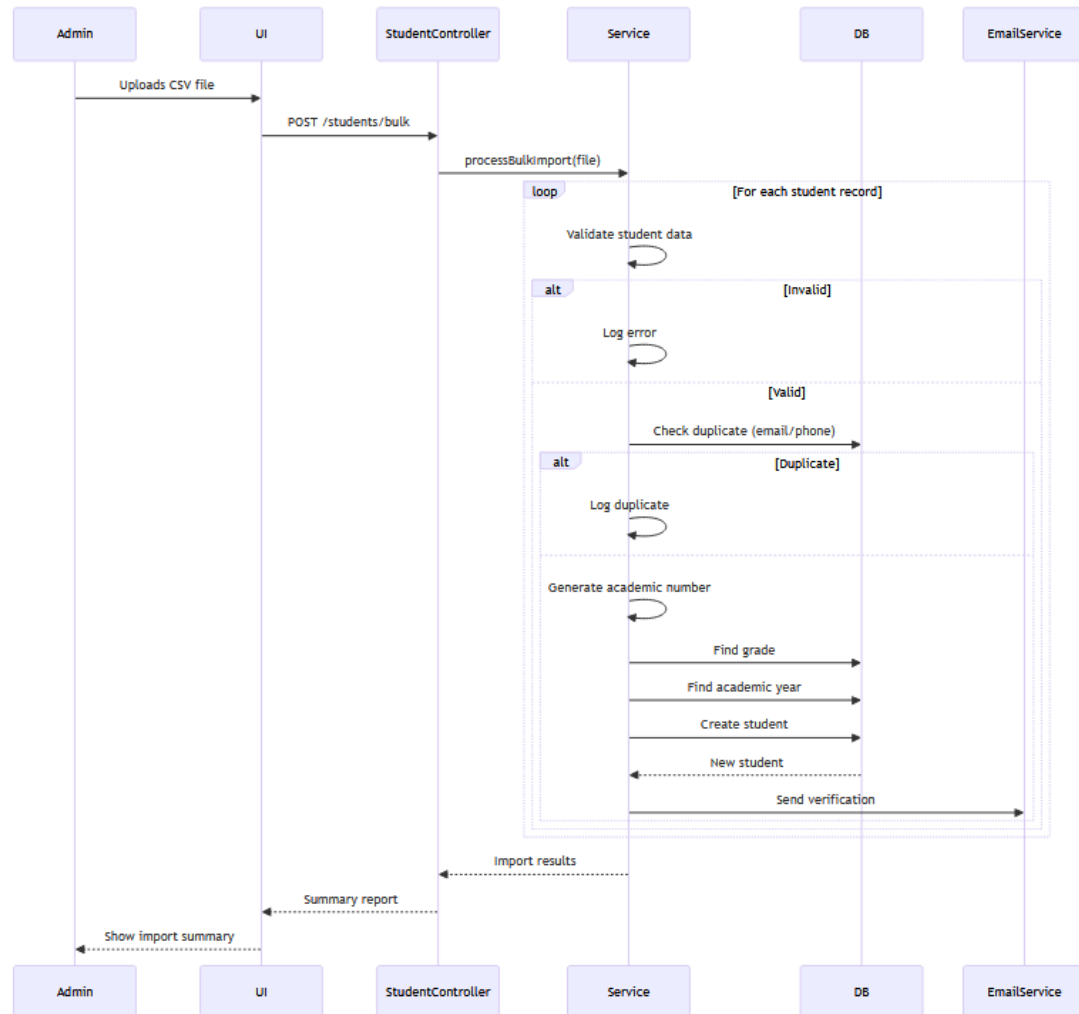
➤ Sequence Diagram : Admin assigning teacher to class



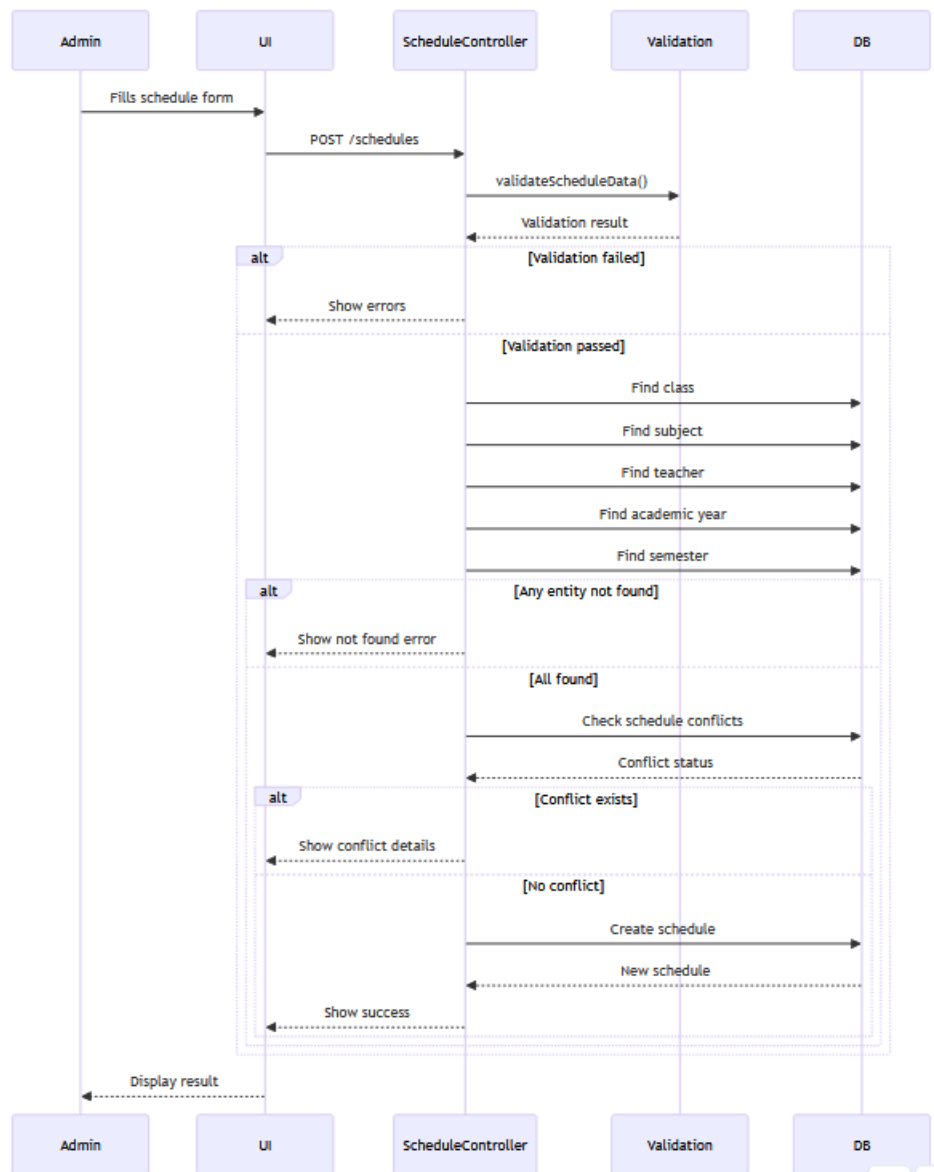
➤ Sequence Diagram : Admin Creating Curriculum Structure (Grade-subject-semester)



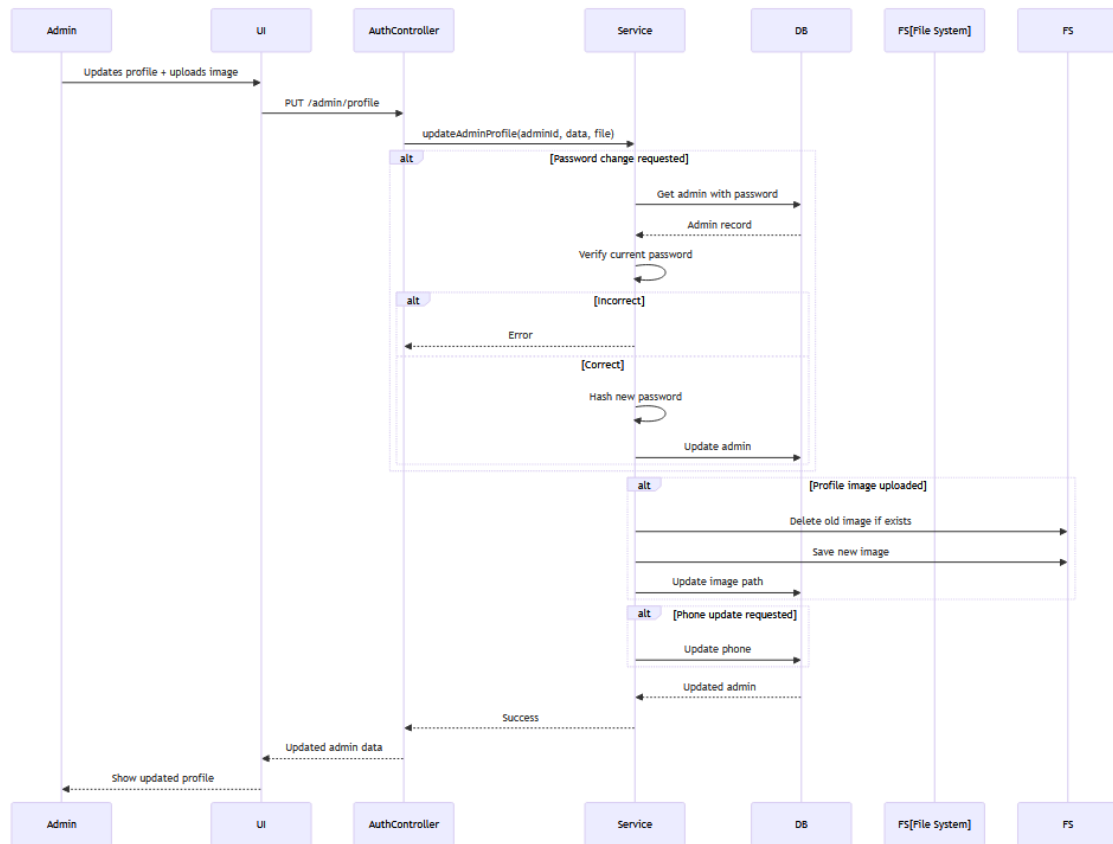
➤ Sequence Diagram : Admin Bulk Student Import



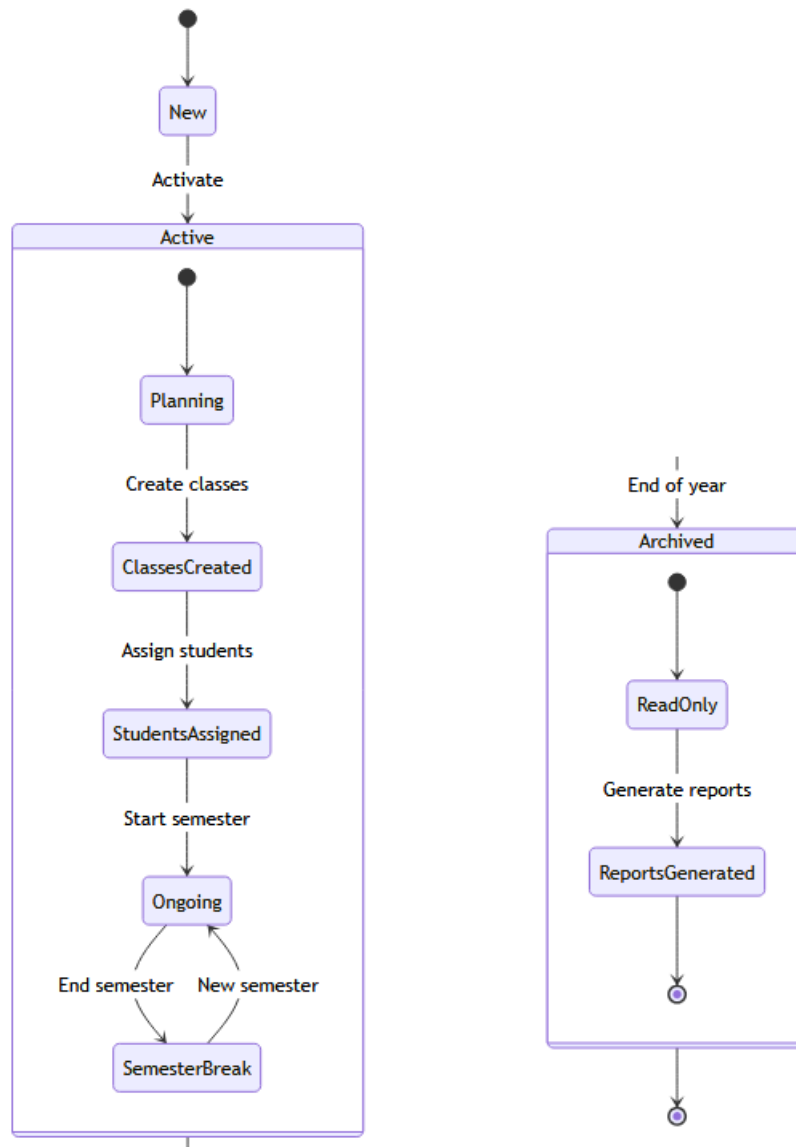
➤ Sequence Diagram : Class Schedule Creation Sequence



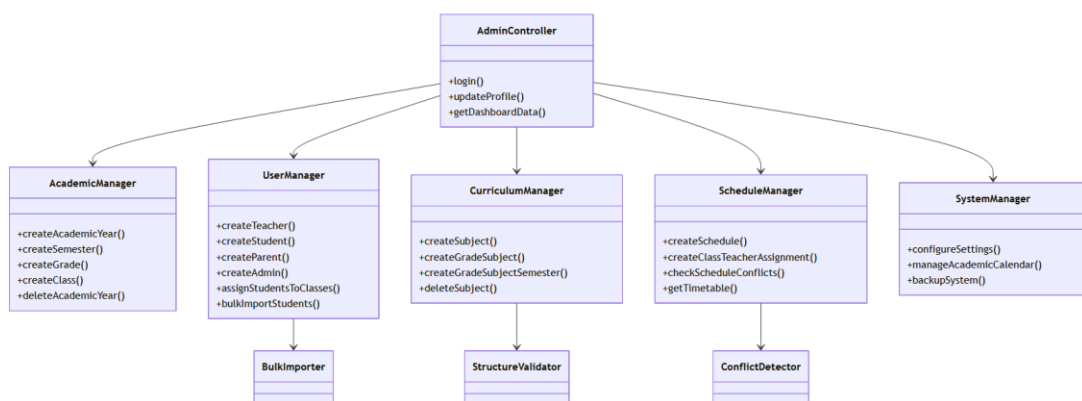
➤ Sequence Diagram : Admin Profile Update



➤ Entity State Diagram : Academic Year Lifecycle

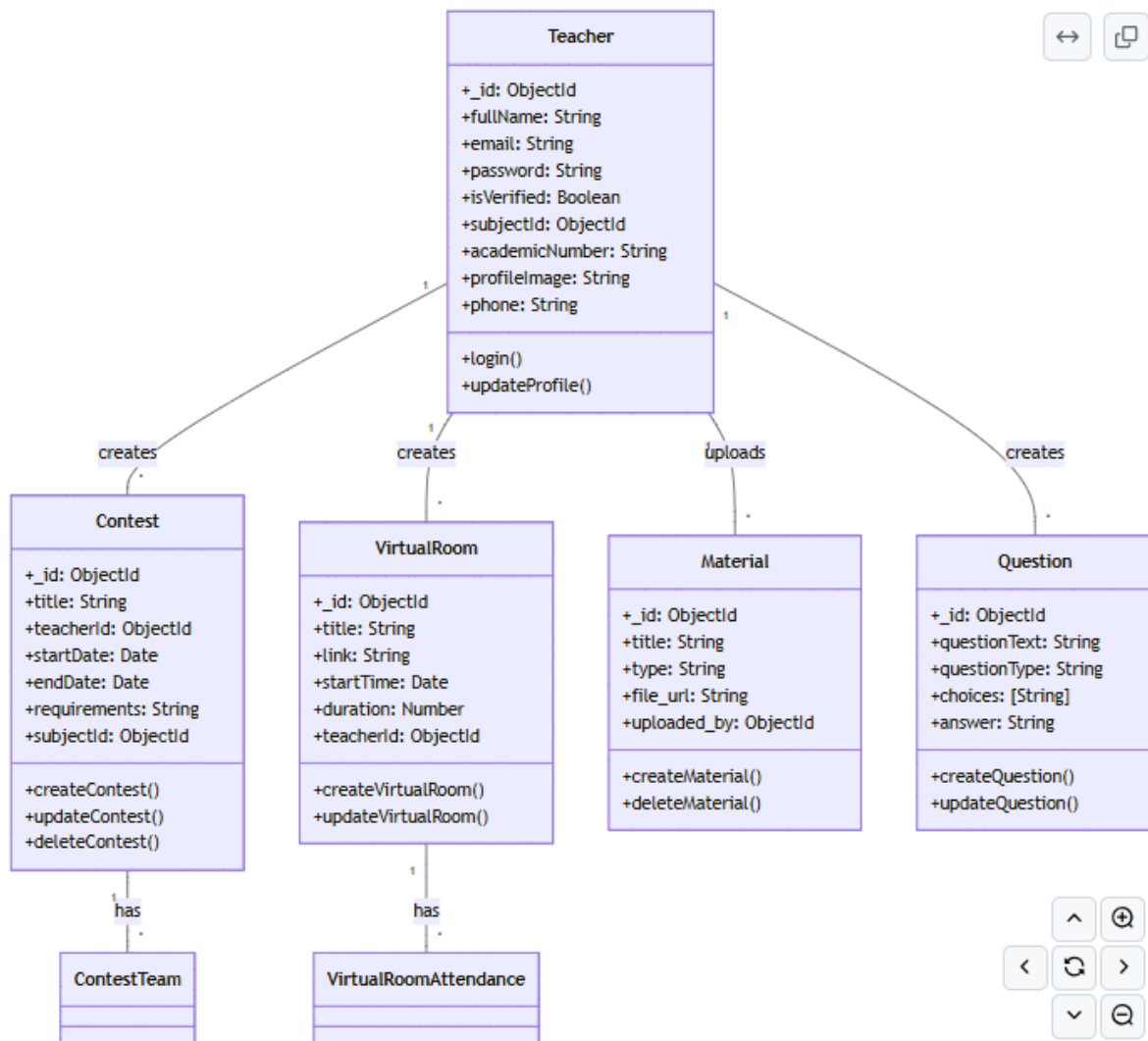


➤ Class Diagram : Core Admin Functionality

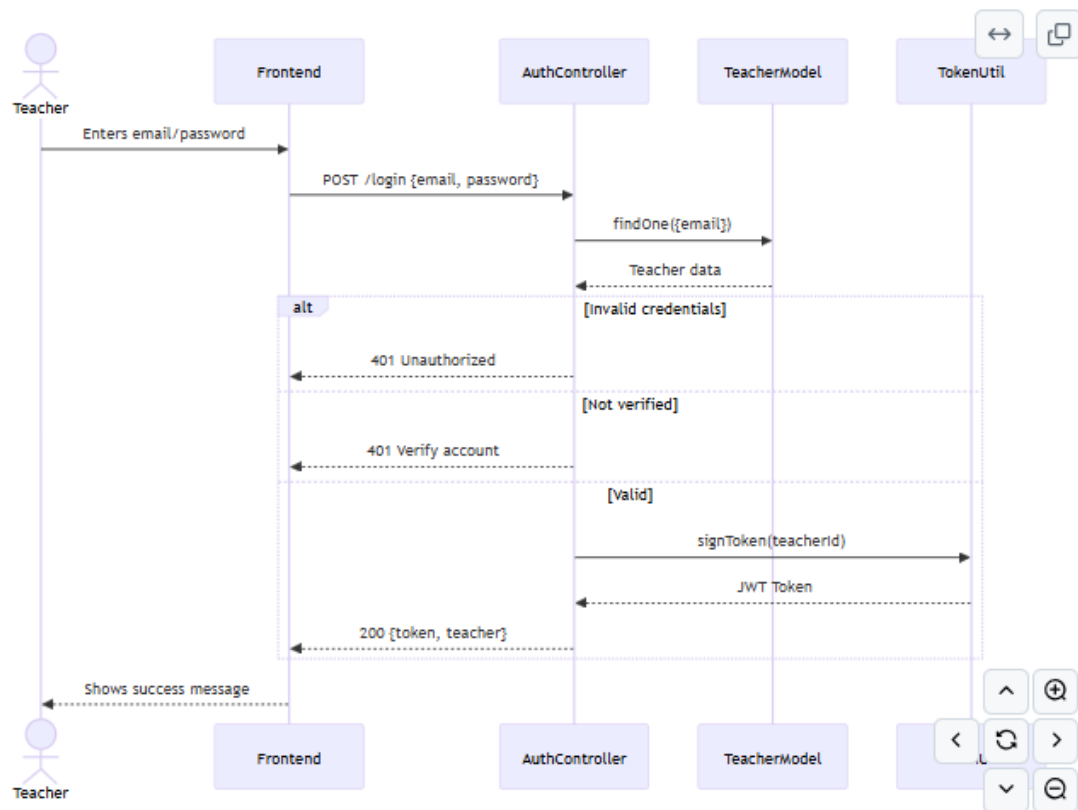


2- Teacher Diagrams :

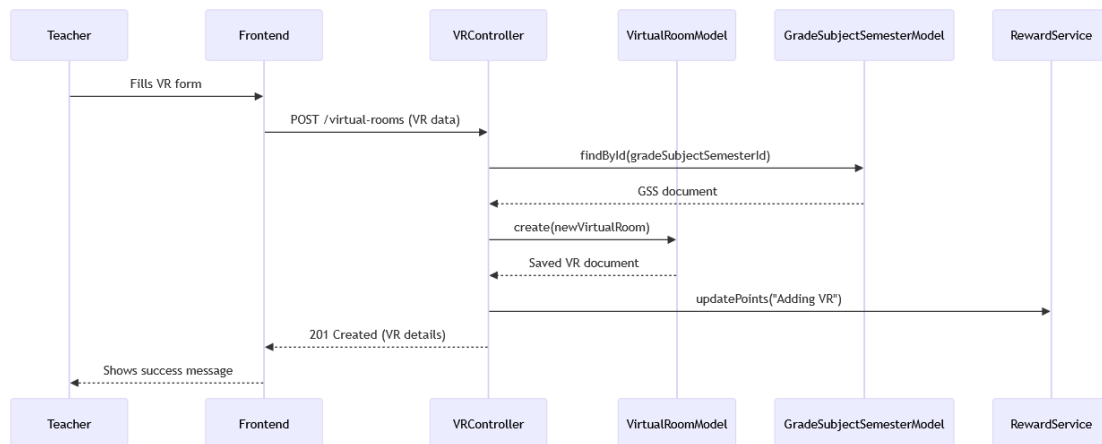
➤ Class Diagram



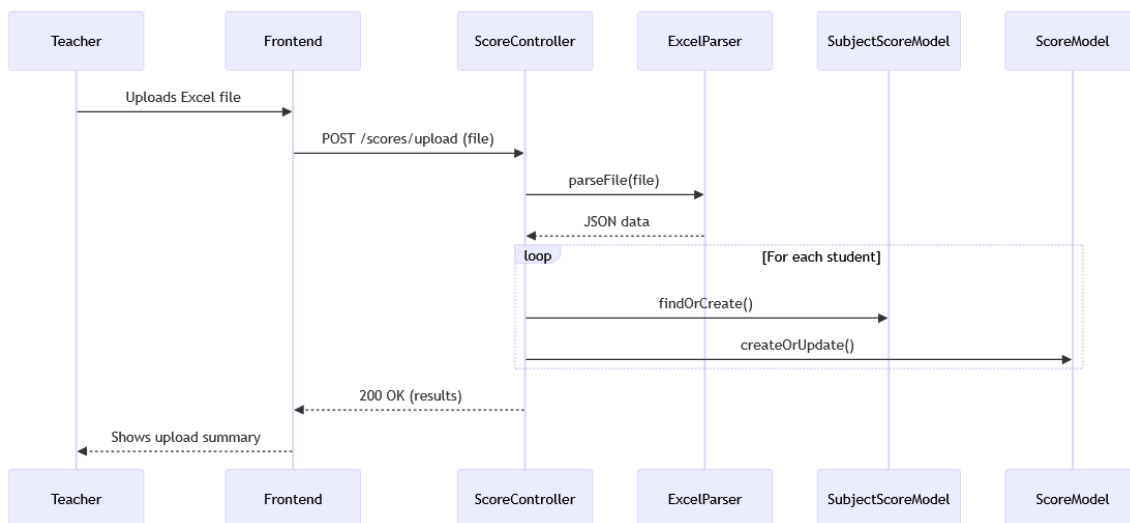
➤ Sequence Diagram: Teacher Login



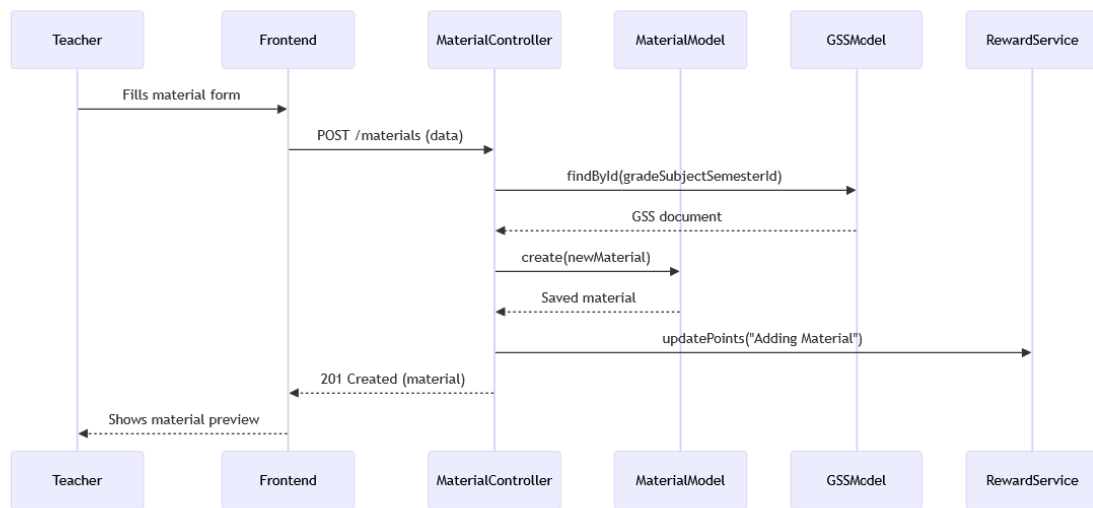
➤ Sequence Diagram: Virtual Room Creation Sequence



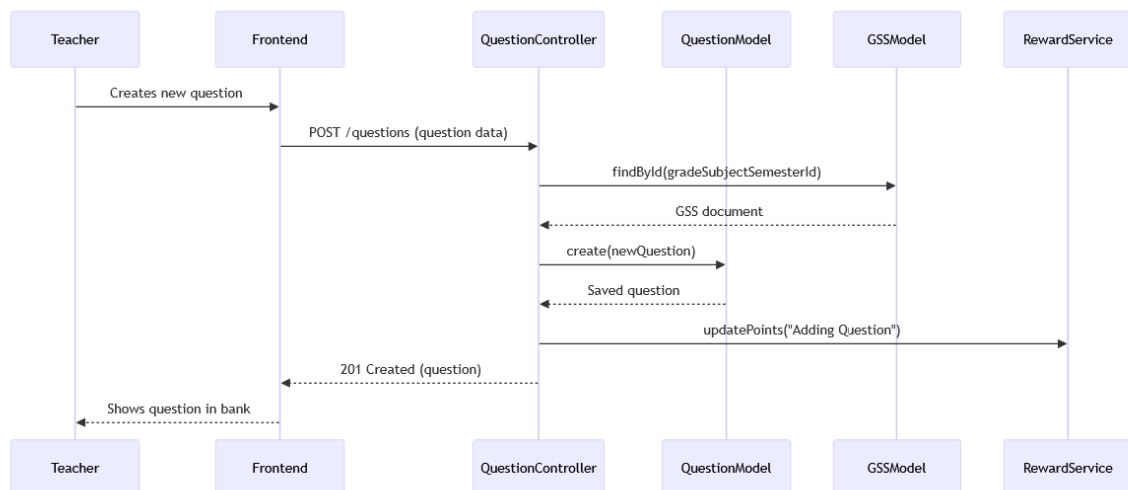
➤ Sequence Diagram: Exam Score Upload Sequence



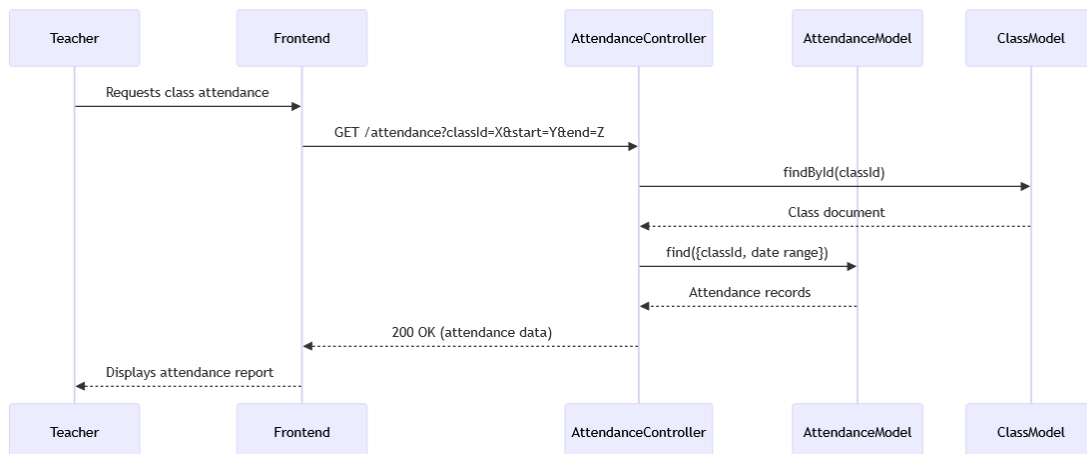
➤ Sequence Diagram: Material Creation Sequence



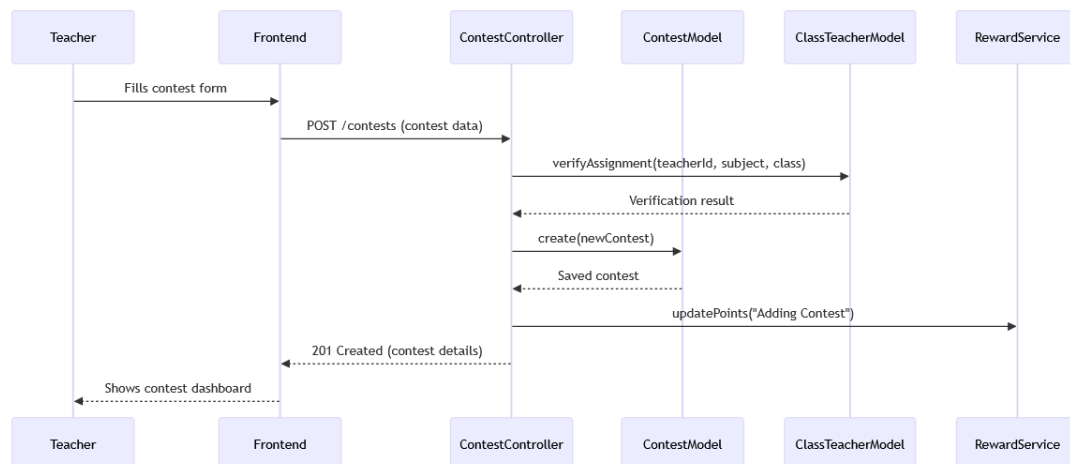
➤ Sequence Diagram: Question Bank Management Sequence



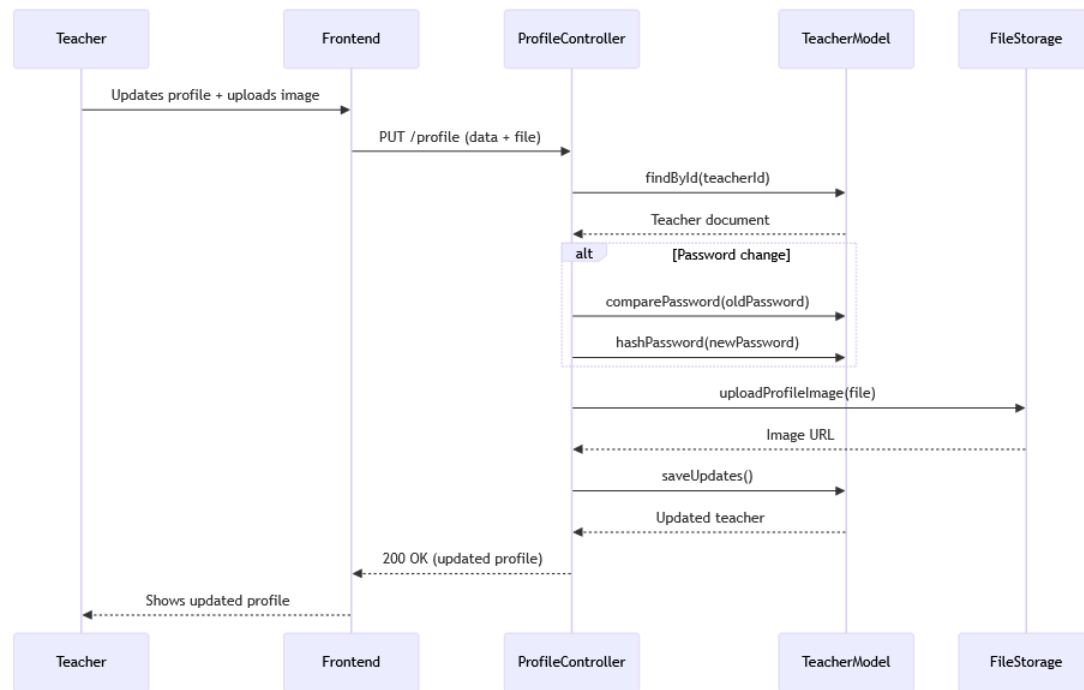
➤ Sequence Diagram: Attendance Management Sequence



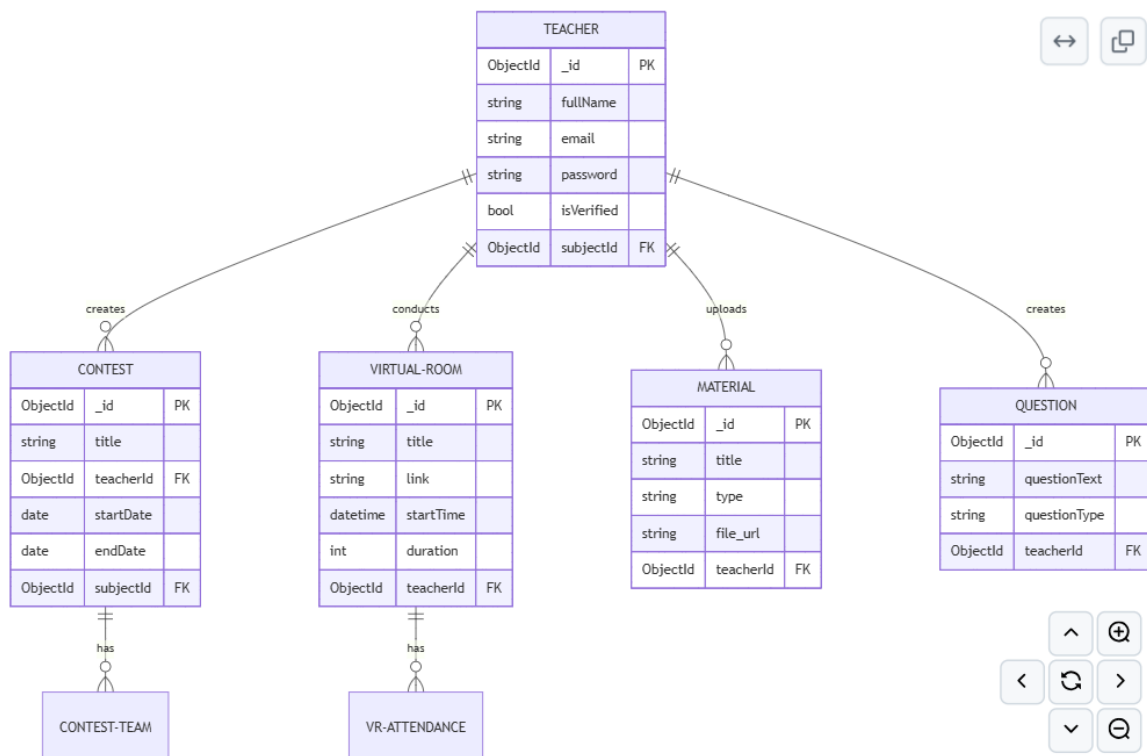
➤ Sequence Diagram: Contest Creation Sequence



➤ Sequence Diagram: Profile Update Sequence

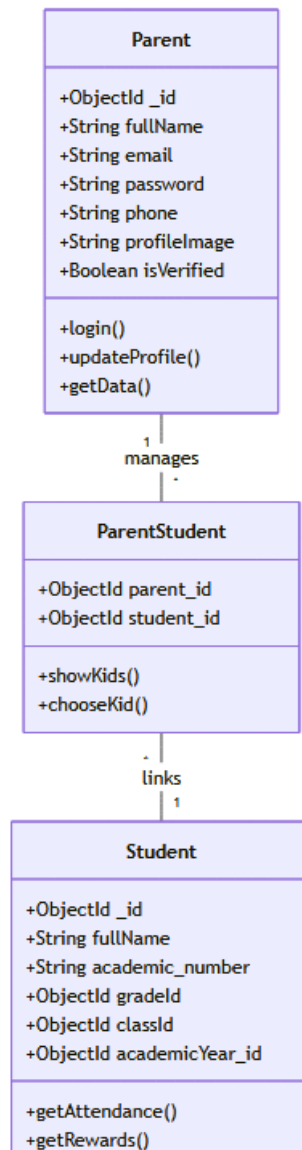


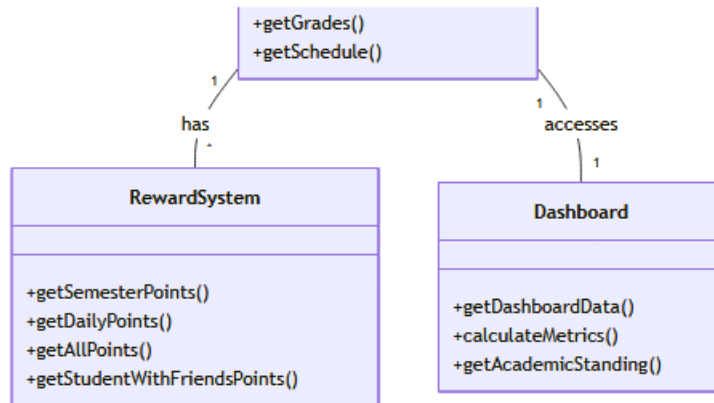
➤ Entity-Relationship Diagram (ERD)



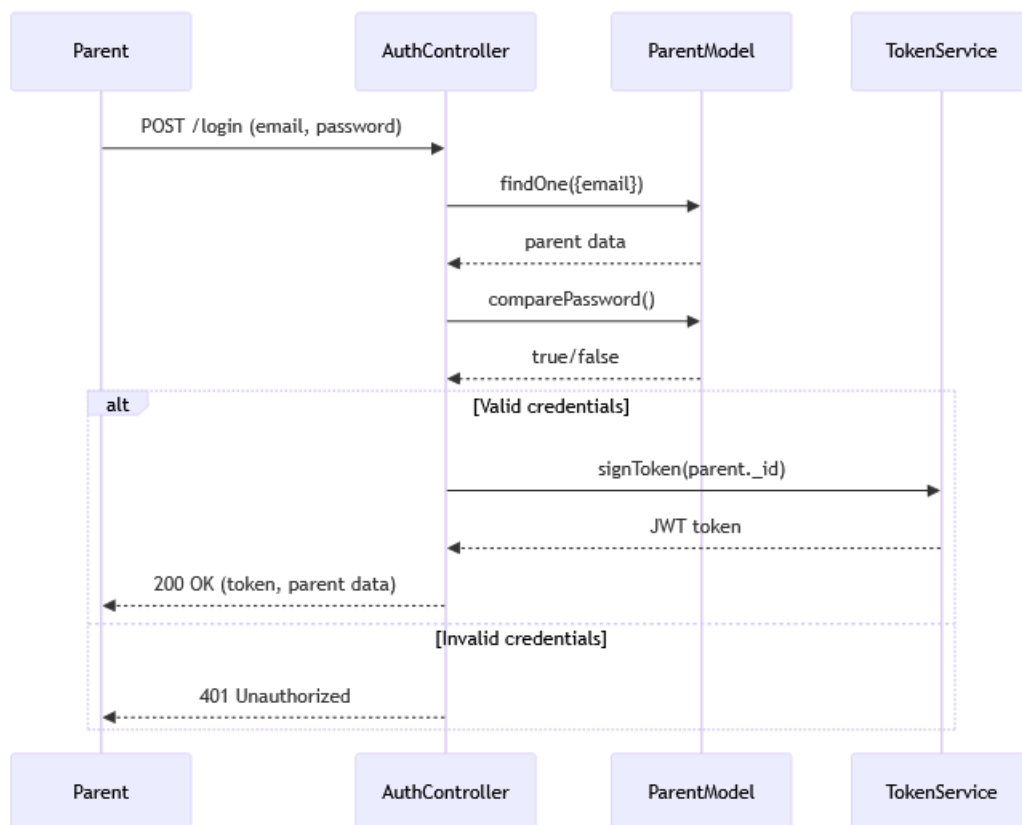
3- Parent Diagrams :

➤ Class Diagram

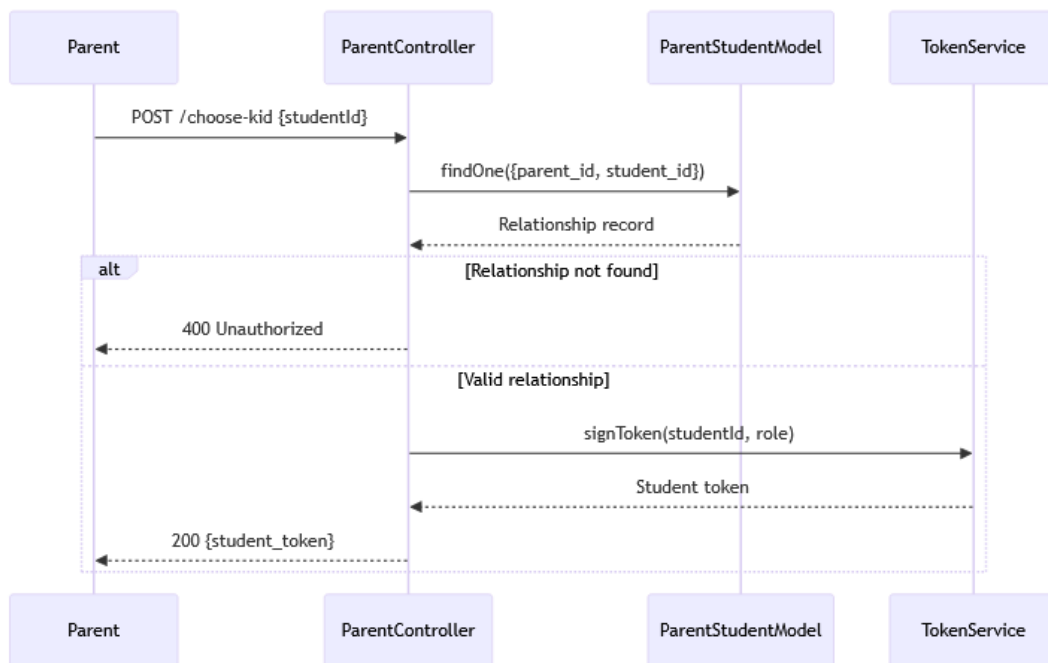




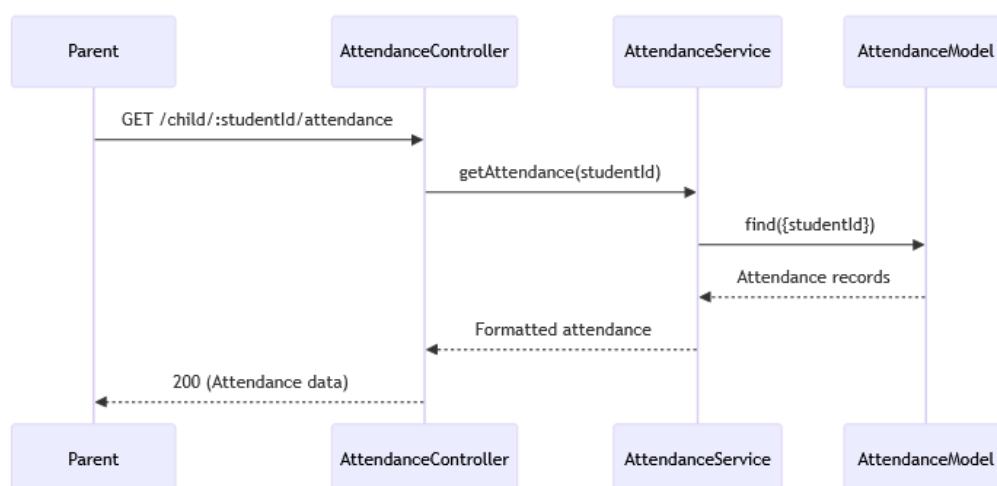
➤ Sequence Diagram: Login Sequence Diagram



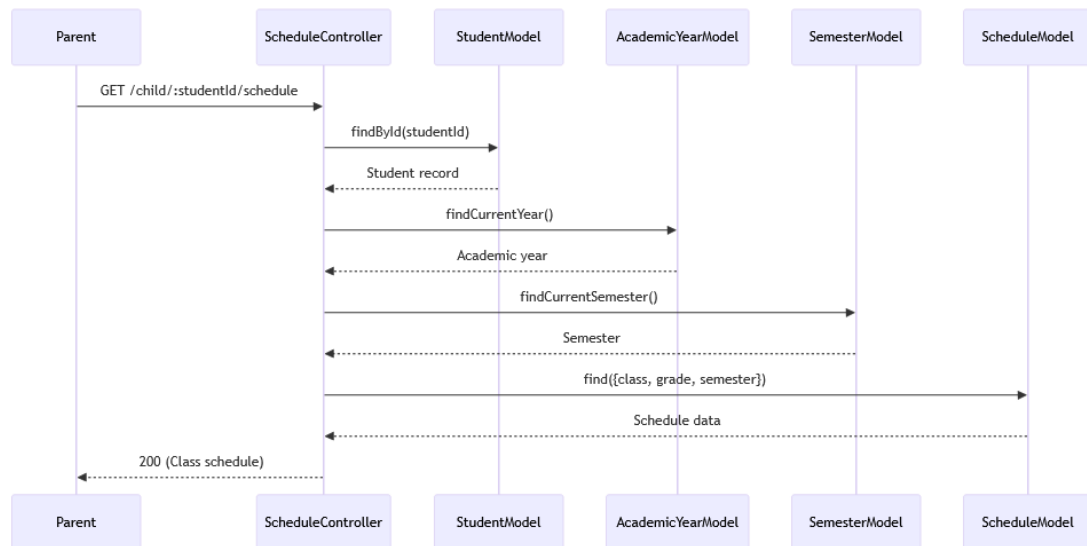
➤ Sequence Diagram: Choose Kid Sequence



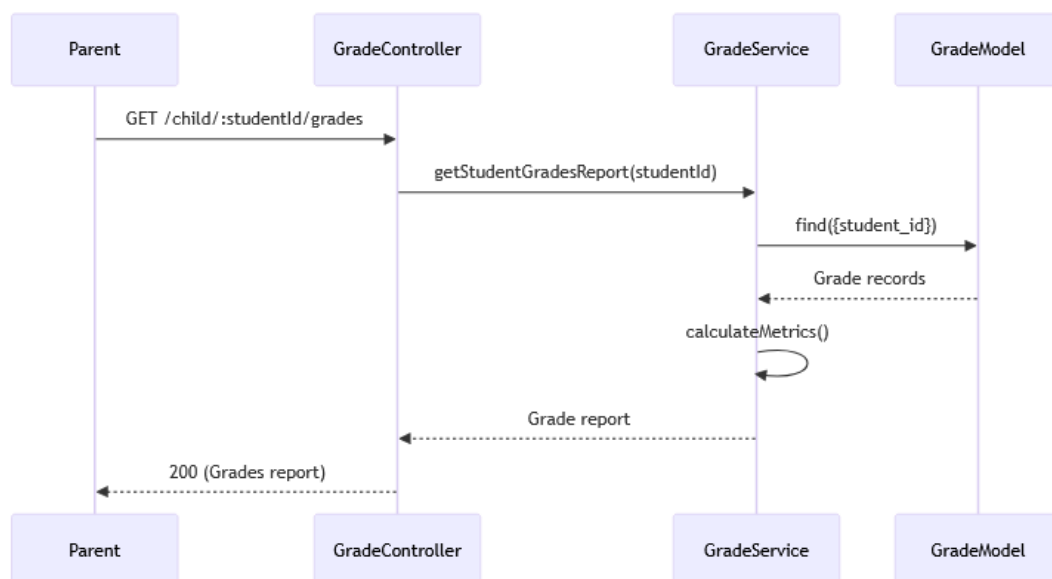
➤ Sequence Diagram: View Child Attendance Sequence



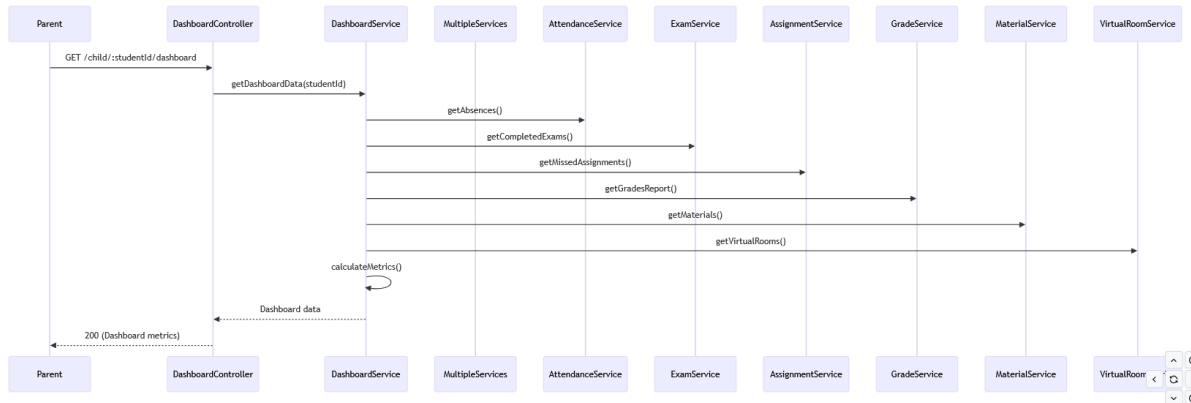
➤ Sequence Diagram: View Child Schedule Sequence



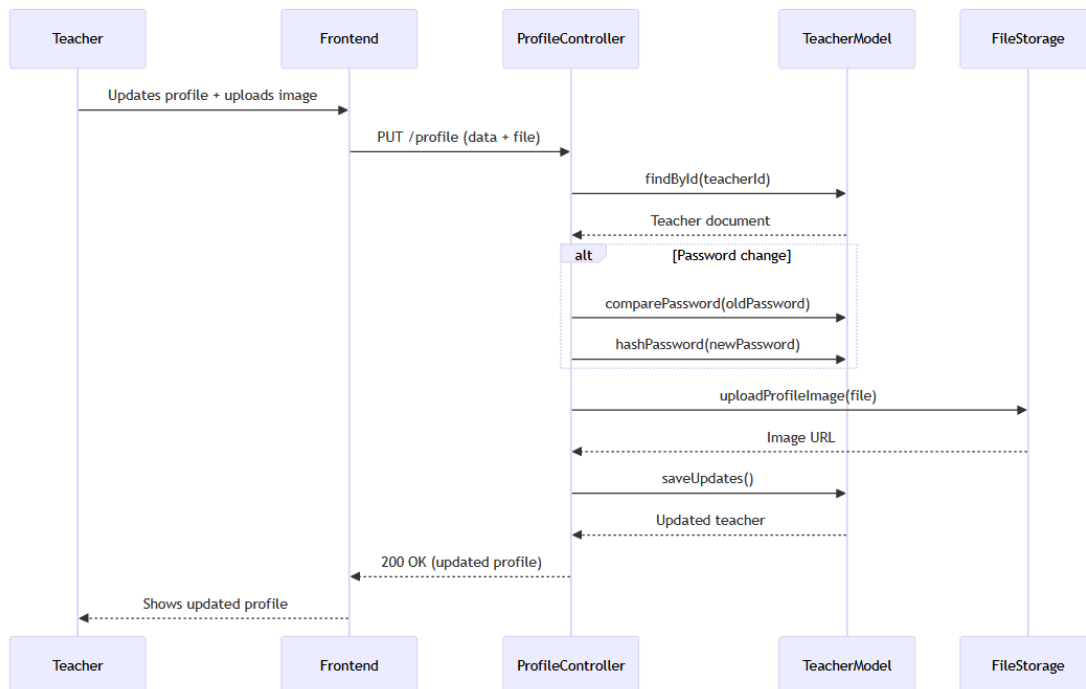
➤ Sequence Diagram: View Child Grades Sequence



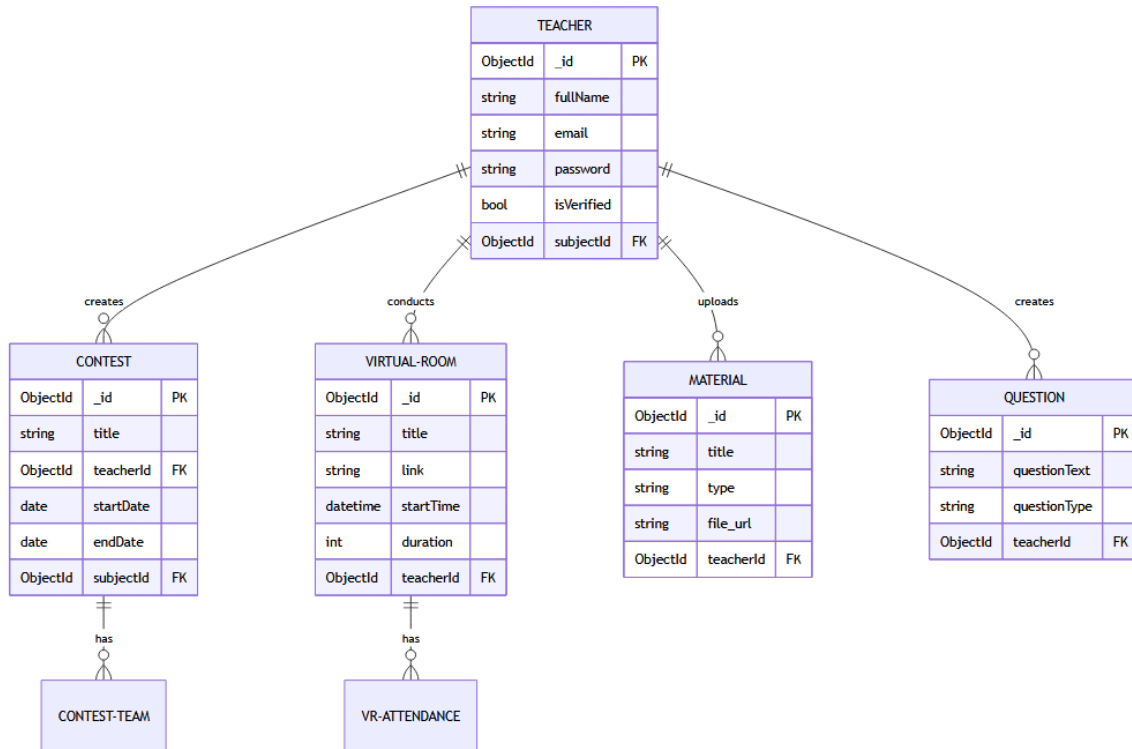
➤ Sequence Diagram: View Child Dashboard Sequence



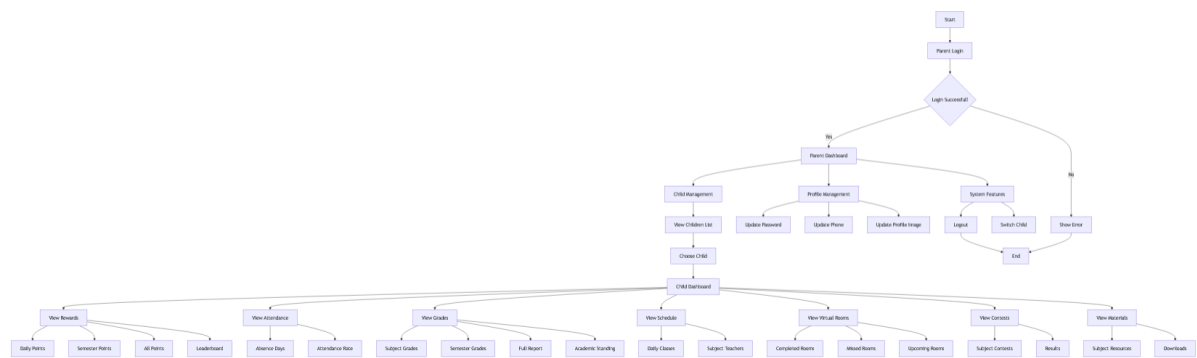
➤ Sequence Diagram: Update Profile Sequence



➤ Entity-Relationship Diagram (ERD)

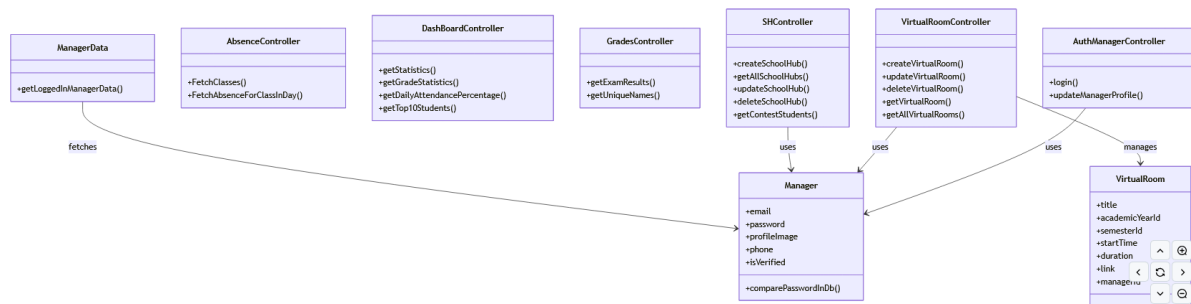


➤ Flowchart

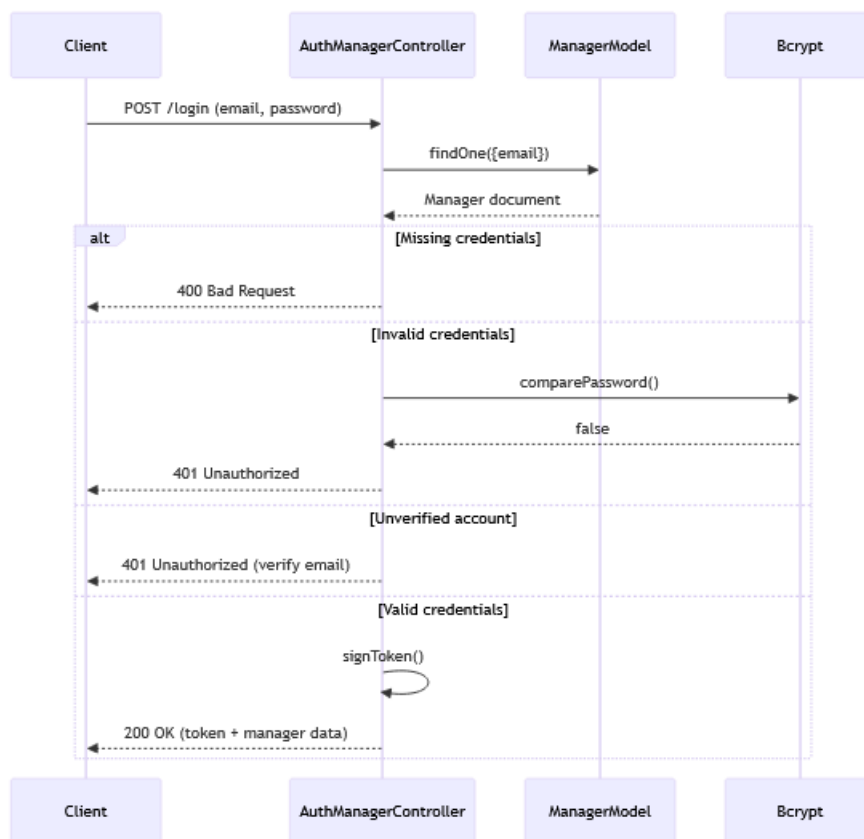


4- manager Diagrams :

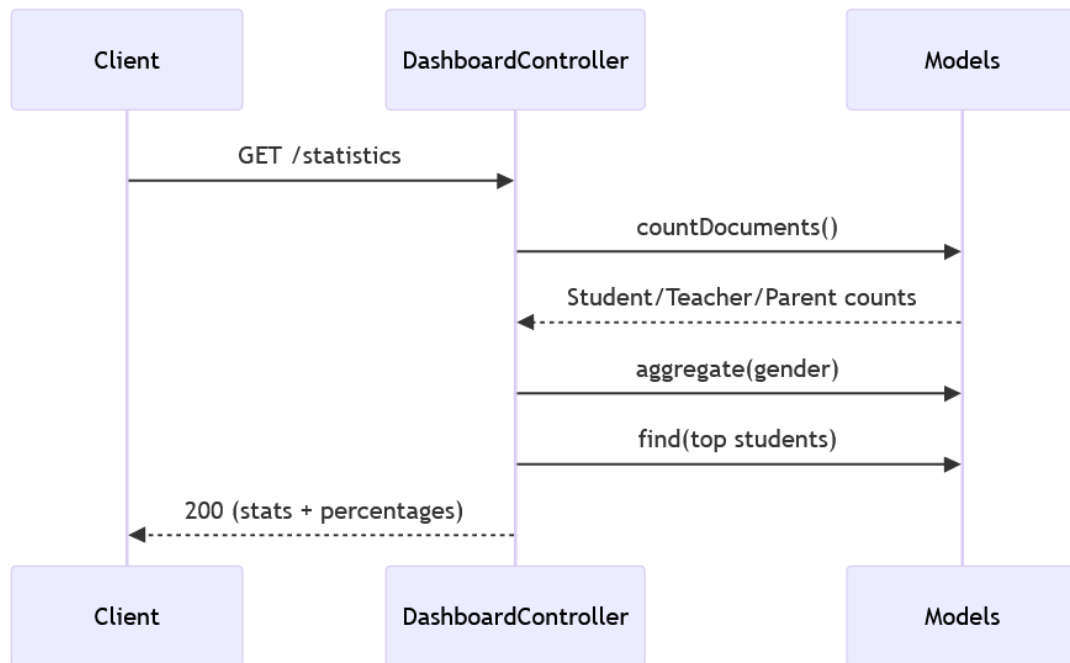
➤ Class Diagram



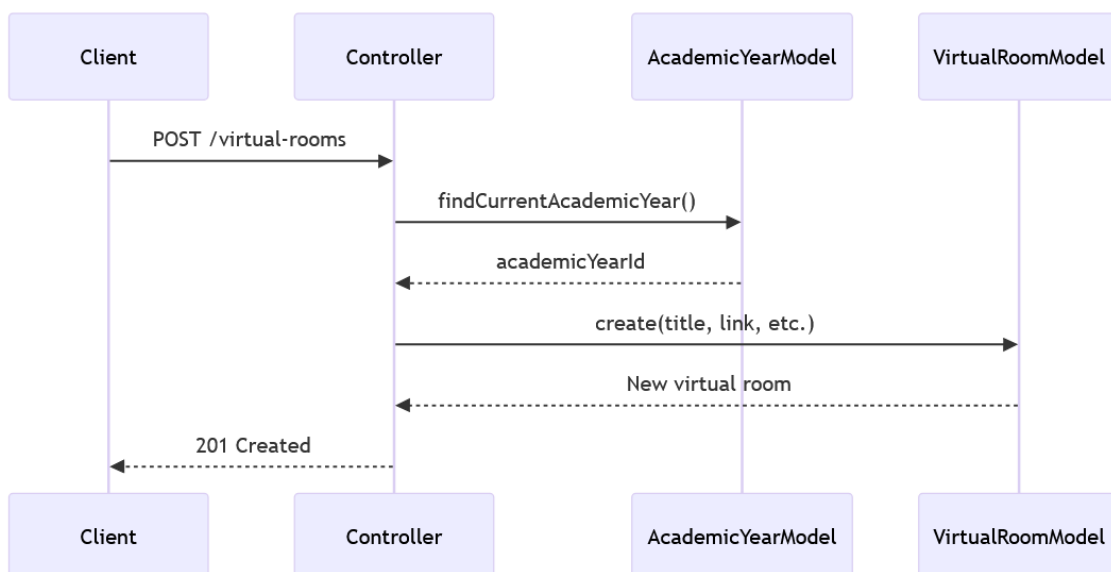
➤ Sequence Diagram: Manager Login



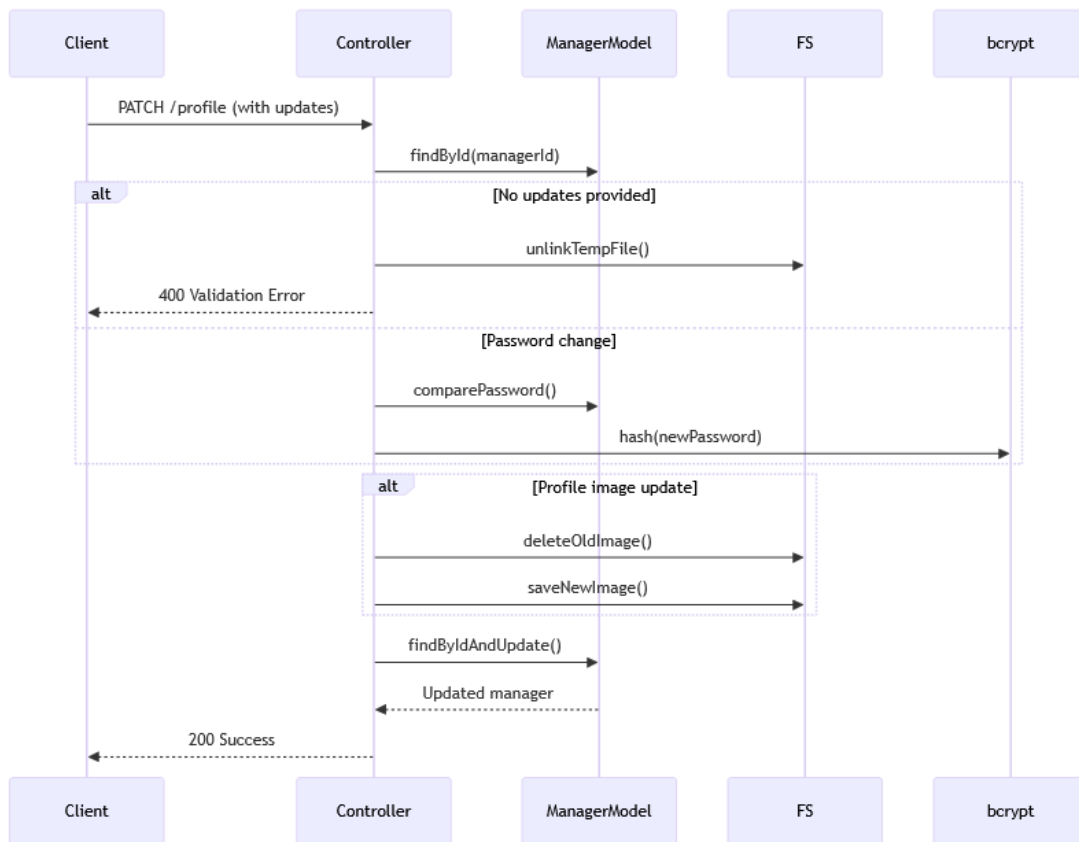
➤ Sequence Diagram: Dashboard Statistics Sequence



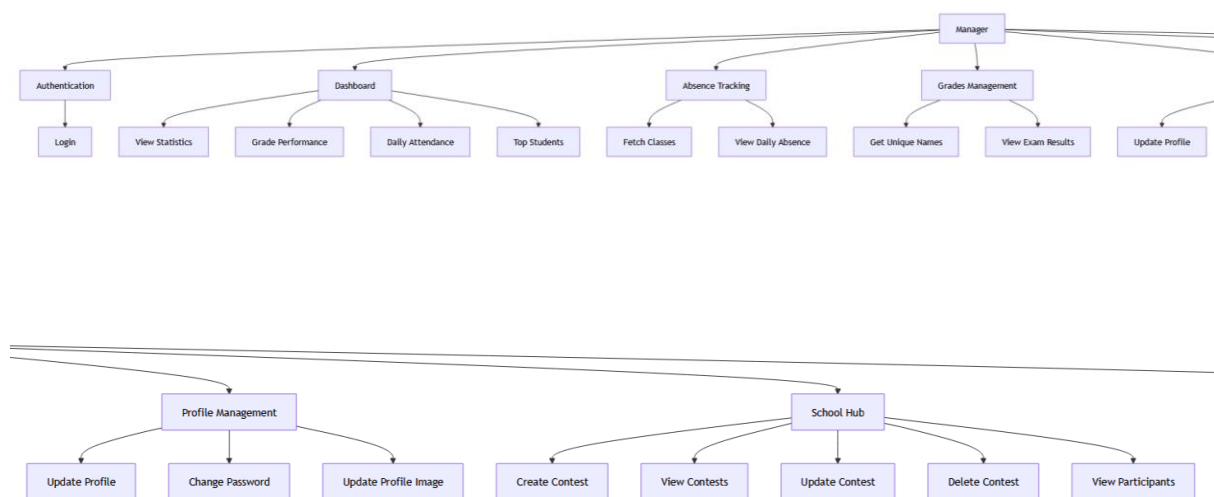
➤ Sequence Diagram: Create Virtual Room Sequence

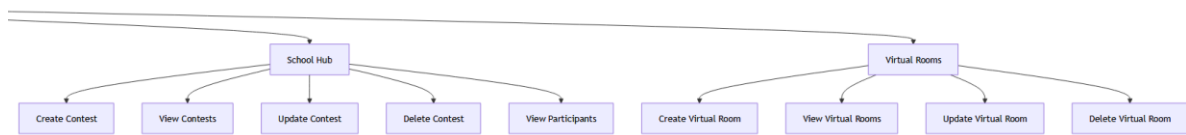


➤ Sequence Diagram: Update Profile Sequence

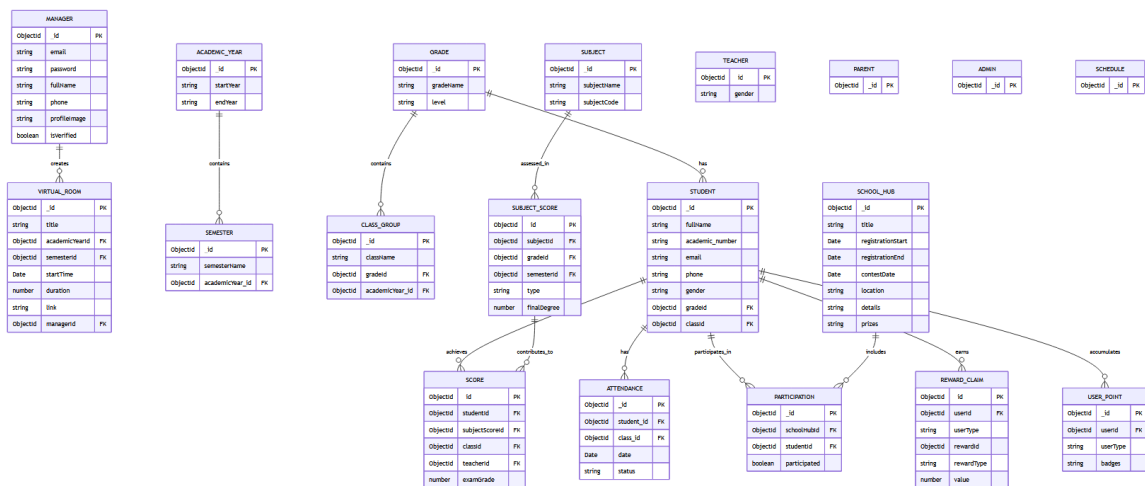


➤ Flowchart



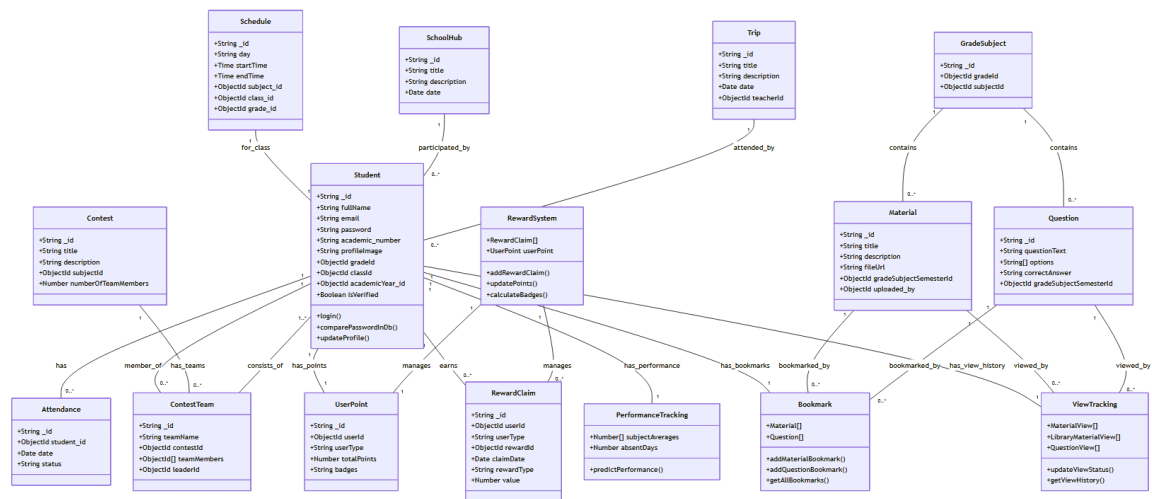


➤ Entity-Relationship Diagram (ERD)

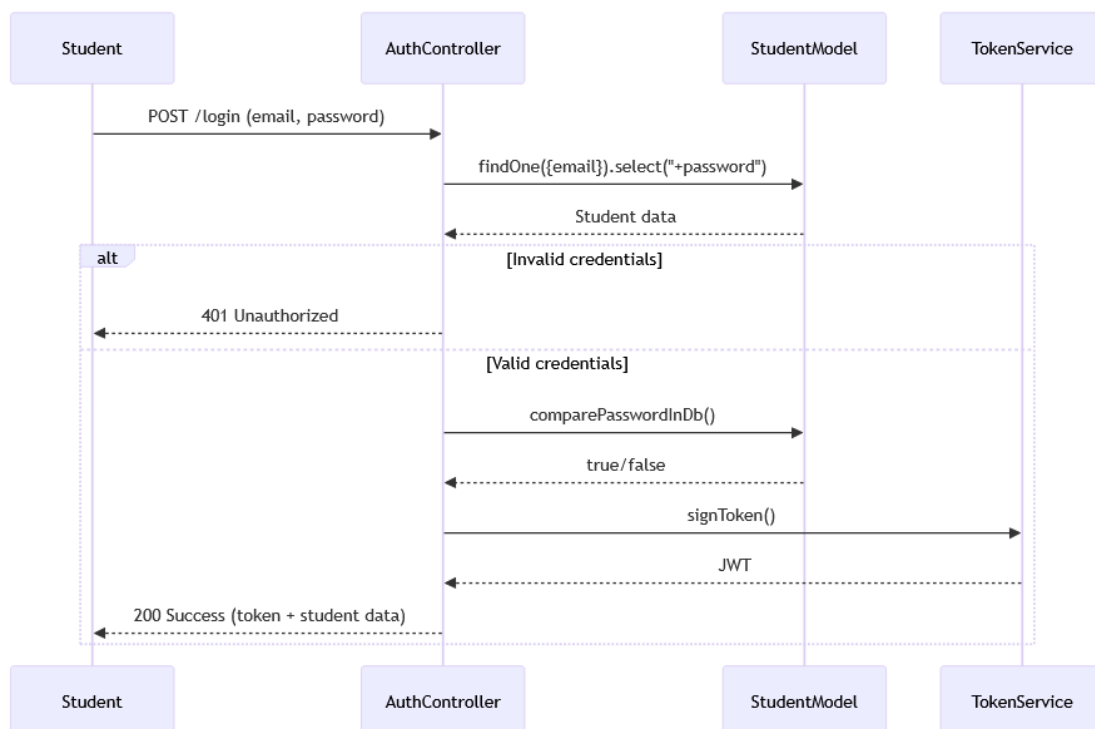


5-Student Diagrams :

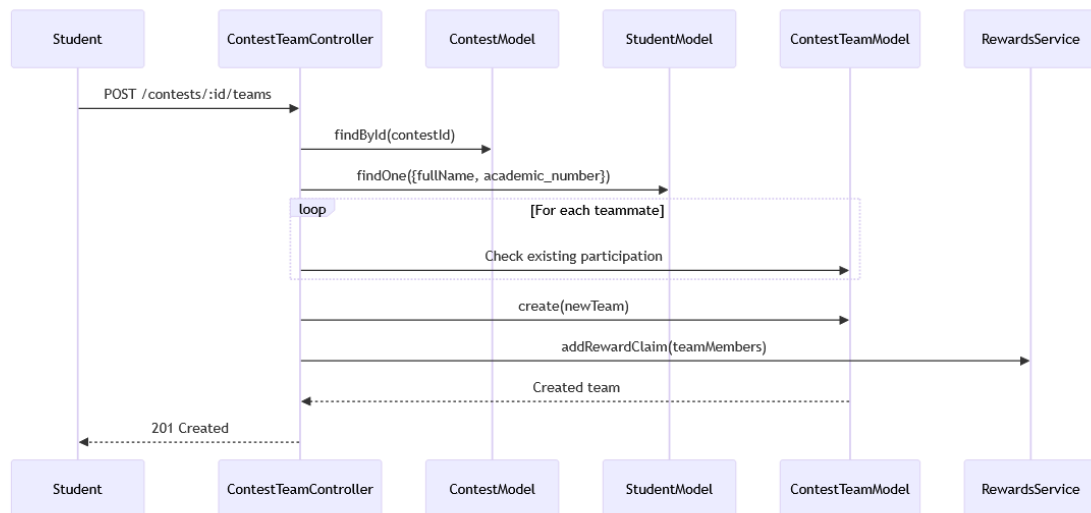
➤ Class diagram



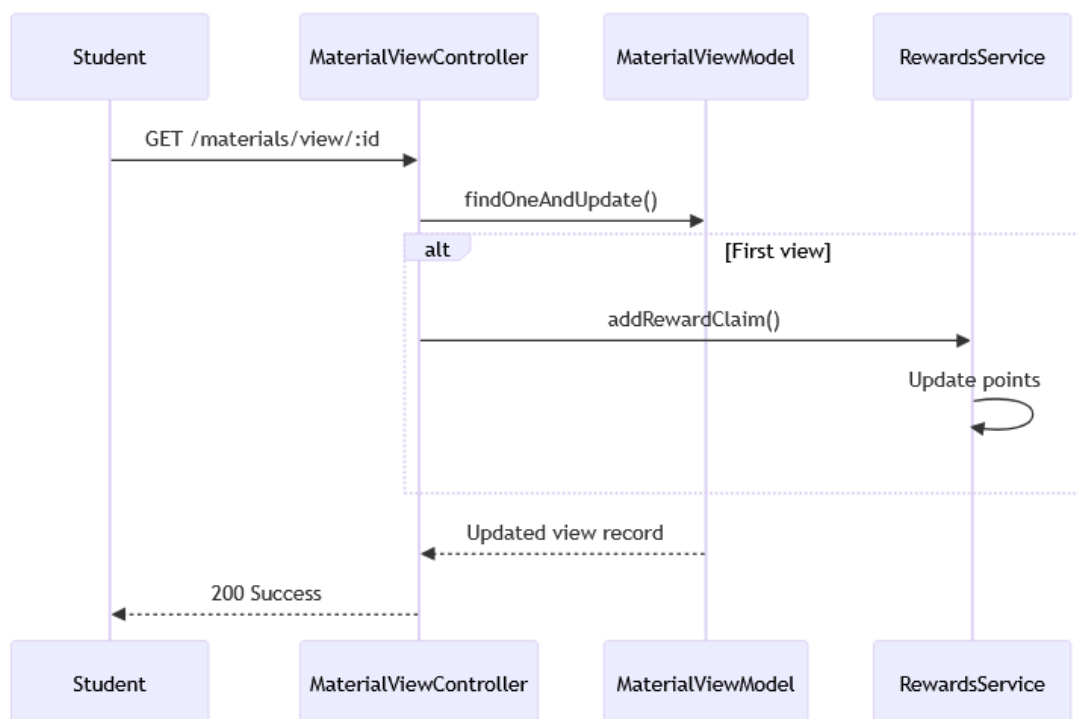
➤ Sequence diagram: Student Login Sequence



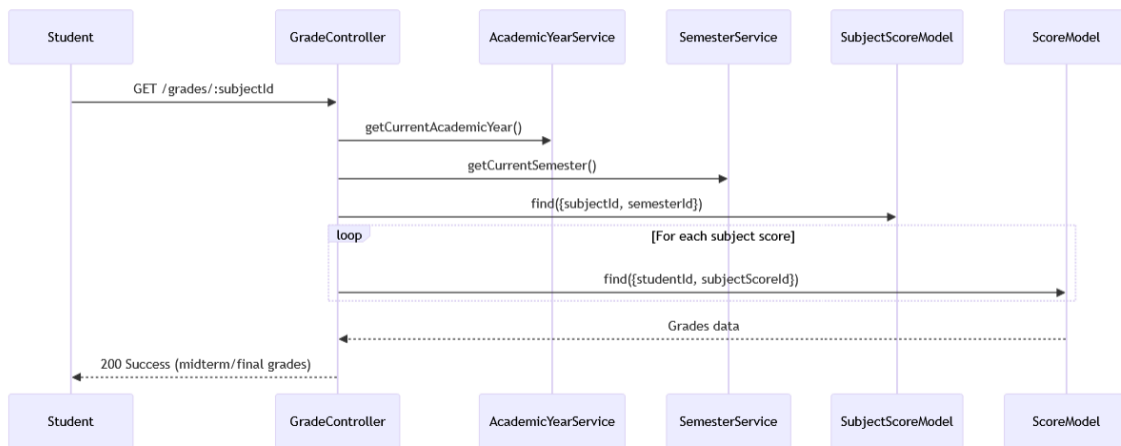
➤ Sequence diagram: Create Contest Team Sequence



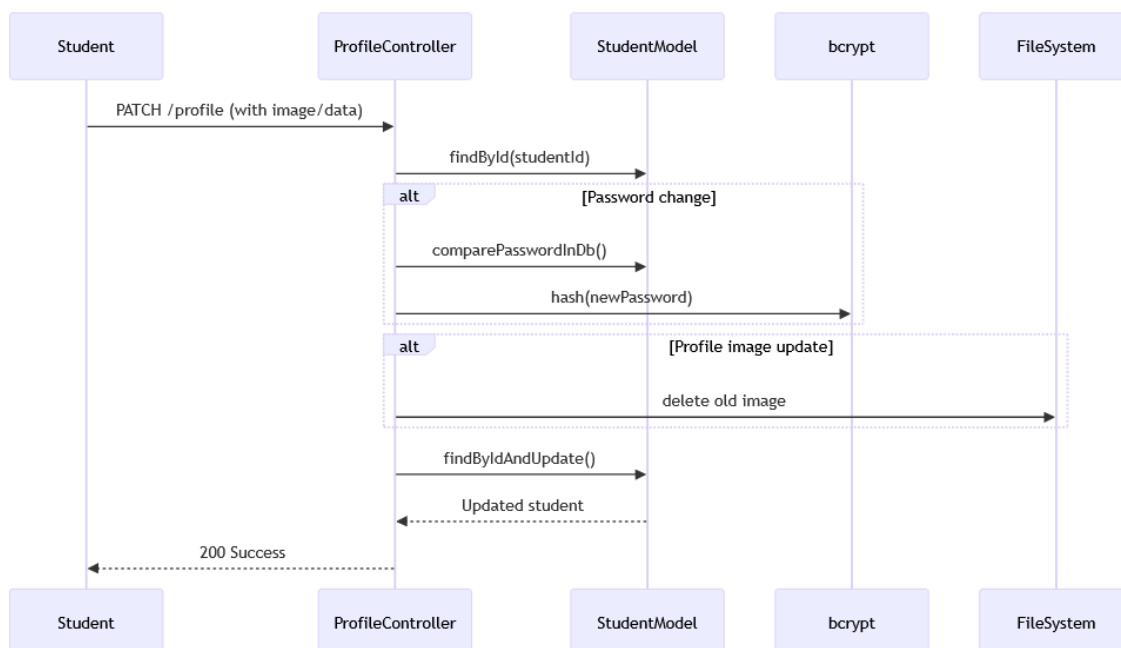
➤ Sequence diagram: View Material Sequence



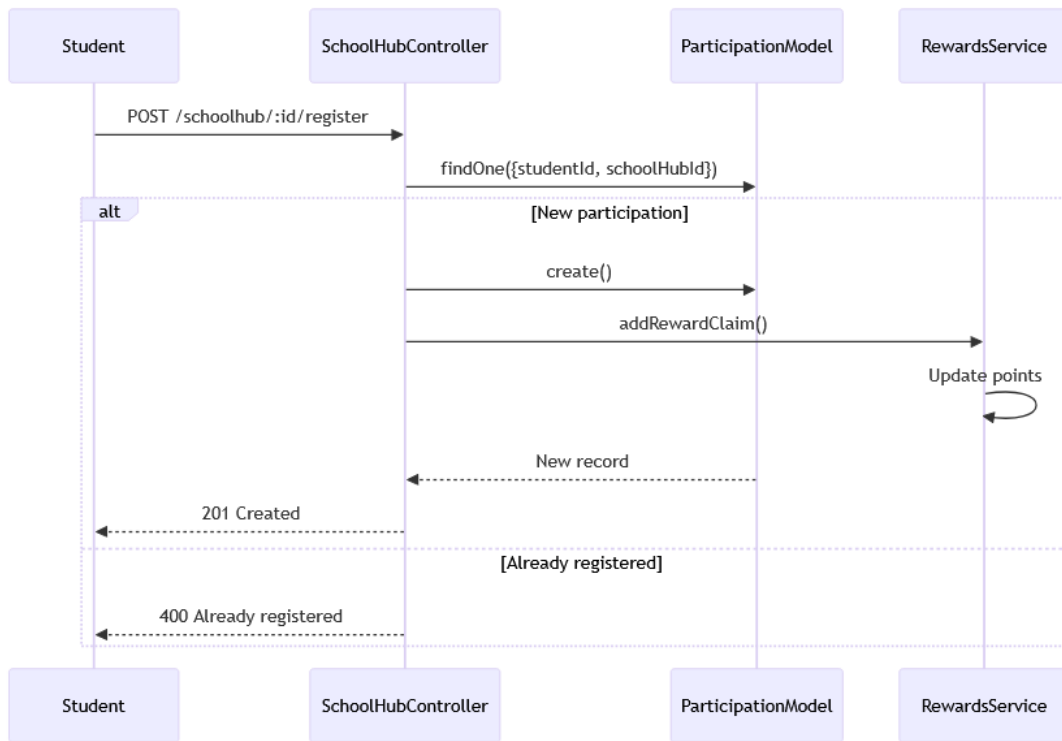
➤ Sequence diagram: Get Grades Sequence



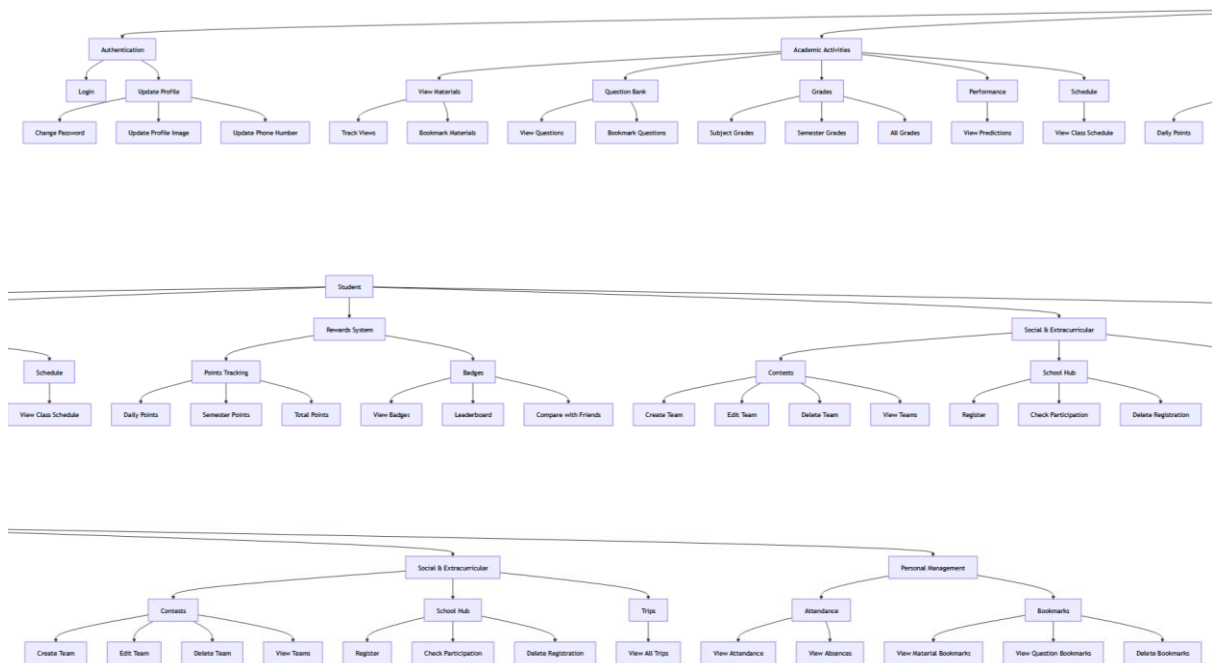
➤ Sequence diagram: Update Profile Sequence



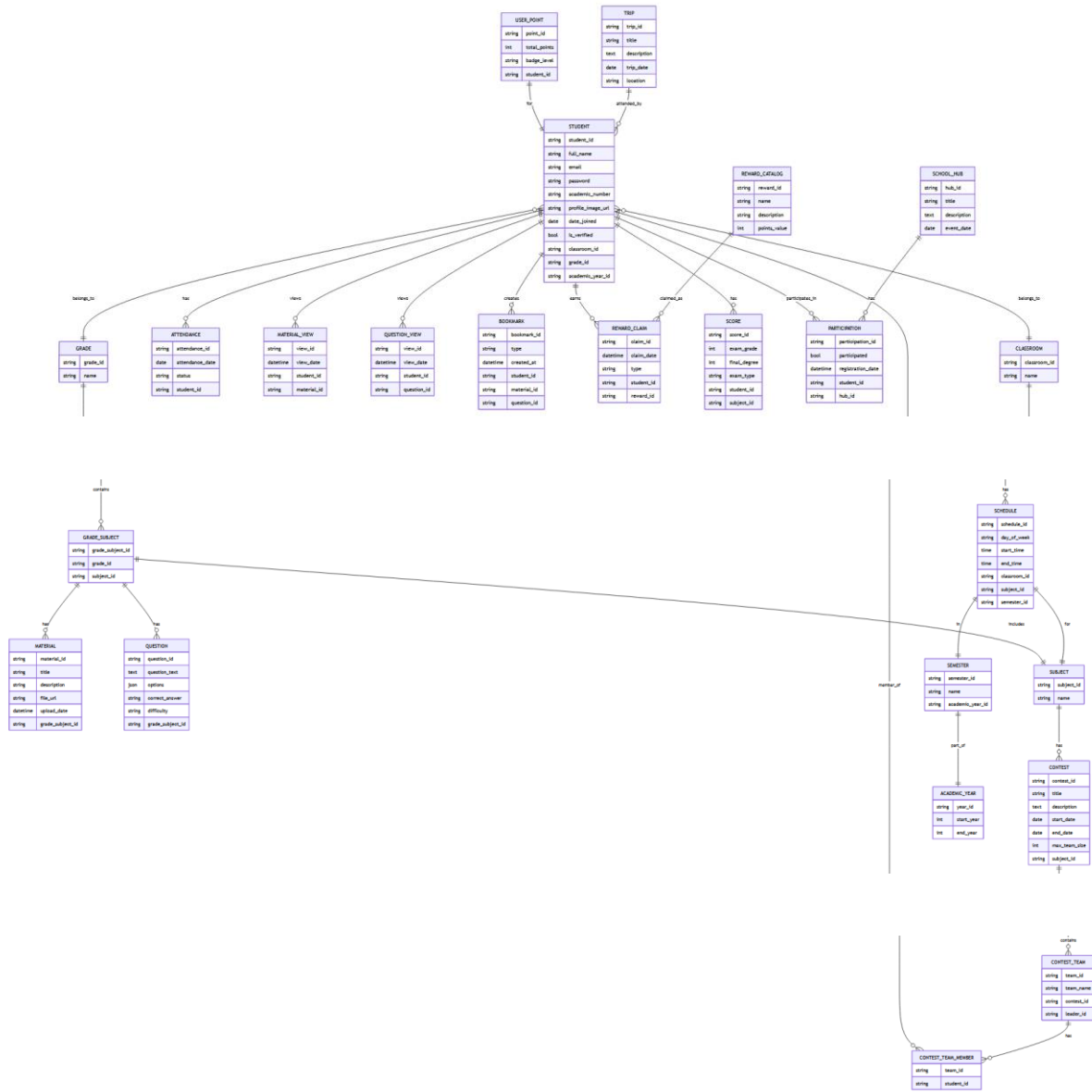
➤ Sequence diagram: Register in School Hub Sequence



➤ Flowchart



➤ Entity-Relationship Diagram (ERD)



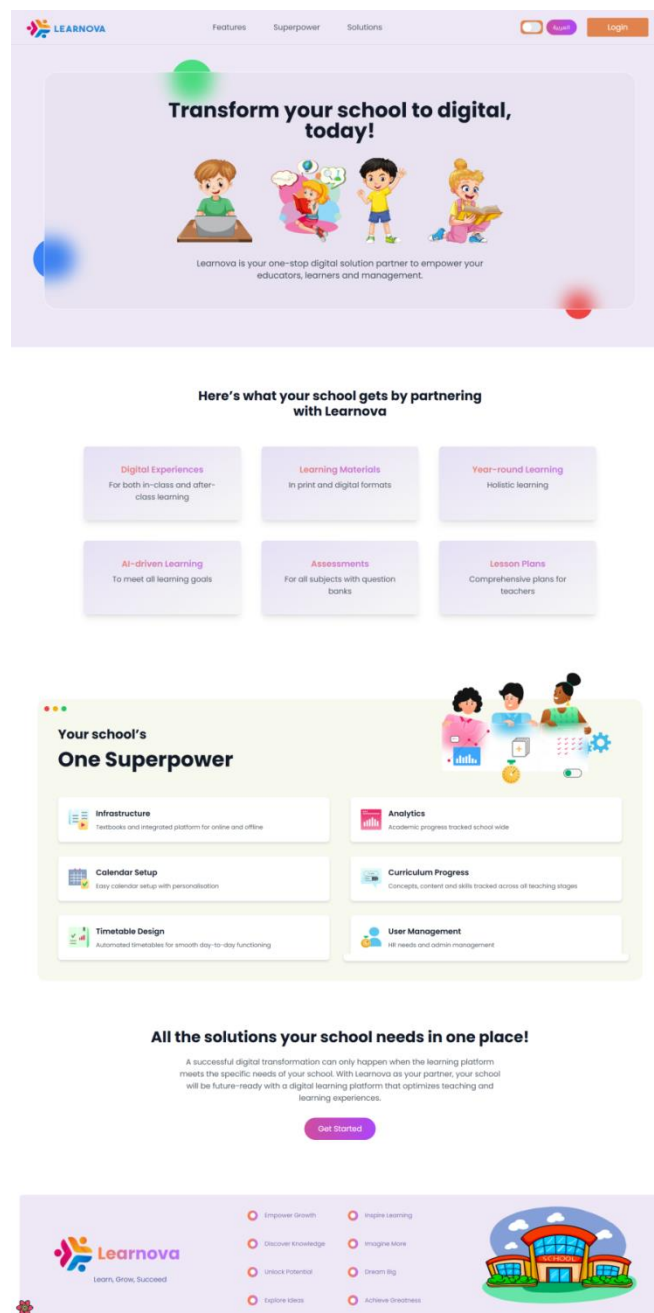
Chapter 5

Implementaion and Result

Onboarding Page

The onboarding page provides a guided introduction to the system for first-time users. It includes a series of short slides or sections explaining the system's purpose, features, and benefits

Outcome: This step helps new users understand the platform before logging in, improving adoption and engagement, especially for students and parents unfamiliar with digital platforms.



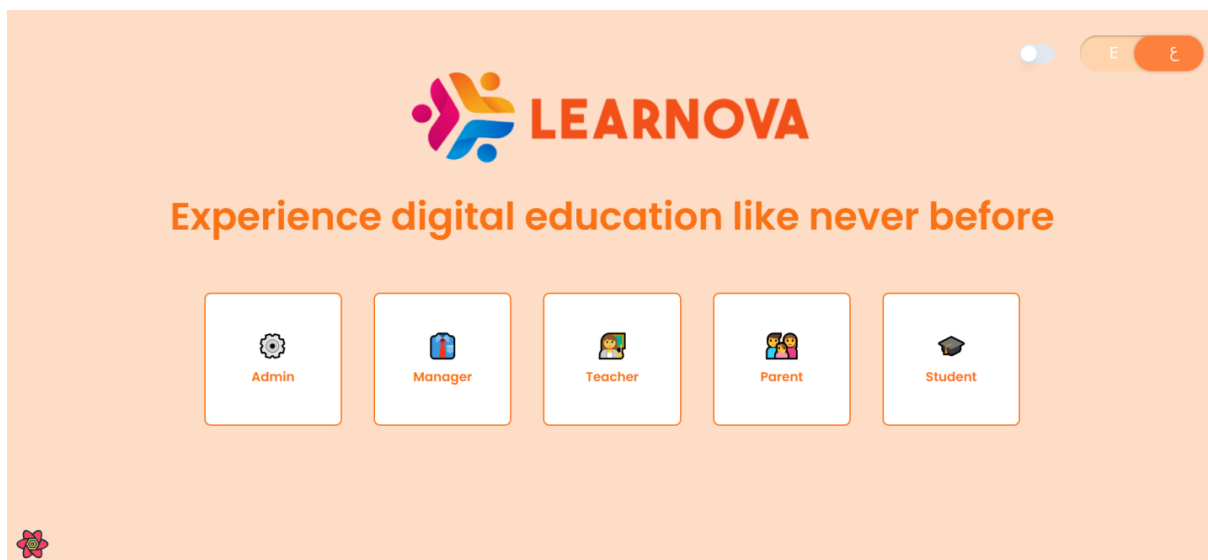
Role Selection Page

After onboarding, users are directed to select their role within the school system. The platform supports multiple roles such as:

- Student
- Teacher
- Parent
- Manager
- Admin

Each role selection personalizes the user experience and controls access to specific features based on their permissions.

Outcome: By selecting their role early, users are redirected to the correct login flow and later see a role-specific dashboard. This helps simplify navigation and improves security by enforcing role-based access control.



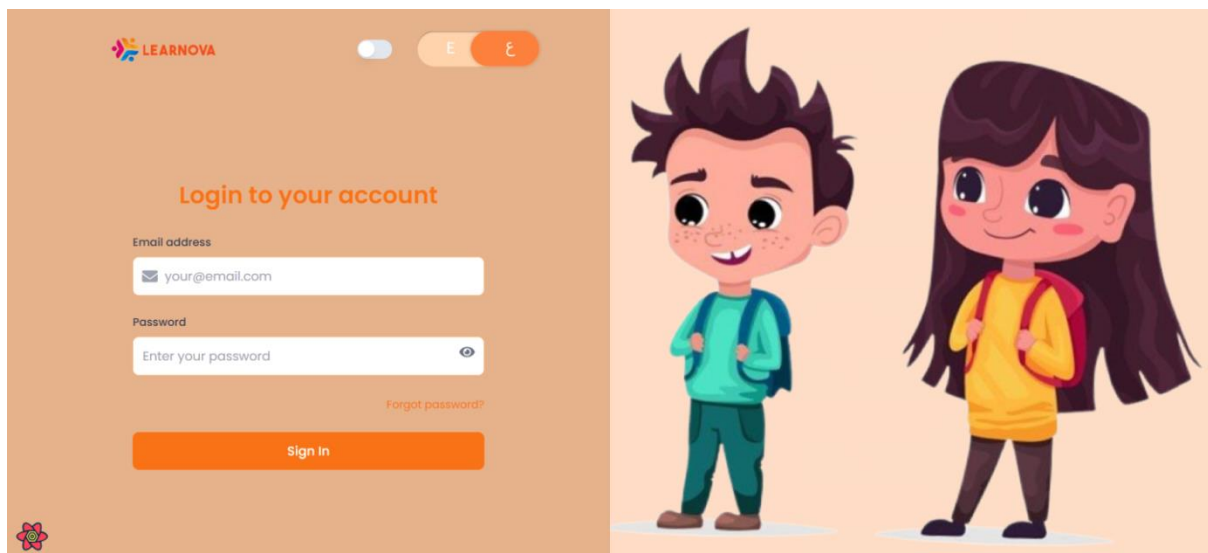
Login Page

The login page allows users to access their accounts using their registered email and password. The interface is clean and responsive, and it provides real-time validation to prevent common input errors. Upon successful login, users are redirected to their dashboard based on their role.

Security Features:

- Input validation (e.g., valid email format)
- Secure password handling
- Feedback for incorrect credentials
- Session handling for authenticated access

Outcome: The login system is secure and reliable, with all authentication requests verified against the backend system. It ensures that only authorized users can access the platform.



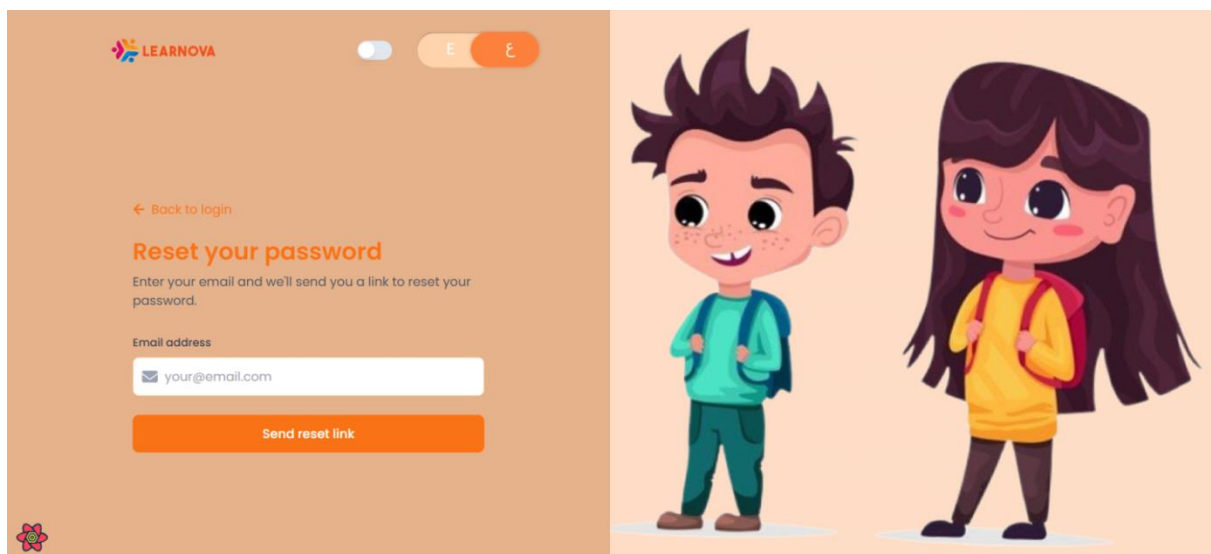
Forget Password Page

For users who forget their password, a dedicated recovery page is available. Users can enter their registered email to receive a reset link via email.

Workflow:

1. User enters their email.
2. A secure, time-limited link is sent.
3. The user is redirected to set a new password.

Outcome: The password reset feature improves user convenience while maintaining system security. It reduces the number of support requests and ensures continued access for users with forgotten credentials.



➤ Admin Role Implementation :

The **Admin** role is one of the most powerful roles in the school system application. It has access to all system features and modules to manage the entire platform effectively

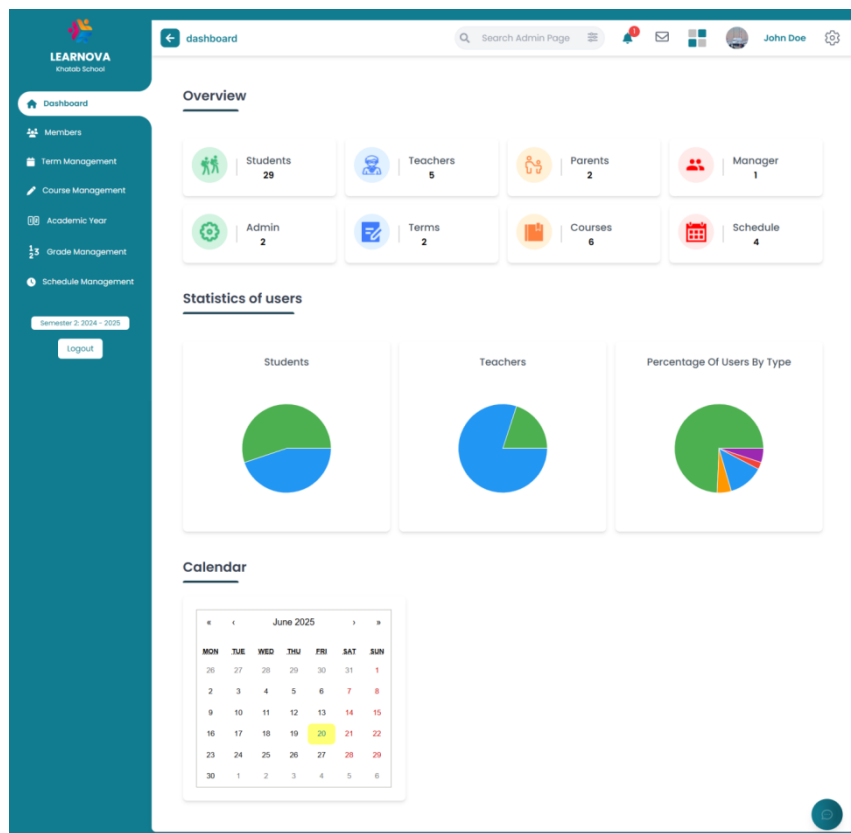
Admin Dashboard

The **Admin Dashboard** is the main control panel for system administrators. It offers a comprehensive overview of the platform's activities and system health. The dashboard is designed with charts, statistics, and quick-action links to important areas of the system.

Features include:

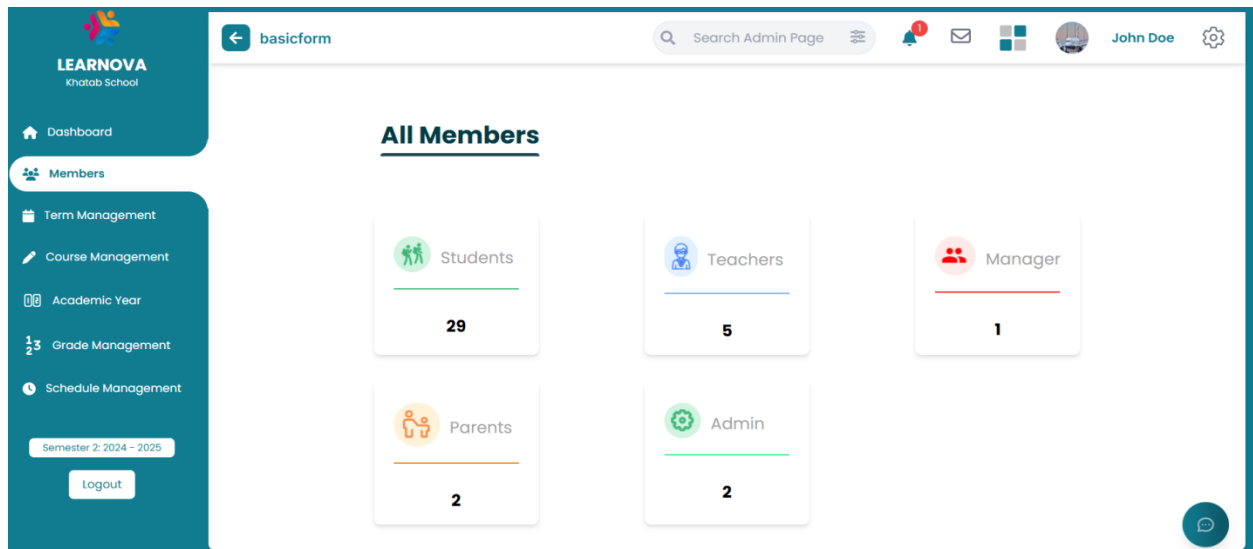
- Total number of users by role (students, teachers, parents, managers, admins)
- Recent system activity and logs
- **Graphs now display statistical numbers and percentage distributions** of users in Learnova

Outcome: The dashboard helps admins monitor and manage the system in real time, enabling quick decision-making and access to system-wide insights.



Members Page

The **Members Page** is where the admin can view, manage, and control all users registered in the system. Each member is categorized by their assigned role. The admin can perform actions such as **viewing details**, **editing information**, **add account** or **delete Account**.



Students

Students are the primary users of the Learnova. Admins can:

View a list of all registered students

Filter by Name or Gender

Add, edit, or remove student accounts

Assign students to Grade

Outcome: The student management feature ensures that all student data is up to date and organized for academic tracking.

- Dashboard
- Members
- Term Management
- Course Management
- Academic Year
- Grade Management
- Schedule Management

Semester 2: 2024 - 2025

Logout

studentform

Search Admin Page

John Doe

Add Student

Full Name

Enter Full Name

Email Address

Enter Email Address

Phone Number

Enter Phone Number

Password

Enter Password

Date Of Birth

mm/dd/yyyy

Gender

Select Gender

Grade

Select Grade

Address

Enter Address

Add Student

Upload Excel File

Choose File | No file chosen

Upload Students

- Dashboard
- Members
- Term Management
- Course Management
- Academic Year
- Grade Management
- Schedule Management

Semester 2: 2024 - 2025

Logout

edit-student

Search Admin Page

John Doe

Edit Student

Full Name

Ahmed Zakaria

Email Address

ahmed@gmail.com

Phone Number

011494031355

Password

.....

Date Of Birth

02/08/2025

Gender

Male

Grade

Grade 1

Update Student

- Dashboard
- Members
- Term Management
- Course Management
- Academic Year
- Grade Management
- Schedule Management

Semester 2: 2024 - 2025

Logout

allstudent

Search Admin Page

John Doe

Students

Export CSV

Add Student

Select Filter

Search for a student by name or gender

Name	Student ID	Email	Class	Gender	Actions
Mahmoud Elframawi	250001	mahmoud@gmail.com	class 1	M	
Ahmed Zakaria	250002	ahmed@gmail.com	class 1	M	
zeko	250003	zeko@gmail.com	class 1	M	
Emma Watson	250006	emmaw@gmail.com	class 1	F	
Sarah Johnson	250007	sarahj22@gmail.com	class 1	F	
John Doe	250008	john.doe@example.com	class 1	M	
Michael Brown	250009	michael.b@example.com	class 1	M	
Olivia Davis	250010	olivia.d@example.com	class 1	F	
James Wilson	250011	james.w@example.com	class 1	M	
Willlaam Taylor	250013	william.t@example.com	class 1	F	

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3

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Page 54 of 131

Parents

Parents can monitor their children's academic progress. Admins can:

- Link parent accounts to student accounts
- Manage parent contact information
- Control parent access rights and communications

Outcome: This strengthens the home-school connection and allows for direct parental engagement.

The screenshot shows the 'Add Parent' form in the LEARNOVA system. The form is titled 'Add Parent' and is located in the 'parentform' section. It contains several input fields: 'Full Name' (with a placeholder 'Enter full name'), 'Email Address' (with a placeholder 'Enter email address'), 'Password' (with a placeholder 'Enter password'), 'Phone Number' (with a placeholder 'Enter phone number'), 'Gender' (a dropdown menu with 'Select Gender'), and 'Student Name 1' (a dropdown menu with 'Select StudentID'). There is also an 'Add Student' button and an 'Add Parent' button at the bottom right of the form. The sidebar on the left shows the 'LEARNOVA Khatib School' logo and navigation links for Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The top bar shows the user 'John Doe' and a search bar.

The screenshot shows the 'Edit Parent' form in the LEARNOVA system. The form is titled 'Edit Parent' and is located in the 'editparentform' section. It contains several input fields: 'Full Name' (with the value 'Ahmed Habib'), 'Email Address' (with the value 'ahmedzakariah@b@gmail.com'), 'Phone Number' (with the value '01060785739'), 'Gender' (a dropdown menu with 'Male'), 'Password' (with a placeholder 'Enter password'), and 'Children' (a dropdown menu with 'zako (250003)' and 'Mahmoud Elframawi (250001)'). There is also an 'Add Student' button and a 'Save Changes' button at the bottom right of the form. The sidebar on the left shows the 'LEARNOVA Khatib School' logo and navigation links for Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The top bar shows the user 'John Doe' and a search bar.

The screenshot shows the 'allparents' page in the LEARNOVA system. The page is titled 'allparents' and is located in the 'allparents' section. It contains a table of parents with columns: Name, Email, Gender, Children, and Actions. There are two rows of data. The first row shows a parent named 'Ahmed Habib' with email 'ahmedzakariah@b@gmail.com', gender 'Male', and children 'zako (250003)' and 'Mahmoud Elframawi (250001)'. The second row shows a parent named 'zako' with email 'ahmadzakar450@gmail.com', gender 'Male', and children 'Williaam Taylor (250013)', 'Ahmed Zakaria (250002)', and 'Emma Watson (250006)'. There is also an 'Add Parent' button and an 'Export CSV' button at the top right of the table. The sidebar on the left shows the 'LEARNOVA Khatib School' logo and navigation links for Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The top bar shows the user 'John Doe' and a search bar.

Name	Email	Gender	Children	Actions
Prof. Ahmed Habib	ahmedzakariah@b@gmail.com	Male	zako (250003) Mahmoud Elframawi (250001)	
zako	ahmadzakar450@gmail.com	Male	Williaam Taylor (250013) Ahmed Zakaria (250002) Emma Watson (250006)	

Admins

Other admins may be created to help manage the system. Super admins can:

- Add or remove admin accounts
- Control admin permissions
- Audit system changes made by other admins

Outcome: Admin-to-admin control improves system transparency and divides workload securely.

adminform

Search Admin Page

John Doe

Add Admin

Full Name
Enter full name

Email Address
Enter email address

Gender
Select Gender

Password
Enter password

Phone Number
Enter phone number

Add Admin

Semester 2: 2024 - 2025

Logout

editadminform

Search Admin Page

John Doe

Edit Admin

Full Name
zeko

Email Address
zeko@gmail.com

Gender
Male

Password
Enter new password (leave blank to keep current)

Phone Number
01060785736

Save Changes

Semester 2: 2024 - 2025

Logout

alladmins

Search Admin Page

John Doe

Admin

Export CSV

Add Admin

Select Filter

Search for a admin by name, email or gender

Name	Email	Gender	Phone	Actions
John Doe	john.doe@example.com	M	01090264016	 
zeko	zeko@gmail.com	M	01060785736	 

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Semester 2: 2024 - 2025

Logout

Managers

Managers may include heads of departments or school-level coordinators. Admins can:

- Assign managerial responsibilities to specific school areas (e.g., curriculum, discipline)
- Oversee department performance
- Track communications or internal reports

Outcome: The manager role helps distribute administrative responsibilities and supports multi-level supervision.

The screenshot shows the 'Add Manager' form in the LEARNOVA Khatib School system. The form is titled 'Add Manager' and contains the following fields: Full Name (text input), Email Address (text input), Gender (dropdown menu), Password (text input), and Phone Number (text input). There is an 'Add Manager' button at the bottom right of the form. The left sidebar shows the navigation menu with options: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The top header shows the user's name 'John Doe' and a search bar.

The screenshot shows the 'Edit Manager' form in the LEARNOVA Khatib School system. The form is titled 'Edit Manager' and contains the following fields: Full Name (text input), Email Address (text input), Gender (dropdown menu), Password (text input), and Phone Number (text input). There is a 'Save Changes' button at the bottom right of the form. The left sidebar shows the navigation menu with options: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The top header shows the user's name 'John Doe' and a search bar.

The screenshot shows the 'allmanagers' page in the LEARNOVA Khatib School system. The page is titled 'Manager' and contains a table of managers. The table has columns: Name, Email, Gender, and Actions. There is a search bar and a filter dropdown at the top. The table shows one manager: John Doe, with email john.doe@example.com and gender M. There is an 'Add Manager' button and an 'Export CSV' button at the top right. The left sidebar shows the navigation menu with options: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The top header shows the user's name 'John Doe' and a search bar.

Name	Email	Gender	Actions
John Doe	john.doe@example.com	M	Edit Delete

Teachers

Teachers are responsible for academic instruction. Admins can:

- View and manage teacher profiles
- Assign teachers to specific subjects and classes
- Review their schedules or course materials
- Enable/disable access or reset accounts

Outcome: This functionality allows admins to efficiently coordinate teaching resources and verify staff involvement

The screenshot shows the 'Add Teacher' form in the LEARNOVA Khatab School admin interface. The interface has a teal sidebar with navigation links: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The main content area is titled 'teacherform' and has a search bar. Below the search bar are two tabs: '1 Personal Data' (active) and '2 Academic Data'. The 'Add Teacher' form is displayed with the following fields: Full Name (text input), Date Of Birth (date picker), Gender (dropdown menu), Address (text input), Phone Number (text input), Email Address (text input), Subject (dropdown menu), and Password (text input). A blue 'Add Teacher' button is at the bottom right of the form.

The screenshot shows the 'Academic Data' form in the LEARNOVA Khatab School admin interface. The interface has a teal sidebar with navigation links: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The main content area is titled 'teacherform' and has a search bar. Below the search bar are two tabs: '1 Personal Data' and '2 Academic Data' (active). The 'Academic Data' form is displayed with the following fields: Teacher (dropdown menu) and Class (dropdown menu). A blue 'Submit Academic Data' button is at the bottom right of the form.



LEARNOVA
Khatat School

Dashboard

Members

Term Management

Course Management

Academic Year

Grade Management

Schedule Management

Semester 2: 2024 - 2025

Logout

edit-teacher

Search Admin Page

John Doe

Edit Teacher

Full Name

Nabil Hanoh

Email Address

nabil@gmail.com

Gender

Male

Password

.....

Phone Number

01028264919

Subject

Science

Save Changes



LEARNOVA
Khatat School

Dashboard

Members

Term Management

Course Management

Academic Year

Grade Management

Schedule Management

Semester 2: 2024 - 2025

Logout

edit-class-teacher

Search Admin Page

John Doe

Edit Class Teacher

Teacher-Subject

ahmed habib - English

Class-Academic Year

class 1 - 2024 - 2025

Save Changes



LEARNOVA
Khatat School

Dashboard

Members

Term Management

Course Management

Academic Year

Grade Management

Schedule Management

Semester 2: 2024 - 2025

Logout

allteachers

Search Admin Page

John Doe

Teachers

Export CSV

Add Teacher

Select Filter

Search for a class teacher by class, subject, teacher, or academic year

Class	Subject	Teacher	Academic Year	Actions
class 1 - Grade 1	Science	Nabil Hanoh	2024 - 2025	 
class 1 - Grade 3	Science	Nabil Hanoh	2024 - 2025	 
class 1 - Grade 5	Science	Nabil Hanoh	2024 - 2025	 
class 1 - Grade 4	Science	Nabil Hanoh	2024 - 2025	 
class 1 - Grade 1	English	ahmed habib	2024 - 2025	 
class 1 - Grade 1	Science	Wessam Mohamed	2024 - 2025	 

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LEARNOVA
Khatat School

Dashboard

Members

Term Management

Course Management

Academic Year

Grade Management

Schedule Management

Semester 2: 2024 - 2025

Logout

allteachers

Search Admin Page

John Doe

Teachers

Export CSV

Add Teacher

Select Filter

Search for a teacher by name, email , gender or subject

Name	Subject	Email	Academic Number	Gender	Actions
 Nabil Hanoh	Science	nabil@gmail.com	TEA-001-Science	M	 
 yasser	Math	yasser@gmail.com	TEA-002-Math	M	 
 yasser1	Science	yasser1@gmail.com	TEA-003-Science	M	 
 ahmed habib	English	ahmadzakr450@gmail.com	TEA-004-English	M	 
 Wessam Mohamed	Science	wessam@gmail.com	TEA-005-Math	F	 

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Schedule Management

Description:

Schedule management allows the admin to create, update, delete and view the weekly class schedules for all grades and classes. This includes assigning subjects to specific time slots, linking teachers, and ensuring no scheduling conflicts occur.

Key Features:

- Define time slots per day (e.g., 8:00–9:00 AM)
- Assign subjects and teachers to each slot
- Separate schedules for each class or grade
- Conflict-checking to prevent double-booking of teachers or rooms
- Visual calendar or table view for easy monitoring

Result:

This module improves classes organization and ensures students and teachers follow a structured timetable with no overlaps.

The screenshot displays the 'Add Schedule' form within the LEARNOVA Khatib School Admin Panel. The interface includes a sidebar with navigation links: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, and Schedule Management. The main content area is titled 'Add Schedule' and contains several dropdown menus for selection: Academic Year, Semester Name, Subject Name, Teacher Name, Grade, Class, Day, and From/To time slots. A 'Logout' button is visible in the sidebar, and an 'Add Schedule' button is at the bottom of the form.

LEARNOVA
Khatib School

Dashboard
Members
Term Management
Course Management
Academic Year
Grade Management
Schedule Management

Semester 2, 2024 - 2025
Logout

← scheduleform Search Admin Page John Doe

Add Schedule

Academic Year: Select Academic Year Semester Name: Select semester
Subject Name: Select subject Teacher Name: Select teacher
Grade: Select grade Class: Select class
Day: Select day From: --:-- -- To: --:-- --
Add Schedule

edit-schedule

Search Admin Page

John Doe

Update Schedule

Academic Year: 2024 / 2025

Semester Name: Semester 2

Subject Name: Science

Teacher Name: Nabil Hanah

Grade: Grade 1

Class: class 1

Day: Monday

From: 09:00 AM

To: 09:45 AM

Update Schedule

allschedules

Search Admin Page

John Doe

Schedule

Export CSV Add Schedule

Select Filter

Search for a schedule by subject, day, or time

Subject Name	Teacher	Grade	Day	From	To	Actions
Science	Nabil Hanah	Grade 1	Monday	09:00	09:45	
Science	yasser1	Grade 1	Tuesday	10:00	11:00	
Science	Nabil Hanah	Grade 3	Thursday	13:00	14:00	
Science	Wessam Mohamed	Grade 3	Tuesday	09:00	10:00	

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Terms Management

Description:

The admin can define and control the academic **terms or semesters** (e.g., Term 1 and Term 2). Each term defines the duration and structure of part of the academic year.

Key Features:

- Set start and end dates for each term
- Enable or disable grading within a specific term
- Assign different course loads per term if needed

Result:

Having flexible term management helps the school align its calendar with ministry or national academic requirements and supports schools with trimester or semester systems.

The screenshot shows the 'Add Term' form in the LEARNOVA Khatolab School admin interface. The left sidebar contains navigation links: Dashboard, Members, Term Management (active), Course Management, Academic Year, Grade Management, and Schedule Management. The main content area has a header 'termform' and a search bar. The form itself has two dropdown menus: 'Term Name' with 'Select Term' and 'Academic Year' with 'Select Year'. An 'Add Term' button is at the bottom right of the form. The footer shows 'Semester 2: 2024 - 2025' and a 'Logout' button.

The screenshot shows the 'Edit Term' form in the LEARNOVA Khatolab School admin interface. The left sidebar is the same as the previous screenshot. The main content area has a header 'edit-term' and a search bar. The form has two dropdown menus: 'Select Term' with 'Semester 1' and 'Select Academic Year' with '2024 - 2025'. An 'Update Term' button is at the bottom right of the form. The footer shows 'Semester 2: 2024 - 2025' and a 'Logout' button.

The screenshot shows the 'allTerms' page in the LEARNOVA Khatolab School admin interface. The left sidebar is the same as the previous screenshots. The main content area has a header 'allTerms' and a search bar. The page displays a list of terms under the heading 'Terms'. There are two terms listed: 'Semester 1' for the academic year '2024 - 2025' and 'Semester 2' for the academic year '2024 - 2025'. Each term has a green icon and a red trash icon. An 'Add Term' button is at the top right. A pagination bar at the bottom shows '< 1 >'. The footer shows 'Semester 2: 2024 - 2025' and a 'Logout' button.

Grade Management

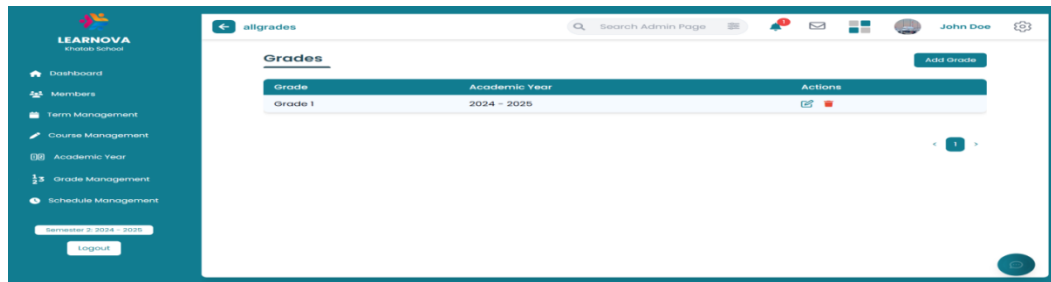
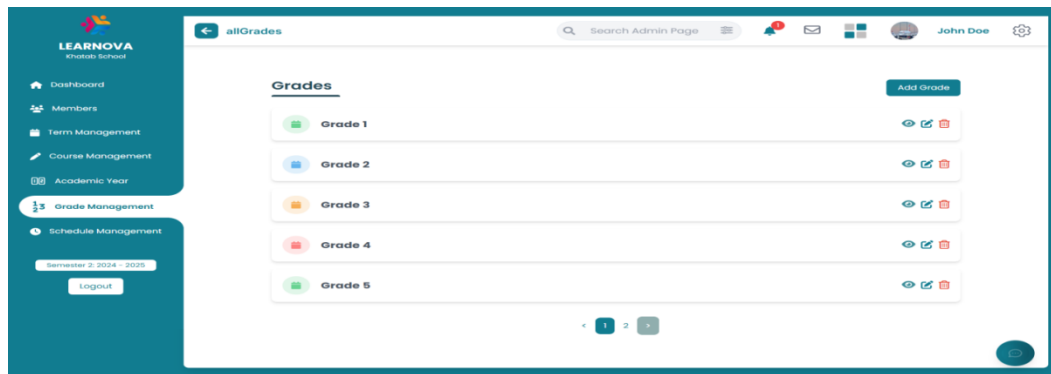
The Grade Management module allows the admin to control academic levels (e.g., Grade 1, Grade 2) within the school system. Admins can:

- **Create** new grades with names, codes, and descriptions
- **Edit** existing grade information
- **Delete** unused or outdated grades
- **View** all grades in a structured table with key details
- **Assign** each grade to a specific **academic year**

The screenshot shows the 'Add Grade' form in the LEARNOVA system. The left sidebar contains the navigation menu with options: Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management (selected), and Schedule Management. The main content area has a header with 'gradeform' and a search bar. Below the header are two tabs: 'Add Grade' (active) and 'Assign Grade'. The 'Add Grade' form has a text input field labeled 'Enter Grade Name' with a placeholder 'Enter grade name' and a blue 'Add Grade' button.

The screenshot shows the 'Assign Grade' form in the LEARNOVA system. The left sidebar is the same as the previous screenshot. The main content area has a header with 'assigngrade' and a search bar. Below the header are two tabs: 'Add Grade' and 'Assign Grade' (active). The 'Assign Grade' form has two dropdown menus: 'Select Grade' with a placeholder '-- Select Grade --' and 'Academic Year' with a placeholder '-- Select Academic Year --'. There is a blue 'Assign Grade' button at the bottom.

The screenshot shows the 'Edit Grade' form in the LEARNOVA system. The left sidebar is the same as the previous screenshots. The main content area has a header with 'editGradeform' and a search bar. Below the header is a tab labeled 'Edit Grade'. The form has a text input field labeled 'Enter Grade Name' with a placeholder 'Grade 1' and a blue 'Save Changes' button.



Academic Year Management

Description:

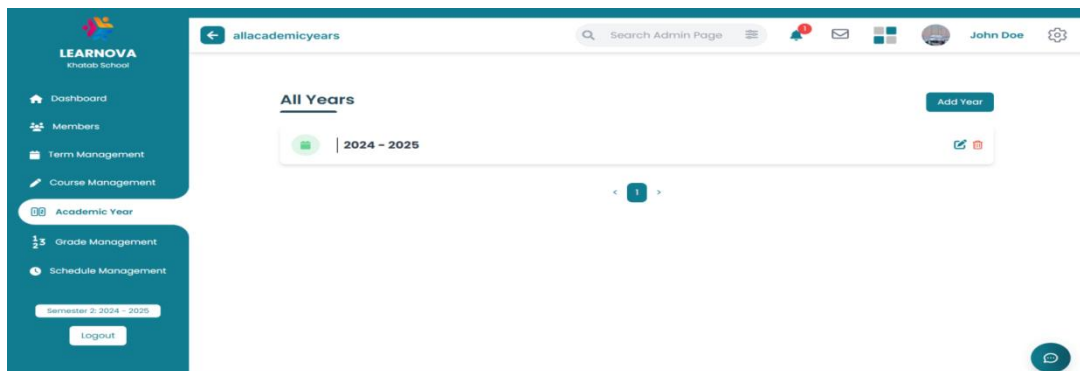
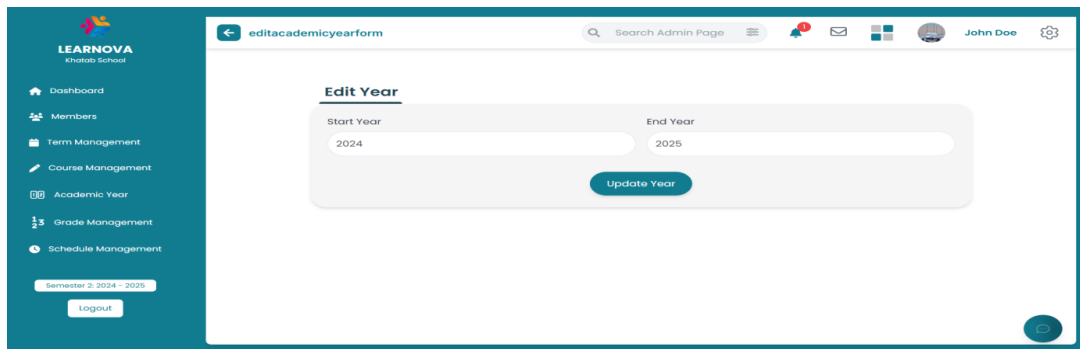
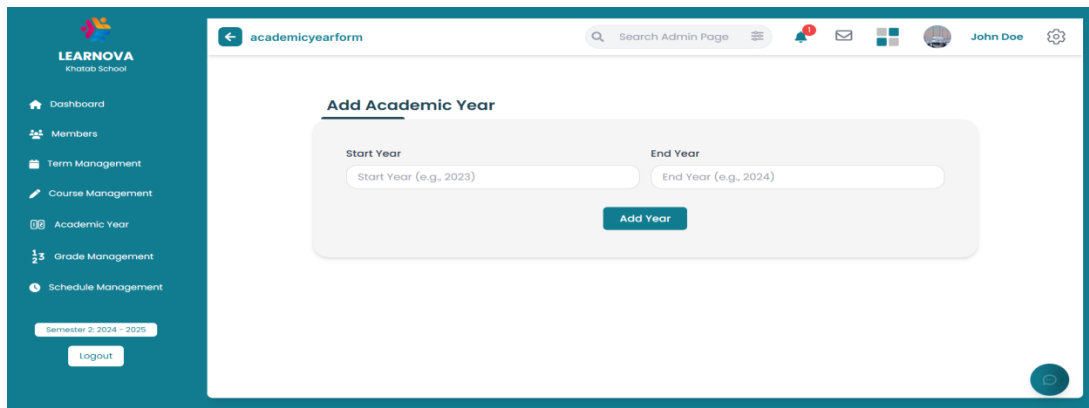
Admins can define academic years and manage their lifecycle — including activating or closing the current year, and archiving old ones.

Key Features:

- Create and name academic years (e.g., 2024–2025)
- Set global start and end dates
- Close academic years to freeze data (e.g., grades, attendance)
- Archive past years for reporting or audits
- Link each academic year to term and course settings

Result:

Academic year control ensures that the system is up to date and prevents data overlap. It also allows historical reporting and safe transitions between years.



Courses Management

Description:

The admin can create and manage all academic courses and subjects offered in the school. This includes defining course details, linking them to grades, assigning semesters, and connecting subjects to specific teachers.

Key Features:

- **Add/Edit/Delete Courses** (e.g., Math, Science, History)
- **Create/Edit/Delete Subjects** under each course

- **Assign Courses and Subjects** to specific **grades** and **semesters**
- **Link Teachers** to each subject or course
- **View Subjects** in a structured format by grade or academic year
- **Enable/Disable Courses** based on the academic year

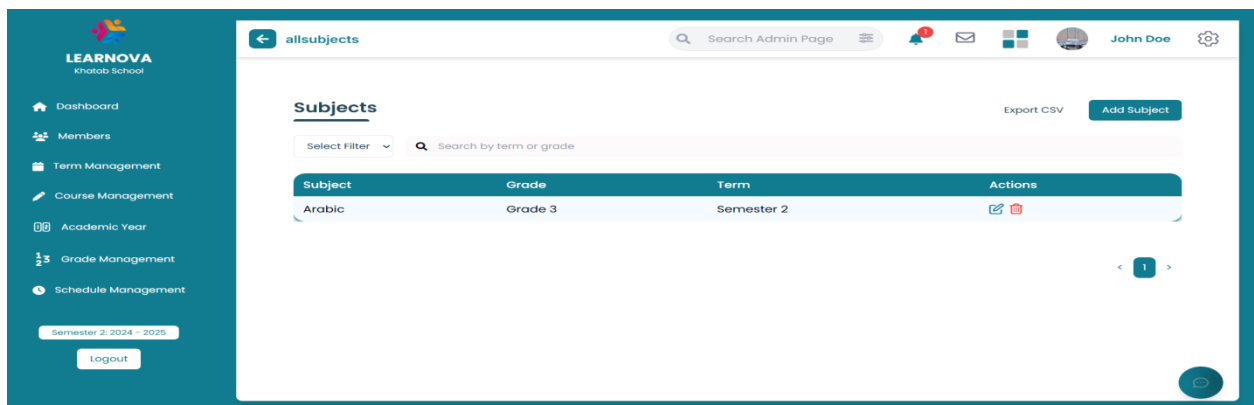
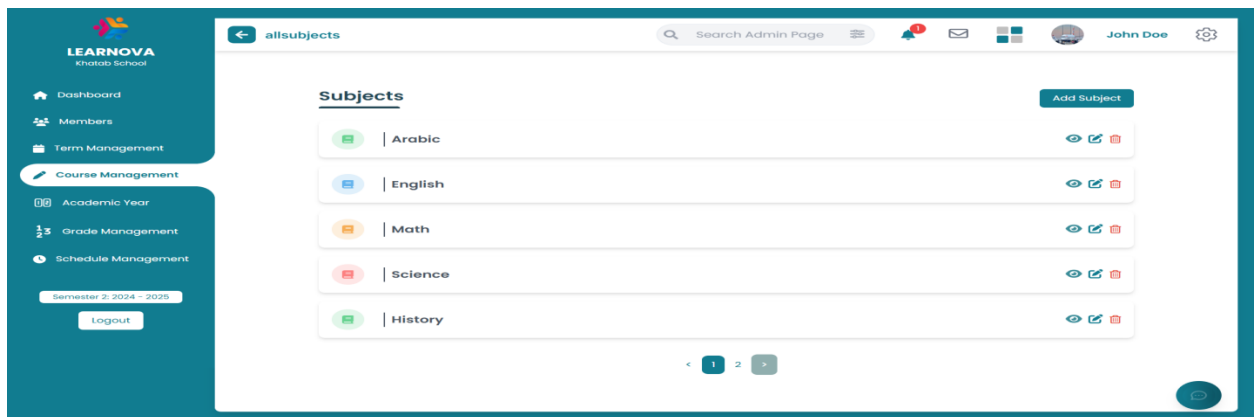
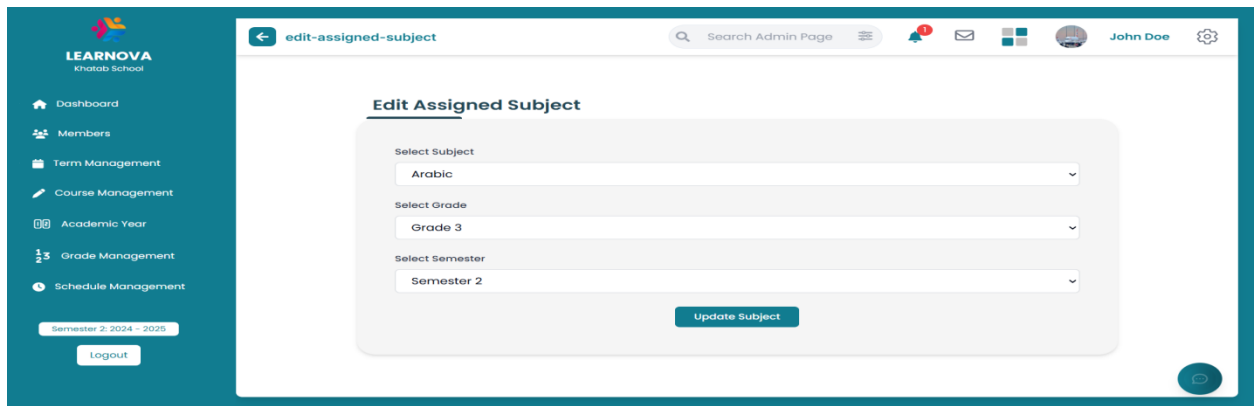
Result:

The system offers full flexibility in managing the school's academic structure. Admins can tailor courses and subjects to match grade levels and term schedules, ensuring proper delivery and oversight of the curriculum.

This screenshot shows the 'Add Subject' form in the LEARNOVA Khatib School Admin Panel. The interface includes a sidebar with navigation links (Dashboard, Members, Term Management, Course Management, Academic Year, Grade Management, Schedule Management) and a top header with a search bar, user profile (John Doe), and settings. The main content area has a breadcrumb 'addsubject' and two tabs: 'Add Subject' (active) and 'Assign Subject'. The 'Add Subject' form contains a text input field labeled 'Enter Subject Name' with a placeholder 'Enter subject name' and an 'Add Subject' button.

This screenshot shows the 'Assign Subject' form in the LEARNOVA Khatib School Admin Panel. The interface is similar to the previous one, but the 'Assign Subject' tab is active. The form contains three dropdown menus: 'Select Subject' (placeholder: '-- Select Subject --'), 'Select Grade' (placeholder: '-- Select Grade --'), and 'Select Semester' (placeholder: '-- Select Semester --'). Below these is an 'Assign Subject' button.

This screenshot shows the 'Edit Subject' form in the LEARNOVA Khatib School Admin Panel. The interface is similar to the previous ones, but the 'edit-subject' breadcrumb and 'Edit Subject' tab are present. The form contains a text input field labeled 'Enter Subject Name' with the value 'Arabic' and a 'Save Changes' button.



➤ **Manager Role Implementation :**

Manager Dashboard

Description:

The Manager Dashboard provides a visual overview of academic and administrative data relevant to the grades or departments managed. It supports data-driven decision-making by presenting key statistics in real time

Key Features:

- **User Overview**

Displays the total number of users in the system, categorized by role (students, teachers, parents, etc.). It helps managers understand the distribution of users they oversee.

- **Percentage of Users by Type**

A pie or bar chart shows the proportion of students, teachers, and parents relative to the total users. This allows managers to assess system usage and role coverage.

- **Calendar Integration**

An interactive calendar highlights important academic dates, including class schedules, exams, meetings, and events. It allows managers to stay organized and anticipate upcoming activities.

- **Top-Performing Students**

A ranking list or chart shows students with the highest academic performance based on grades. This helps managers identify high achievers and track progress across terms.

- **Absence Statistics**

Attendance data is presented visually to help managers monitor student participation. Trends in absenteeism can be observed by grade, subject, or date range.

Result:

The dashboard enables managers to efficiently monitor academic operations, student progress,

and staff involvement. It centralizes critical data into one accessible view, enhancing coordination, responsiveness, and planning accuracy.



Exam Schedule Management

Description:

Managers can create, edit, and manage **exam schedules** for each grade .ensuring time efficiency and coordination across subjects and teachers.

Key Capabilities:

- **Add** exam schedules for specific grades and terms
- **Edit** exam details such as date, time, and assigned subject
- **Delete** outdated or incorrect exam entries
- **View** all scheduled exams in a structured format
- **Avoid conflicts** between overlapping subjects or staff
- **Publish** exam schedules to students and parents for visibility

Result:

This feature enables managers to create accurate, conflict-free exam timetables. It ensures clear communication and smooth coordination across academic staff and student groups.

The screenshot displays the 'Add Exam Schedule' form within the LEARNOVA Khatata School management interface. The form is titled 'Add Exam Schedule' and is located on the 'create-exam-schedule' page. The left sidebar contains navigation links: Dashboard, School Hubs, Absence, Schedule, Grades For Students, and Virtual Classrooms. The main form fields include: Academic Year (dropdown), Semester (dropdown), Grade (dropdown), Subject (dropdown), Exam Date (text input with a calendar icon), Start Time (time input), and End Time (time input). There are also buttons for 'Add other subjects' and 'Upload'. The top navigation bar shows the user's name 'John Doe' and a search bar.

**LEARNOVA**
Khatib School

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023-2024

Logout

← update-exam-schedule

Search Manager Page

John Doe

← Edit Exam Schedule

Subject

Arabic

Exam Date

09/01/2025

Start Time

10:00 AM

End Time

11:00 AM

Subject

English

Exam Date

09/02/2025

Start Time

10:00 AM

End Time

11:01 AM

Subject

Science

Exam Date

09/03/2025

Start Time

10:00 AM

End Time

11:00 AM

Add other subjects

Update

**LEARNOVA**
Khatib School

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023-2024

Logout

← get-exam-schedules

Search Manager Page

John Doe

1 Weekly Schedule

2 Exams Schedule

← Exam Schedules 2024-2025

Add Exam Schedule

Grade	Semester	Start Time
Grade 1	Semester 2	9/1/2025
Grade 2	Semester 2	1/1/2025
Grade 3	Semester 2	4/10/2025
Grade 4	-	No exam schedule available for current academic year
Grade 5	Semester 2	4/4/2025
Grade 5	Semester 1	6/19/2025
Grade 6	-	No exam schedule available for current academic year

**LEARNOVA**
Khatib School

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023-2024

Logout

← get-exam-schedule

Search Manager Page

John Doe

← Exam Schedule Details

Edit Exam Schedule

Delete Schedule

Academic Year	Grade	Semester
2024 - 2025	Grade 1	Semester 2

Subject	Exam Date	Start Time	End Time
Arabic	9/1/2025	10:00	11:00
English	9/2/2025	10:00	11:01
Science	9/3/2025	10:00	11:00

Weekly Schedule Management

Description:

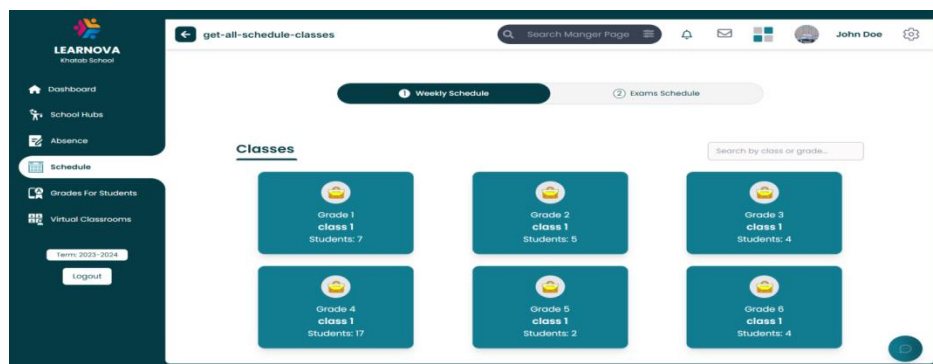
Managers are able to **view** and manage weekly class schedules for the grades or classes assigned to them. This helps maintain an organized academic flow across subjects and days.

Key Capabilities:

- **View** weekly schedules by **class ID only**
- See subject distribution across days and periods
- Ensure that each class has a balanced and complete timetable
- Monitor teacher assignments to avoid scheduling conflicts
- Display schedules in calendar or table format for easy review

Result:

This feature helps managers maintain visibility over class timetables, ensuring efficient subject coverage and teacher coordination. While limited to viewing by class ID, it supports streamlined class-level oversight.



The screenshot displays the 'get-weekly-schedule' page for Semester 2 (2024-2025). It features a table with columns for days of the week and rows for time slots. The table shows that most slots are empty, except for one slot on Tuesday at 10:00, which is occupied by 'Science Yasser Grade 1 class 1 1 hour'. There is an 'Export as PDF' button in the top right corner of the table area.

Time	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
08:00	No class	No class	No class	No class	No class	No class	No class
09:00	No class	No class	No class	No class	No class	No class	No class
10:00	No class	No class	Science Yasser Grade 1 class 1 1 hour	No class	No class	No class	No class
11:00	No class	No class	No class	No class	No class	No class	No class
12:00	No class	No class	No class	No class	No class	No class	No class
13:00	No class	No class	No class	No class	No class	No class	No class
14:00	No class	No class	No class	No class	No class	No class	No class

Absence Management

Description:

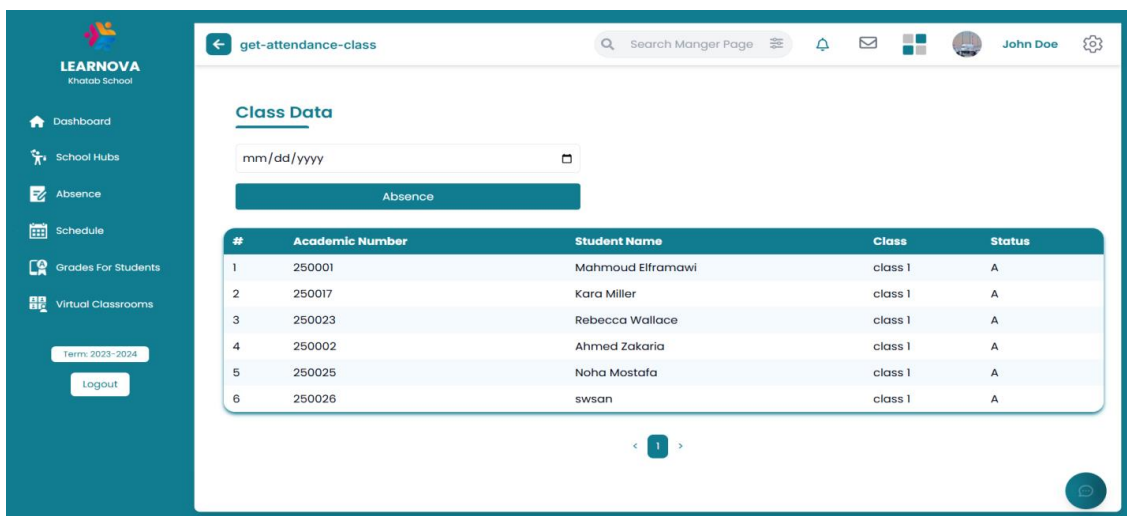
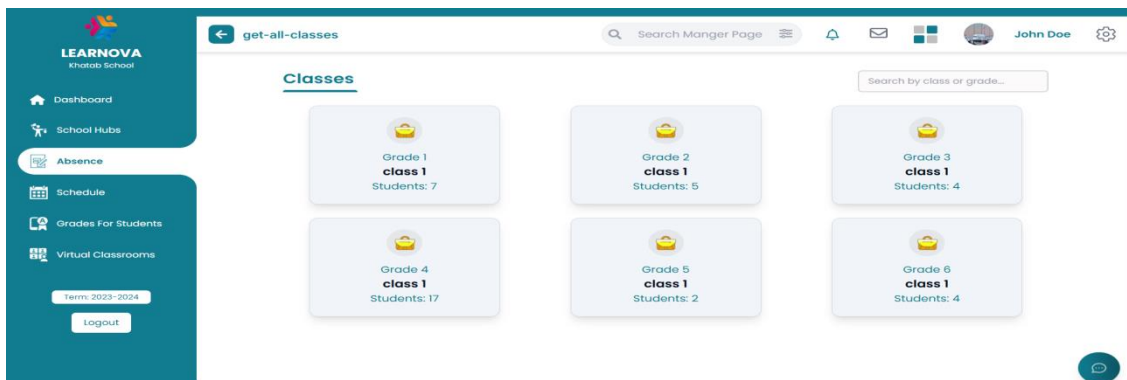
Managers can view and track student **absences** by filtering data based on **class**, **grade**, and **specific date or day**.

Key Capabilities:

- Filter absences by class or grade
- Select a specific day to view absence records
- Identify frequently absent students
- Use data to follow up with parents or staff

Result:

Improves student monitoring and supports early intervention by providing real-time access to attendance data.



Students Grades Access

Description:

Managers can view student grades filtered by grade level, class, and subject. This supports performance evaluation and academic oversight.

Key Capabilities:

- Filter grades by grade, class, and subject
- View term-based and final exam results
- Track performance trends for individuals or groups

Result:

Helps managers analyze student achievement and identify areas that need academic support or intervention.

The screenshot displays the 'Grades' management page within the LEARNOVA Khatab School system. The interface features a teal sidebar on the left with navigation options: Dashboard, School Hubs, Absence, Schedule, Grades For Students (highlighted), and Virtual Classrooms. Below these are buttons for 'Term: 2023-2024' and 'Logout'. The main content area has a header with a back arrow, the text 'grade', a search bar labeled 'Search Manger Page', and user information for 'John Doe'. The 'Grades' section is titled and contains a 'Search for Results' box with three dropdown menus for 'Grade' (labeled 'Select grade'), 'Class' (labeled 'Select class'), and 'Subject' (labeled 'Select subject'). A 'Search' button is positioned below these filters. The bottom right corner of the main area includes a circular chat icon.

Virtual Classrooms

Description:

The **Virtual Classrooms** feature allows managers to monitor and review online classes held by teachers. Each session includes specific details to ensure proper planning and tracking of virtual learning.

Key Capabilities:

- View a list of scheduled or completed virtual classes
- Each class includes:
 - **Title** of the session
 - **Start time**
 - **Duration** (in minutes or hours)
 - **Meeting link** (e.g., Zoom, Google Meet, or internal platform)
- Monitor teacher and student attendance in sessions
- Ensure sessions align with the academic schedule

Result:

This feature helps managers oversee remote learning by ensuring classes are scheduled properly, start on time, and are accessible to students. It also supports follow-up on missed or rescheduled sessions.

The screenshot displays the 'Upload Virtual Rooms' form within the LEARNNOVA Khatab School interface. The left sidebar contains navigation links: Dashboard, School Hubs, Absence, Schedule, Grades For Students, and Virtual Classrooms. The main content area shows a form with the following fields: Title (with a red asterisk), Start Time (with a date-time picker showing mm/dd/yyyy --:-- --), Duration, and Link. An 'Upload' button is located at the bottom right of the form. The top navigation bar includes a search bar, notification icons, and the user's name 'John Doe'.

edit-virtual-room

Search Manger Page

Edit Virtual Rooms

Title *
testing manager

Start Time
04/01/2025 10:00 AM

Duration
60

Link
https://www.google.com/

Update

virtual-room

Search Manger Page

Virtual Classrooms

Upload Virtual Rooms

Title	Start Time	Duration	Link	Actions
testing manager	2025-04-01 (12:00 PM)	60 Minutes	View File	Edit Delete
meet 2	2025-03-17 (06:30 PM)	60 Minutes	View File	Edit Delete
meet 3	2025-03-18 (07:32 PM)	60 Minutes	View File	Edit Delete
meet 5	2025-06-30 (05:54 PM)	80 Minutes	View File	Edit Delete
Mr.	2025-06-20 (05:59 PM)	13 Minutes	View File	Edit Delete

School Hub Management

Description:

The School Hub enables managers to organize and oversee various school-related activities such as events, clubs, competitions, and collaborative projects. It provides a centralized space for managing the details and participation of these activities.

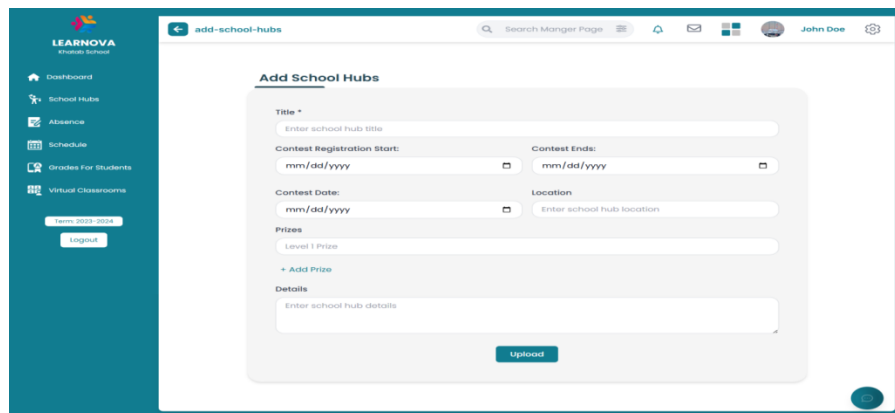
Key Capabilities:

- Add, edit, or delete school activities or sub-events

- Manage detailed information for each activity, including:
- Title and description
- Location where the activity takes place
- Level indicating difficulty or group classification
- Prize or rewards associated with the activity
- View participants involved in each activity
- Track participation across different user roles
- Access full details of each activity for planning and reporting

Result:

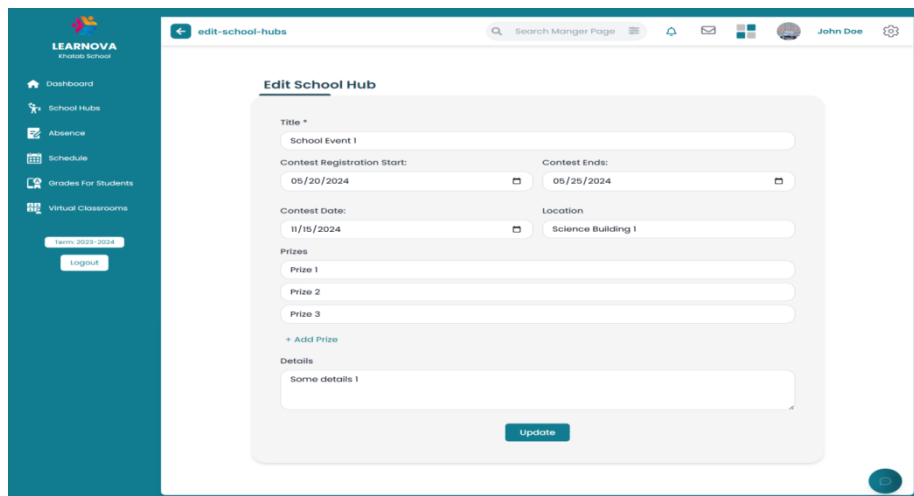
The School Hub feature improves school organization by enabling structured planning and monitoring of all internal activities. It supports engagement, coordination, and visibility across students, staff, and administration.



The screenshot shows the 'Add School Hubs' form in the LEARNOVA Khatab School system. The form is titled 'Add School Hubs' and includes the following fields:

- Title ***: Enter school hub title
- Contest Registration Start**: mm/dd/yyyy
- Contest Ends**: mm/dd/yyyy
- Contest Date**: mm/dd/yyyy
- Location**: Enter school hub location
- Prizes**: Level 1 Prize, + Add Prize
- Details**: Enter school hub details

An 'Upload' button is located at the bottom right of the form.



The screenshot shows the 'Edit School Hub' form in the LEARNOVA Khatab School system. The form is titled 'Edit School Hub' and includes the following fields:

- Title ***: School Event 1
- Contest Registration Start**: 05/20/2024
- Contest Ends**: 05/25/2024
- Contest Date**: 11/15/2024
- Location**: Science Building 1
- Prizes**: Prize 1, Prize 2, Prize 3, + Add Prize
- Details**: Some details 1

An 'Update' button is located at the bottom right of the form.

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023 - 2024

Logout

school-hubs

Search Manger Page

John Doe

Participants

Academic Number	Name	Phone	Email	Grade	Class
250026	swsan	01234468789	swsan@gmail.com	Grade 1	class 1
250002	Ahmed	011494031357	ahmed@gmail.com	Grade 1	class 1

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023 - 2024

Logout

school-hubs

Search Manger Page

John Doe

Activities

Location

Science Building 1

Details

Some details 1

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023 - 2024

Logout

school-hubs

Search Manger Page

John Doe

Activities

Level 1

Prize 1

Level 2

Prize 2

Level 3

Prize 3

Dashboard

School Hubs

Absence

Schedule

Grades For Students

Virtual Classrooms

Term: 2023 - 2024

Logout

school-hubs

Search Manger Page

John Doe

School Hubs

Add School Hubs

School Event 1

Contest Registration Start: 9/20/2024, 3:00:00 AM

Contest End: 9/25/2024, 3:00:00 AM

Contest Date: 11/15/2024, 3:00:00 AM

Details

Prizes

School Event 2

Contest Registration Start: 9/12/2025, 3:00:00 AM

Contest End: 9/18/2025, 3:00:00 AM

Contest Date: 9/24/2025, 3:00:00 AM

Details

Prizes

School Event 3

Contest Registration Start: 1/30/2025, 3:00:00 AM

Contest End: 2/10/2025, 3:00:00 AM

Contest Date: 2/12/2025, 3:00:00 AM

Details

Prizes

School Event1

Contest Registration Start: 8/20/2024, 3:00:00 AM

Contest End: 8/25/2024, 3:00:00 AM

Contest Date: 11/15/2024, 3:00:00 AM

Details

Prizes

Nam sint laboris ten

Contest Registration Start: 3/13/2025, 3:00:00 AM

Contest End: 9/13/2025, 3:00:00 AM

Contest Date: 9/24/2025, 3:00:00 AM

Details

Prizes

Odit qui ut amet vo

Contest Registration Start: 8/24/1999, 3:00:00 AM

Contest End: 8/29/2025, 3:00:00 AM

Contest Date: 9/24/2025, 3:00:00 AM

Details

Prizes

Quo ipsum quisquam c

Contest Registration Start: 3/1/1995, 2:00:00 AM

Contest End: 9/4/2025, 3:00:00 AM

Contest Date: 11/4/2025, 2:00:00 AM

Details

Prizes

Noble elit eveniet

Contest Registration Start: 10/2/1995, 2:00:00 AM

Contest End: 1/21/2025, 3:00:00 AM

Contest Date: 4/10/2025, 2:00:00 AM

Details

Prizes

Asperiores sed dolor

Contest Registration Start: 10/20/2006, 2:00:00 AM

Contest End: 2/22/2025, 3:00:00 AM

Contest Date: 9/11/2025, 3:00:00 AM

Details

Prizes

Nostrud lure et sint

Contest Registration Start: 1/27/1990, 2:00:00 AM

Contest End: 9/4/2025, 3:00:00 AM

Contest Date: 9/8/2025, 3:00:00 AM

Details

Prizes

School Event

Contest Registration Start: 9/20/2024, 3:00:00 AM

Contest End: 9/25/2024, 3:00:00 AM

Contest Date: 11/15/2024, 3:00:00 AM

Details

Prizes

➤ **Teacher Role Implementation :**

Teacher Dashboard

The **Teacher Dashboard** serves as the teacher's main workspace within the school system. It provides quick access to all relevant teaching resources, real-time data, and personal performance tools.

Key Features:

- **Current Courses**

Displays a list of courses currently assigned to the teacher. Each course includes subject information, class or grade level, and direct access to teaching tools and student data.

- **Virtual Classrooms Access**

Teachers can view and manage their virtual sessions, including:

- Scheduled session details
- Meeting links and duration
- Attendance tracking tools
- Integration with the weekly class schedule.

- **System Notifications**

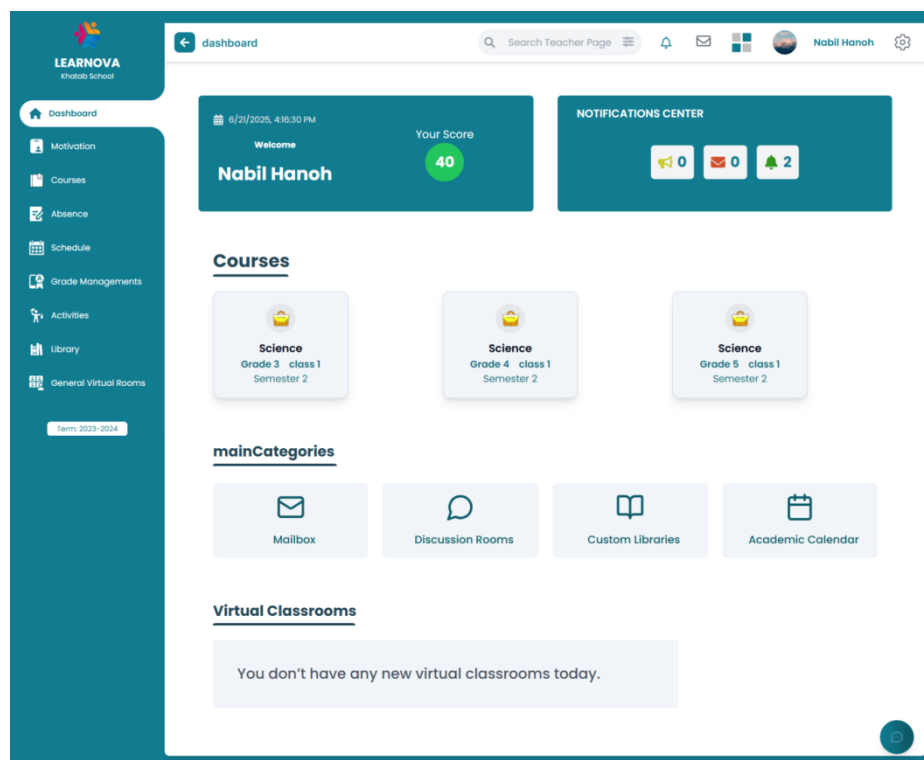
important alerts from school management, such as exam schedules or staff updates.

- **Motivation Summary Preview**

a snapshot of the teacher's current points and recognition status.

Result:

The dashboard simplifies academic responsibilities by centralizing all teaching, scheduling, and communication functions in one place. It supports both physical and virtual teaching environments efficiently.



Motivation Page

Description:

The **Motivation Page** is designed to recognize and reward teachers based on their performance and engagement throughout each academic term and year. It provides a summary of their earned points, performance ranking, and motivational level.

Key Features:

- **Summary of Score:**

Displays the total points earned by the teacher based on classroom activities, student results, attendance, engagement, and system contributions.

- **Motivational Card System:**

- Teachers start each semester with a basic status.
- At the end of each semester, their performance is evaluated and they are awarded a recognition card:
 - **Green Card:** For basic contribution

- **Golden Card:** For higher engagement
- **Diamond Card:** For exceptional performance
- **Score Breakdown:**
 - Teachers can view scores per term
 - Performance tracked across all terms in the academic year
 - Transparent point system encourages continuous improvement
- **Top Teacher Ranking:**
 - A leaderboard showcases top-performing teachers
 - Promotes healthy competition and recognition among staff

Result:

The Motivation Page encourages professional growth and rewards dedication. It helps foster a culture of excellence and appreciation within the academic staff.

LEARNOVA
Khatatib School

motivation

Search Teacher Page

Nabil Hanoh

Green
40

Score
for all semesters
40

Points Summary

Points Earned Today **0**

Your Score for this Semester **40**

Score for all semesters **40**

Summary Of Your Score

Every member starts his/her journey with a green membership card. In each semester, you will start earning points from the first day. Your final score at the end of the semester will determine the type of card you deserve to use throughout the next semester as recognition for your efforts.

- If your points are between **0 and 250** in your school, you will be eligible for the Learnova **Green** Card.
- If your points are between **251 to 400** in your school, you will be eligible for the Learnova **Golden** Card.
- If your points are **401 or more** in your school, you will be eligible for the Learnova **Diamond** Card.

Top Teachers

Full Name	Academic Number	Total Points	Badge	Subject
Nabil Hanoh	TEA-001-Science	40	Green	Science
ahmed habib	TEA-004-English	10	Green	English
yasser	TEA-002-Math	0	Green	Math
yasserl	TEA-003-Science	0	Green	Science
Wessam Mohamed	TEA-005-Math	0	Green	Science

Student Attendance Management

Description:

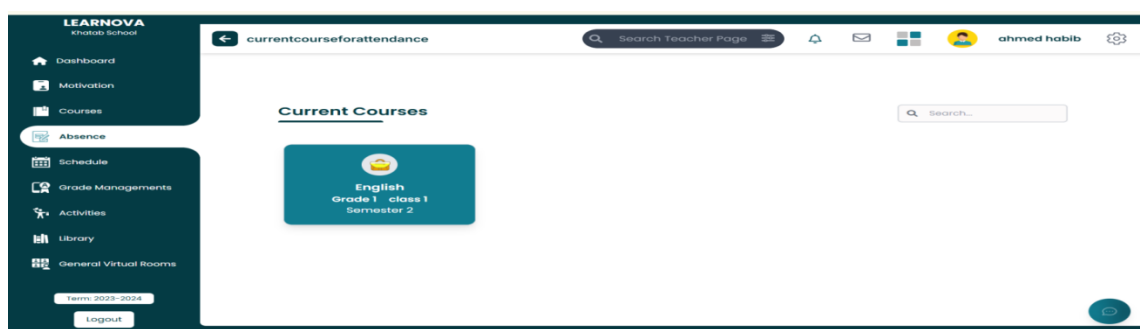
Teachers have the ability to record student attendance directly through the system. Attendance is taken based on the teacher's assigned classes and subjects and is organized by academic period and semester.

Key Capabilities:

- Filter Students before taking attendance using:
 - Grade Subject Semester ID
 - Class ID
- Take Absence:
 - Teachers mark students as present or absent for a specific period
 - Attendance is recorded per session, class, and subject
- Generate Attendance Reports:
 - Teachers can view detailed attendance history for each student
 - Reports are organized by period, date range, class, and subject
- Useful for performance tracking, parent communication, and term reviews

Result:

This feature ensures accurate, daily tracking of student attendance by the assigned teacher. It promotes accountability and provides a complete record for academic follow-up and reporting.



LEARNOVA

Khatab School

Dashboard

Motivation

Courses

Absence

Schedule

Grade Managements

Activities

Library

General Virtual Rooms

Term: 2023-2024

Logout

takeattendance

Search Teacher Page

ahmed habib

1 Take Attendance

2 Attendance Report

#	Student Name	Academic Number	Grade	Class	Status
1	Mahmoud Elframawi	250001	class 1	Grade 1	☐
2	Ahmed Zakaria	250002	class 1	Grade 1	☐
3	zako	250003	class 1	Grade 1	☐
4	Kara Miller	250017	class 1	Grade 1	☐
5	Rebecca Wallace	250023	class 1	Grade 1	☐
6	Noha Mostafa	250025	class 1	Grade 1	☐
7	swsan	250026	class 1	Grade 1	☐
8	Drake Stuart	250032	class 1	Grade 1	☐

Submit Attendance

LEARNOVA

Khatab School

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Library

General Virtual Rooms

Term: 2023-2024

Logout

attendancereport

Search Teacher Page

ahmed habib

1 Take Attendance

2 Attendance Report

Attendance Summary

Select Class

mm/dd/yyyy

mm/dd/yyyy

Generate Report

#	Academic Number	Student Name	Class	Status
1	250001	Mahmoud Elframawi	class 1	11
2	250002	Ahmed Zakaria	class 1	8
3	250017	Kara Miller	class 1	8
4	250023	Rebecca Wallace	class 1	9
5	250025	Noha Mostafa	class 1	7
6	250026	swsan	class 1	7
7	250032	Drake Stuart	class 1	1

LEARNOVA

Khatab School

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Term: 2023-2024

Logout

student-attendance-details

Search Teacher Page

ahmed habib

Attendance Details for Ahmed Zakaria

Academic Number: 250002

Class: class 1

Total Absences: 8

Total Attendances: 10

Date	Status
2025-02-08	Absent
2025-02-09	Absent
2025-02-10	Present
2025-02-12	Absent
2025-02-15	Absent
2025-02-16	Absent

Weekly Schedule View

Description:

Teachers have access to their personalized **weekly schedule**, showing all assigned teaching sessions across subjects, classes, and periods for the academic week.

Key Capabilities:

View full weekly timetable

Displayed in a clear calendar or table format showing:

- Days of the week
- Periods or time slots
- Assigned classes and subjects

Session Details

Each scheduled session includes subject name, class or grade, and location (or virtual link if online).

Real-Time Updates

Automatically reflects any changes made by school management or schedule coordinators.

Result:

This feature helps teachers plan their teaching week efficiently, avoid time conflicts, and stay organized across multiple classes and subjects.

Time	Sunday	Wednesday	Tuesday	Wednesday	Thursday	Friday	Saturday
08:00	No class	No class	No class	No class	No class	No class	No class
09:00	No class	No class	No class	No class	No class	No class	No class
10:00	No class	No class	No class	No class	No class	No class	No class
11:00	No class	No class	No class	No class	No class	No class	No class
12:00	No class	No class	No class	No class	No class	No class	No class
13:00	No class	No class	No class	No class	No class	No class	No class
14:00	No class	No class	No class	No class	No class	No class	No class

Grades Management

Teachers can manage student grading based on academic performance. The system allows filtering students by multiple criteria, exporting lists, entering grades, and managing previously submitted scores.

Key Capabilities:

- **Get Students by Filter**

Teachers can retrieve students based on:

- **Grade Subject Semester ID**
- **Class ID**

- **Export Students List**

Teachers can export the filtered list of students for use in offline grading or record-keeping.

- **Assign Grades**

Teachers can enter academic grades for each student within the selected group. Grading may be done per subject and per term or assessment type.

- **View Student Grades**

Teachers can review all previously submitted grades organized by subject, class, and semester.

- **Edit or Delete Grades**

Teachers can update or delete submitted grades if changes are needed, in accordance with system permissions.

Result:

This grading system gives teachers full control over academic evaluations within their assigned classes, ensuring flexibility, accuracy, and transparency in the grading process.

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Library

General Virtual Rooms

Terms: 2023-2024

current-courses-for-grades

Search Teacher Page

Nabil Hanah

Current Courses

Search ...

Science

Grade 1 Class: class 1

Semester 2

Science

Grade 3 Class: class 1

Semester 2

Science

Grade 4 Class: class 1

Semester 2

Science

Grade 5 Class: class 1

Semester 2

LEARNOVA

Khatib School

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Library

General Virtual Rooms

Terms: 2023-2024

exam-score

Search Teacher Page

Nabil Hanah

Exam Scores

Academic Year: 2024 - 2025

Grade: Grade 1

Subject: Science

Export to CSV

Go to Upload File

LEARNOVA

Khatib School

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General Virtual Rooms

Terms: 2023-2024

exam-score

Search Teacher Page

Nabil Hanah

Grades For Students

Upload Grades File

Choose a file (Excel, CSV, or PDF)

Upload

Get Students Grades

Select exam type

Get Students Grades

LEARNOVA

Khatib School

Dashboard

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Activities

Library

General Virtual Rooms

Terms: 2023-2024

exam-results

Search Teacher Page

Nabil Hanah

Student Exam Results

Type: final

Final Degree: 100

Choose a file

Upload & Update

Export Data

Delete All Data

Library Section

Description:

The **Library** is a dedicated section where teachers can access and manage educational materials. It is divided into three tabs: **Library**, **General Library**, and **Teacher Library** — all offering the same core actions (add, view, delete), but serving different purposes.

Library Tab

Purpose:

The main access point for navigating the educational resources. It acts as a container that brings together both the **General Library** and the **Teacher Library**.

Functionality:

- View and switch between general and personal resources
- Unified access to all available materials
- Consistent interface for managing content

General Library Tab

Purpose:

A shared space where teachers can access school-wide materials provided by administrators or other staff.

Functionality:

- **View** shared materials (PDFs, videos, lesson files, etc.)
- **Add** new resources (if permission is granted)
- **Delete** only resources uploaded by the current teacher (if applicable)
- Used for collaboration and resource sharing across departments

Teacher Library Tab

Purpose:

A private area for teachers to manage their own teaching materials and keep them organized

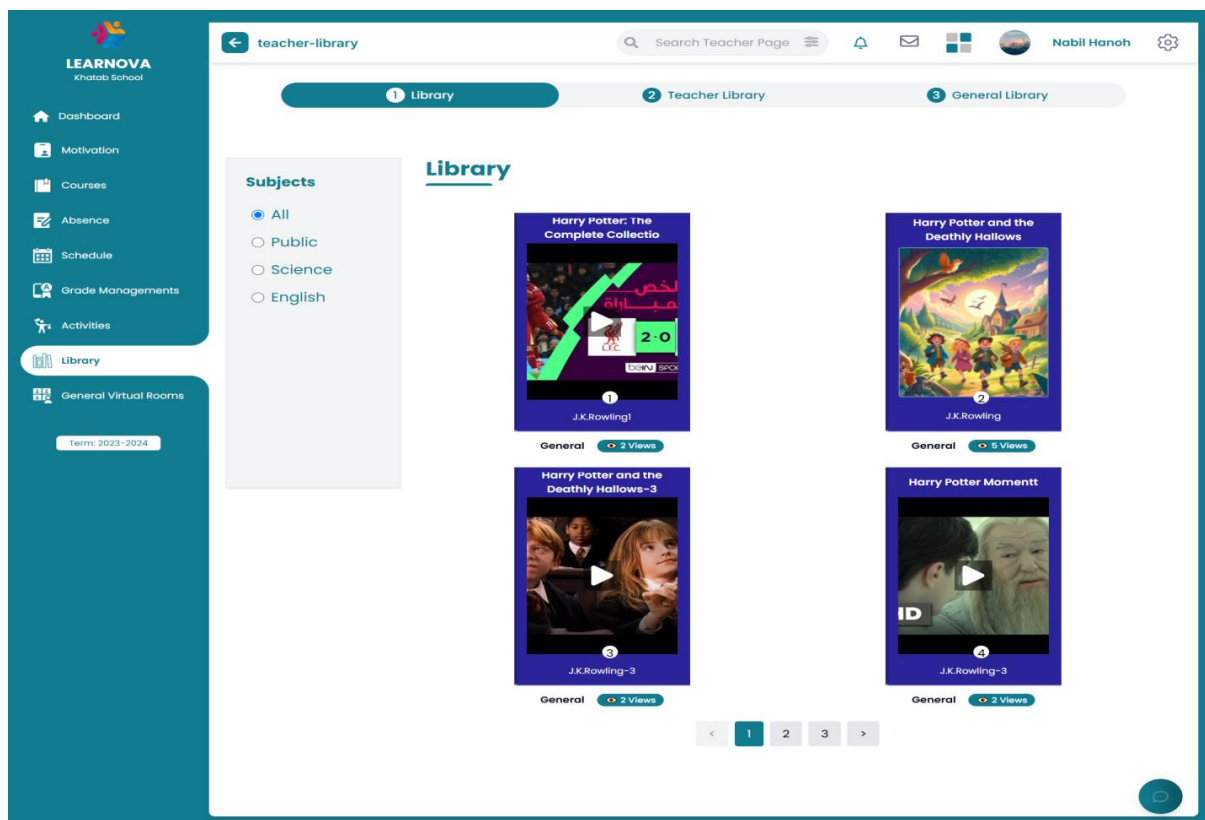
by subject or class.

Functionality:

- **Add** personal teaching resources
- **View** and organize content for repeated use
- **Delete** or update any of their uploaded items
- Used for storing lesson plans, documents, and custom content

Result:

All three tabs support the same core actions — **add, view, delete** — but are separated by ownership and access level. This ensures efficient organization of teaching content, with a balance between private materials and shared school-wide resources.



LEARNOVA

Khatib School

Dashboard

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Absence

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Activities

Library

General Virtual Rooms

Term: 2023-2024

library

Search Teacher Page

Nabil Hanoh

Harry Potter Momentt

Back

AuthorJ.K.Rowling-3

TypeVideo

Uploaded ByNabil Hanoh

DownloadDownload Material

Harry Potter and the Deathly Hallows: Part 2 (4/5) Movie ...

Watch LaterShare

HD

Watch on YouTube

LEARNOVA

Khatib School

Dashboard

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Activities

Library

General Virtual Rooms

Term: 2023-2024

Logout

all-subjects-library

Search Teacher Page

Nabil Hanoh

1 Library

2 Teacher Library

3 General Library

All Courses

Search...

Science

Grade 1 Semester 2

Science

Grade 3 Semester 2

Science

Grade 5 Semester 2

English

Grade 1 Semester 2

Science

Grade 4 Semester 2

LEARNOVA

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Activities

Library

General Virtual Rooms

Term: 2023-2024

Logout

library-form

Search Teacher Page

Nabil Hanoh

Add Library Item

Subjectslibraryitem.enterTitle

FileUrllibraryitem.enterUrl

Descriptionlibraryitem.enterDescription

Type

libraryitem.selectType

libraryitem.gradeSubjectSemester

libraryitem.selectId

Upload

LEARNOVA

Khatatb School

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Library

General Virtual Rooms

Term: 2023-2024

Logout

all-materials-library

Search Teacher Page

Nabil Hanoh

All Materials in Library

Add Item

Item Details	Description	Type	Grade	Subject	Actions
Introduction to Algebra	A comprehensive guide to understanding basic algebra concepts.	PDF	Grade 1	Science	

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update-item-library

Search Teacher Page

Nabil Hanoh

Edit Material

Item Details *

Harry Potter: The Complete Collectio

Author *

J.K.Rowlingl

FileUrl *

https://www.youtube.com/watch?v=Y8yB6YJXab4

Type

Video

Update

LEARNOVA

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Term: 2023-2024

Logout

library-item-form

Search Teacher Page

Nabil Hanoh

Add Library Item

Item Details

libraryitem.enterTitle

Author

libraryitem.enterAuthor

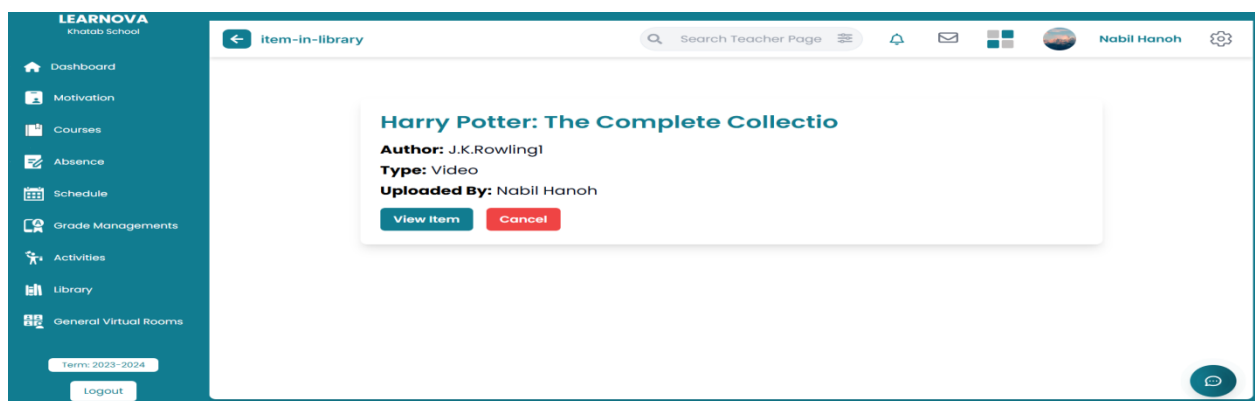
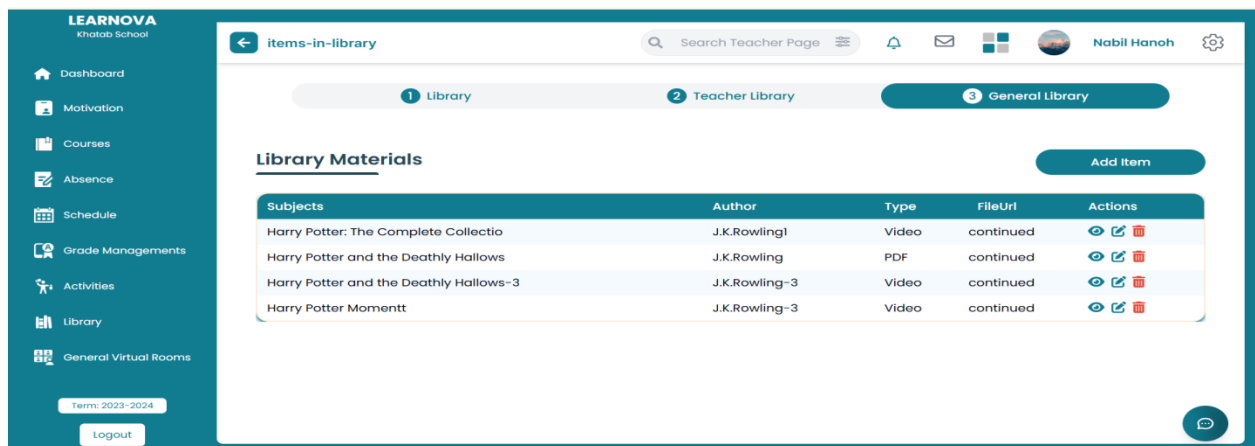
FileUrl

libraryitem.enterUrl

Type

libraryitem.selectType

Upload



Virtual Classrooms

Description:

The **Virtual Classroom** section allows teachers to manage and monitor their online teaching sessions. The interface is organized into three tabs to help track class status efficiently: **All**, **Missed**, and **Completed**.

📁 All Tab

Purpose:

Displays all scheduled virtual sessions assigned to the teacher, regardless of their current status.

Functionality:

- View all upcoming, ongoing, missed, or completed sessions

- Access session details: title, class, subject, date, start time, duration, and meeting link
- Launch or join sessions directly

☐ **Missed Tab**

Purpose:

Lists virtual classes that were scheduled but not conducted or marked as incomplete.

Functionality:

- Helps track missed or unheld sessions
- Allows teachers to follow up with students or reschedule if needed
- Supports accountability and session tracking

☐ **Completed Tab**

Purpose:

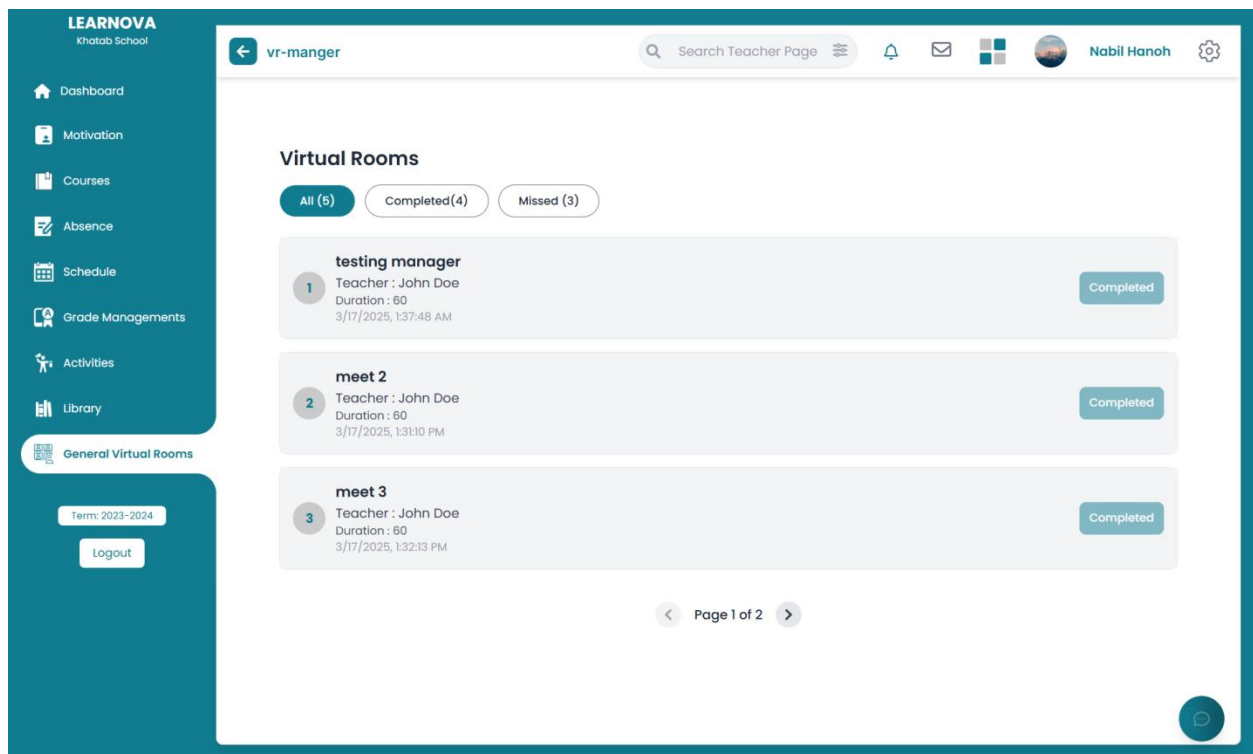
Shows all virtual classes that have been successfully conducted and marked as completed.

Functionality:

- View completed session history by subject or class
- Review attendance records (if integrated)
- Useful for reporting and performance evaluation

Result:

The tabbed structure of the Virtual Classroom module helps teachers stay organized, ensures no session is overlooked, and supports smooth delivery of online classes.



School Hubs

Description:

The **School Hubs** feature enables teachers to actively contribute to the school community by organizing extracurricular activities, events, and student group engagements. Teachers can manage their own hubs or participate in shared ones.

Key Capabilities:

- **Create and Manage School Hub Events**

Teachers can initiate new events such as competitions, clubs, field trips, or workshops.

- Define event name, purpose, schedule, and target grade/class
- Add supporting details such as location, level, and prizes

- **Add Activities**

- Create specific activities within the hub (e.g., reading challenge, science fair)
- Organize by subject, theme, or student age group
- Manage activity schedule and format (individual, group, etc.)

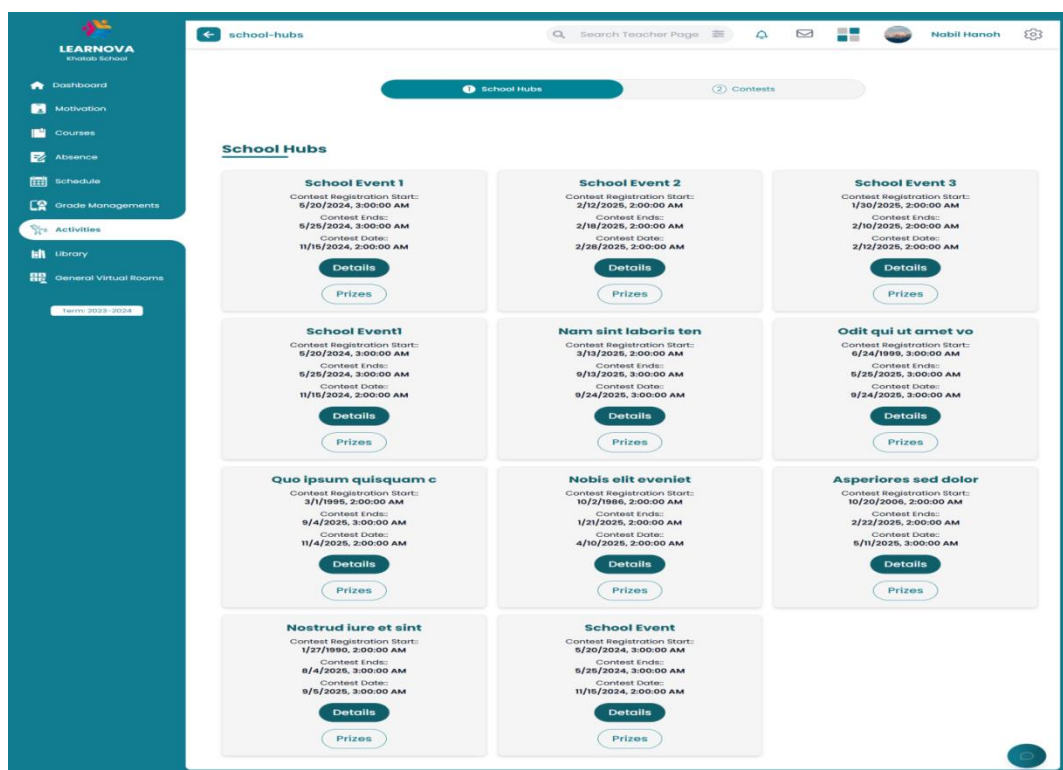
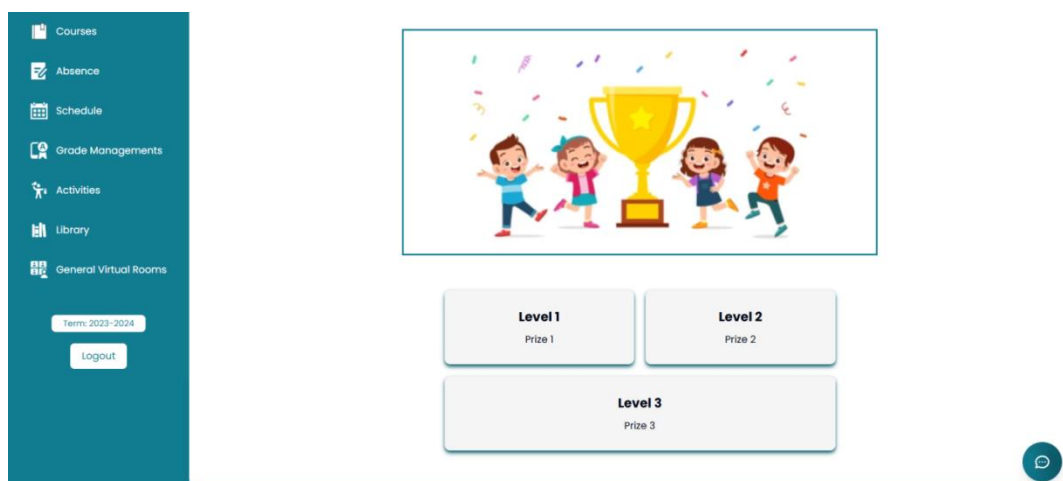
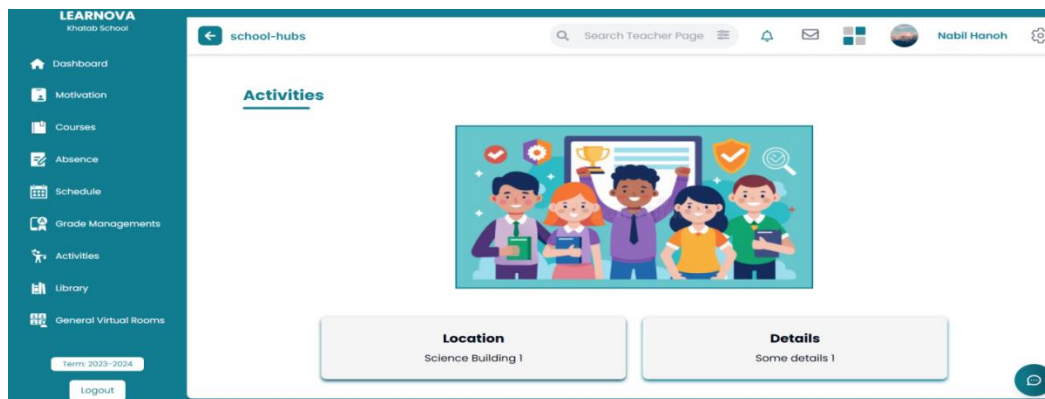
- **Delete Activities or Events**
 - Remove outdated or canceled events and activities
 - Maintain relevance and clarity within the hub
- **View Hub Details**
 - See all event information including description, date, location, and assigned organizers
 - Track progress or updates related to each activity
- **View Participants**
 - Access a list of students and staff participating in specific activities or events
 - Monitor involvement for attendance or recognition purposes

Activities May Include:

- Academic competitions (e.g., math contests, spelling bees)
- Creative projects (e.g., art exhibitions, storytelling)
- Sports or fitness events
- Club meetings (e.g., robotics, debate, science club)
- Volunteering or school service initiatives
- Theme days or cultural even

Result:

This feature empowers teachers to engage students beyond the classroom, fostering a vibrant school culture. It also provides tools for tracking participation, organizing logistics, and supporting well-rounded student development.



LEARNOVA

Khatatb School

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General Virtual Rooms

Term: 2023 - 2024

Logout

contests

Search Teacher Page

Nabil Hanoh

Add Contest

Title *

Start Date

mm/dd/yyyy

End Date

mm/dd/yyyy

Subject

Select Subject

Class

Select Class

Grade

Select Grade

Num of team members

Enter number of team members

Requirements

Write here your requirements...

Upload

LEARNOVA

Khatatb School

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Term: 2023 - 2024

Logout

contests

Search Teacher Page

Nabil Hanoh

School Hubs

Contests

Activities

Add Contest

Title	Teacher	Subject	Start Date	End Date	Num of team members	Requirements	Action
Science Contest 12	Nabil Hanoh	Science	2025-06-19	2025-06-20	5	Each team must have a science major...	Edit Delete
Science Contest 2	Nabil Hanoh	Science	2025-02-16	2025-02-20	10	Each team must have a science major...	Edit Delete
Science Contest	Nabil Hanoh	Science	2024-06-01	2024-06-12	4	Each team must have a science major...	Edit Delete
Science Contest1	Nabil Hanoh	Science	2024-06-01	2024-06-12	5	Each team must have a science major...	Edit Delete
Science Contest1	Nabil Hanoh	Science	2024-06-01	2024-06-12	5	Each team must have a science major...	Edit Delete
Science Contest3	Nabil Hanoh	Science	2024-06-01	2024-06-12	5	Each team must have a science major...	Edit Delete

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Khatatb School

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Term: 2023 - 2024

Logout

contests

Search Teacher Page

Nabil Hanoh

Participants

Team Name	(Leader)	Teammates
Code Masters	swsan	1. swsan (250026) 2. Mahmoud Elframawi (250001) 3. Ahmed Zakaria (250002)
Innovators	Noha Mostafa	1. Noha Mostafa (250025) 2. Kara Miller (250017)

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Logout

contests

Search Teacher Page

Nabil Hanoh

Edit Contest

Title *

Science Contest 12

Start Date

06/19/2025

End Date

06/20/2025

Subject

Science

Class

class 1

Grade

Grade 1

Num of team members

5

Requirements

Each team must have a science major...

Update

Course Management

Teachers have access to a Courses section that allows them to manage and interact with subjects they are teaching. This section is divided into Current Courses (actively assigned) and All Courses (including past terms).

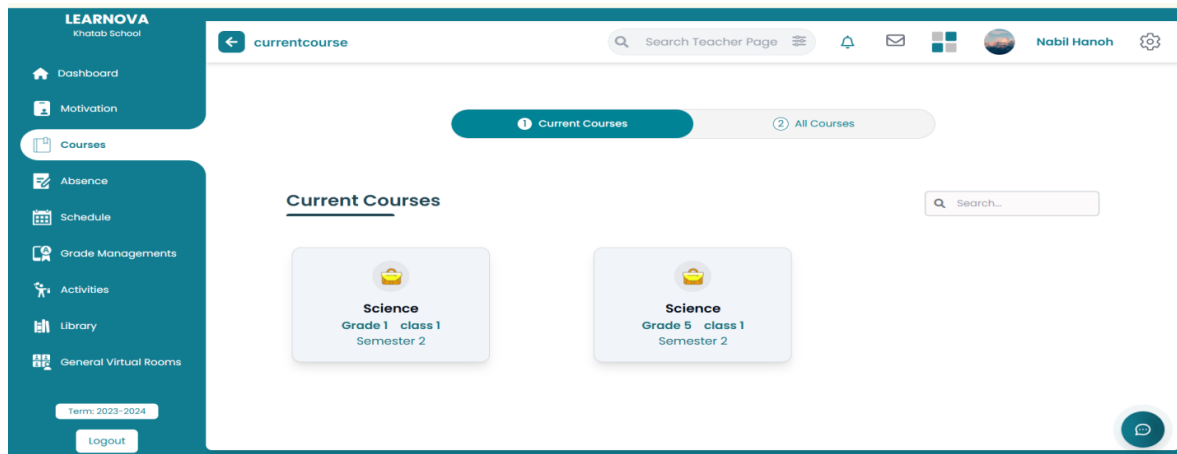
Current and All Courses

Description:

- Current Courses: Displays subjects the teacher is currently assigned to in the active semester.
- All Courses: Shows a history of all courses the teacher has taught or is involved in.

Functionality:

- Access related content, students, and tools
- Track progress and manage course resources



LEARNOVA

Khatib School

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General Virtual Rooms

Term: 2023 - 2024

Logout

addmaterial

Search Teacher Page

Nabil Hanoh

Science - Grade 1

Video Lectures

Total: 3

+

Course Material

Total: 5

+

Virtual Room

Total: 9

+

Question Bank

Total: 6

+

Assignments

Total: 4

+

Exams

Total: 5

+

LEARNOVA

Khatib School

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Library

General Virtual Rooms

Term: 2023 - 2024

Logout

materialform

Search Teacher Page

Nabil Hanoh

UploadMaterial

Title *

Description

Type

PDF

FileUri *

Upload

LEARNOVA

Khatib School

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General Virtual Rooms

Term: 2023 - 2024

Logout

see-material

Search Teacher Page

Nabil Hanoh

All Materials

Title	Description	Type	FileUri	Actions
unit three	updated	Video	View File	 
Science unit five	This is for Unit 5	Video	View File	 
Science unit five	This is for Unit 5	PDF	View File	 
Science unit five	This is for Unit 5	PDF	View File	 
unit four	content for unit 4	Video	View File	 
Science unit four	This is for Unit 4	PDF	View File	 
Science unit four	This is for Unit 4	PDF	View File	 
Science unit four	This is for Unit 4	PDF	View File	 

LEARNOVA

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Term: 2023 - 2024

Logout

update-material

Search Teacher Page

Nabil Hanoh

Edit Material

Title *

unit three .

Description

updated

Type

Video

FileUri *

https://drive.google.com/file/d/1_fXJ2dIlh4c3ZytkViMno2sTIFyCe6n-/view?usp=drive_link

Update

update-material

Edit Material

Title *
unit three

Description
updated

Type
Video

FileUri *
https://drive.google.com/file/d/1_TXJ2dLh4c3ZyfkVIMno2sTIFyCe6n-/view?usp=drive_link

Update

materialform

UploadMaterial

Title *

Description

Type
PDF

FileUri *

Upload

see-material

All Materials

Title	Description	Type	FileUri	Actions
unit three	updated	Video	View File	✓ ✗
Science unit five	This is for Unit 5	Video	View File	✓ ✗
Science unit five	This is for Unit 5	PDF	View File	✓ ✗
Science unit five	This is for Unit 5	PDF	View File	✓ ✗
unit four	content for unit 4	Video	View File	✓ ✗
Science unit four	This is for Unit 4	PDF	View File	✓ ✗
Science unit four	This is for Unit 4	PDF	View File	✓ ✗
Science unit four	This is for Unit 4	PDF	View File	✓ ✗

Virtual Room (VR)

Description:

Teachers can create live virtual classrooms for real-time teaching.

Functionality:

- Schedule and start virtual sessions
- Include details like meeting link, duration, and title
- Mark attendance or record sessions (if supported)

Upload Virtual Rooms

Title *

Start Time
mm/dd/yyyy --:--:--

Duration

Link

Upload

Virtual Classrooms

Title	Start Time	Duration	Link	Actions
Science Revision Session 10	2025-02-13 (01:40 PM)	68	View File	
science session 5	2025-02-28 (11:55 PM)	50	View File	
Weekly Meet	2025-03-04 (10:00 AM)	60	View File	
Midterm Revision	2025-02-28 (06:08 AM)	60	View File	
Quick Minute	2025-02-20 (10:05 PM)	5	View File	
Quick Revision2	2025-02-20 (10:04 PM)	5	View File	
test VR	2025-02-26 (05:22 PM)	1	View File	
vr 5 meeting	2025-03-13 (02:00 PM)	80	View File	
vr44	2025-06-18 (05:30 PM)	60	View File	

Edit Virtual Rooms

Title *

Science Revision Session 10

Start Time
02/13/2025 11:40 AM

Duration
68

Link
http://google.com

Update

Question Bank

Description:

A space where teachers build and manage reusable questions for assessments.

Functionality:

- Create multiple-choice, short-answer, or essay questions
- Categorize questions by subject, chapter, or difficulty
- Store and reuse in exams and quizzes

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Khatob School

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Term: 2023-2024

Logout

all-question-bank

Search Teacher Page

Nabil Hanoh

1 My questions

2 All Questions

All Questions For This Subject

#	Question	Type	Answer	Actions
1	What is the H ₂ O?	Essay	it's water	 
2	Plague is caused by ?	MCQ	Bacteria	 
3	Insects are animals?	True/False	False	 
4	What force keeps us on the ground?	Short Answer	Gravity	 
5	What is the mpemba effect ?	Essay	Hot water freezing faster than cold	 
6	What is the chemical formula for water?	Essay	H ₂ O	 

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Term: 2023-2024

Logout

my-question-bank

Search Teacher Page

Nabil Hanoh

1 My questions

2 All Questions

My questions

#	Question	Type	Answer	Actions
1	What is the H ₂ O?	Essay	it's water	 
2	Plague is caused by ?	MCQ	Bacteria	 
3	Insects are animals?	True/False	False	 
4	What force keeps us on the ground?	Short Answer	Gravity	 
5	What is the mpemba effect ?	Essay	Hot water freezing faster than cold	 
6	What is the chemical formula for water?	Essay	H ₂ O	 

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Logout

edit-question

Search Teacher Page

Nabil Hanoh

Edit Question

question Type:

Essay

Question:

What is the H₂O?

Answer:

it's water

Update

LEARNOVA

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Term: 2023-2024

question-bank-form

Search Teacher Page

Nabil Hanoh

Upload Question

question Type :

Multiple Choice

Question:

Choices:

Choice 1

Choice 2

Choice 3

Choice 4

Correct Answer:

Select Correct Answer

Upload



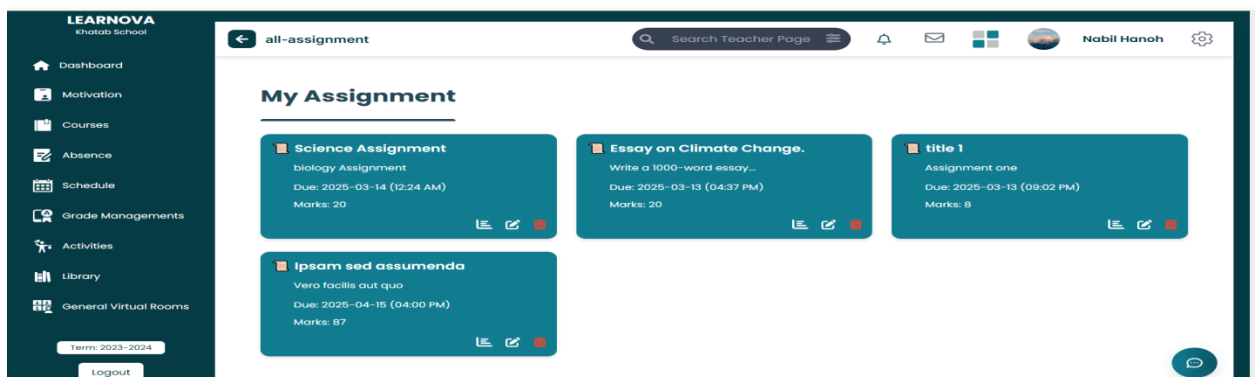
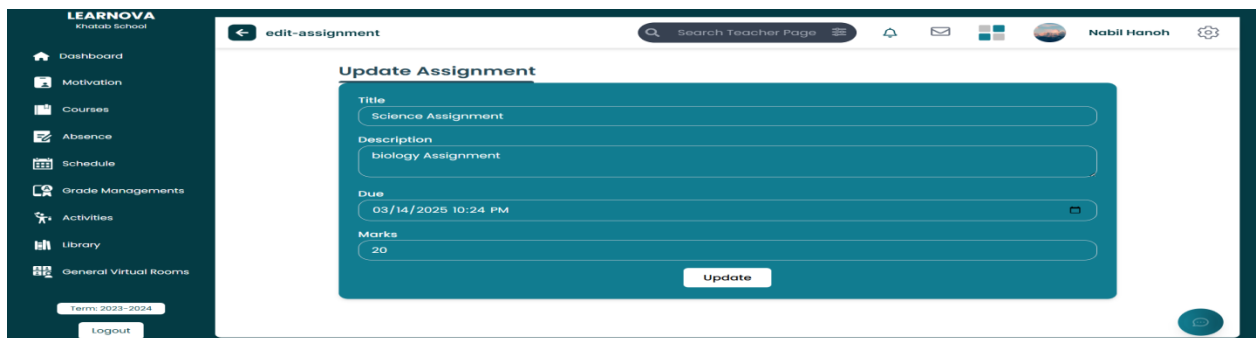
Assignments

Description:

Teachers can create assignments for individual or group submission.

Functionality:

- Define assignment instructions, deadline, and file types
- Allow file uploads or text answers
- Review and grade submitted work



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Khatab School

← assignment-form

Search Teacher Page

Nabil Hanoh

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Term: 2023-2024

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Upload Assignment

Title

Description

Due

mm/dd/yyyy --:-- --

Marks

Upload

LEARNOVA

Khatab School

← my-exams

Search Teacher Page

Nabil Hanoh

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Term: 2023-2024

Logout

My Exams

ghjkl

tyyulop

Duration: 80 minutes

Start Time: 2025-06-25 (04:01 PM)

Type: Offline

Marks: 80

End Time: 2025-07-01 (04:01 PM)

Est at ut rem quam

Magni expedita persp

Duration: 1 minutes

Start Time: 2025-04-15 (10:09 AM)

Type: Online

Marks: 30

End Time: 2026-06-07 (05:11 AM)

Aut et iste rerum mo

Ea mollit consectetur

Duration: 1 minutes

Start Time: 2025-04-15 (10:00 AM)

Type: Online

Marks: 20

End Time: 2026-07-17 (02:08 PM)

Final exam

final exam 2025

Duration: 60 minutes

Start Time: 2025-03-14 (08:25 PM)

Type: Offline

Marks: 110

End Time: 2025-07-15 (08:25 PM)

Final arabic exam for prep

This exam covers topics from biology and physics.

Duration: 3 minutes

Start Time: 2025-03-02 (11:00 AM)

Type: Online

Marks: 80

End Time: 2025-03-10 (01:00 PM)

Exams

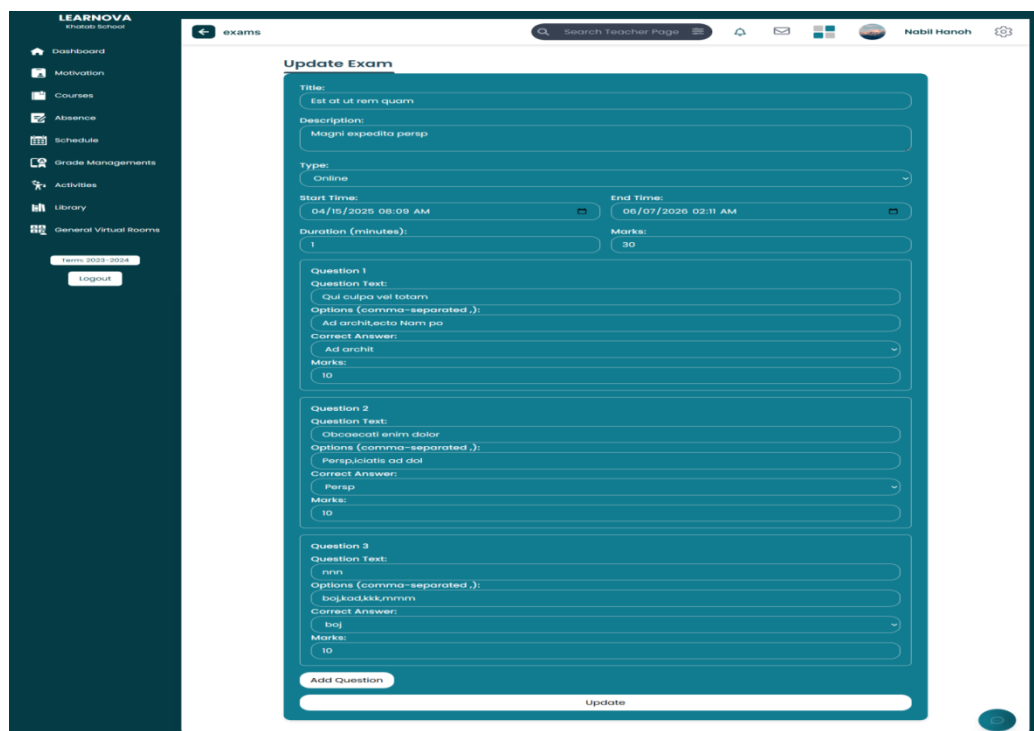
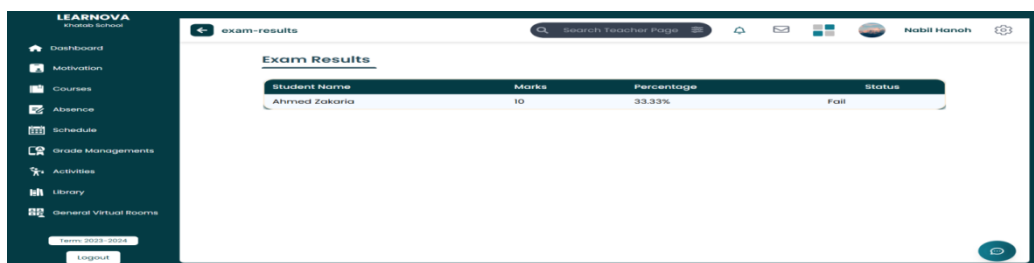
Description:

Teachers can create and assign exams to students through the course.

Functionality:

- Set exam title, duration, and time window
- Add questions from the question bank
- Configure grading rules or automatic scoring

Here Courses section offers a complete suite of tools for managing the learning process — from teaching content to evaluation — allowing teachers to deliver high-quality, interactive education.



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exam-form

Search Teacher Page

Nabil Hanoh

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General Virtual Rooms

Term: 2023-2024

Logout

Upload Exam

Title:

Description:

Type:

Online

Start Time:

mm/dd/yyyy --:-- --

End Time:

mm/dd/yyyy --:-- --

Duration (minutes):

0

Marks:

0

Question 1

Question Text:

Options (comma-separated ,):

Option 1, Option 2, Option 3, Option 4

Correct Answer:

Select correct answer

Marks:

0

Add Question

Create Exam

LEARNOVA

Khatab School

exam-details

Search Teacher Page

Nabil Hanoh

Dashboard

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General Virtual Rooms

Term: 2023-2024

Logout

Est at ut rem quam

General Information

Description: Magni expedita persp

Type: Online

Start Time: 4/15/2025, 10:09:00 AM

End Time: 6/7/2026, 5:11:00 AM

Duration: 1 minutes

Marks: 30

Subject: Science

Grade: Grade 1

Exam Questions

Question 1: Qui culpa vel totam

Type: MCQ

Options:

Ad archit

ecto Nam po

Correct Answer: Ad archit

Marks: 10

Question 2: Obcaecati enim dolor

Type: MCQ

Options:

Persp

iclatia ad dol

Correct Answer: Persp

Marks: 10

Question 3: nnn

Type: MCQ

Options:

boj

kod

kkk

mmmm

Correct Answer: boj

Marks: 10

➤ **Student Role Implementation :**

Student Dashboard

Description:

The student dashboard is the central page students see upon logging in. It gives an overview of their academic status and provides quick access to key sections.

Key Features:

- View recent announcements and updates
- Access enrolled courses and upcoming deadlines
- Quick links to attendance, grades, schedule, and motivation
- Alerts for missing work or exam reminders

Result:

Helps students stay organized and aware of their responsibilities in one unified interface.

Motivation Page

Description:

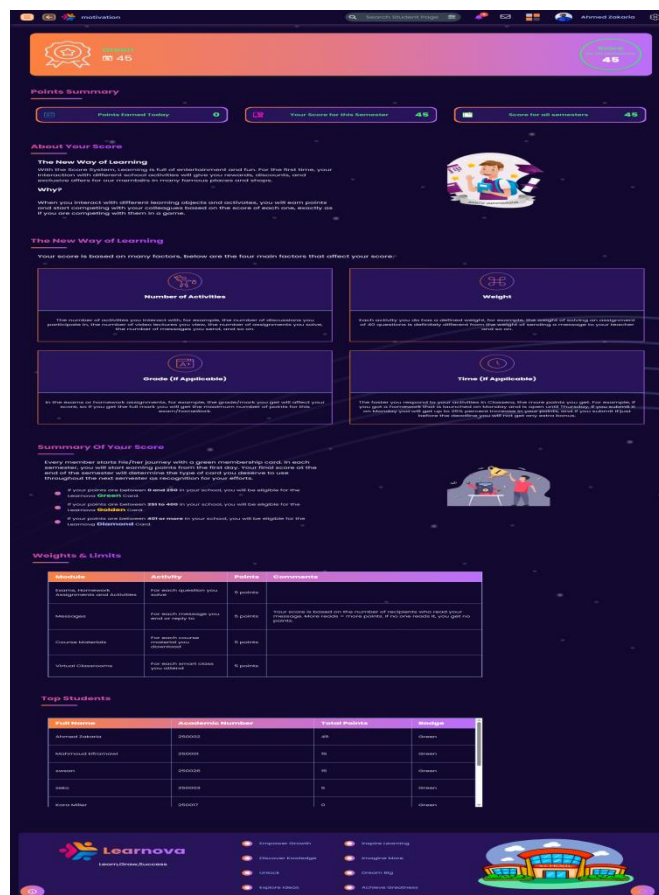
Students can track their motivational score based on their academic and behavioral performance.

Key Features:

- Display of current score level
- Breakdown of points earned per term
- Visual representation of progress
- Badge or card (e.g., green, gold, diamond) reflecting performance tier

Note:

Scores and recognition types are calculated based on data entered by teachers and system logic.



School Hubs and Activities

Description:

Students can view and participate in school-organized hubs and activities. These include extracurricular events, academic clubs, competitions, and collaborative projects created by teachers or managers.

Key Features:

View Available Activities:

Students can access a list of activities assigned to them or open for participation, including:

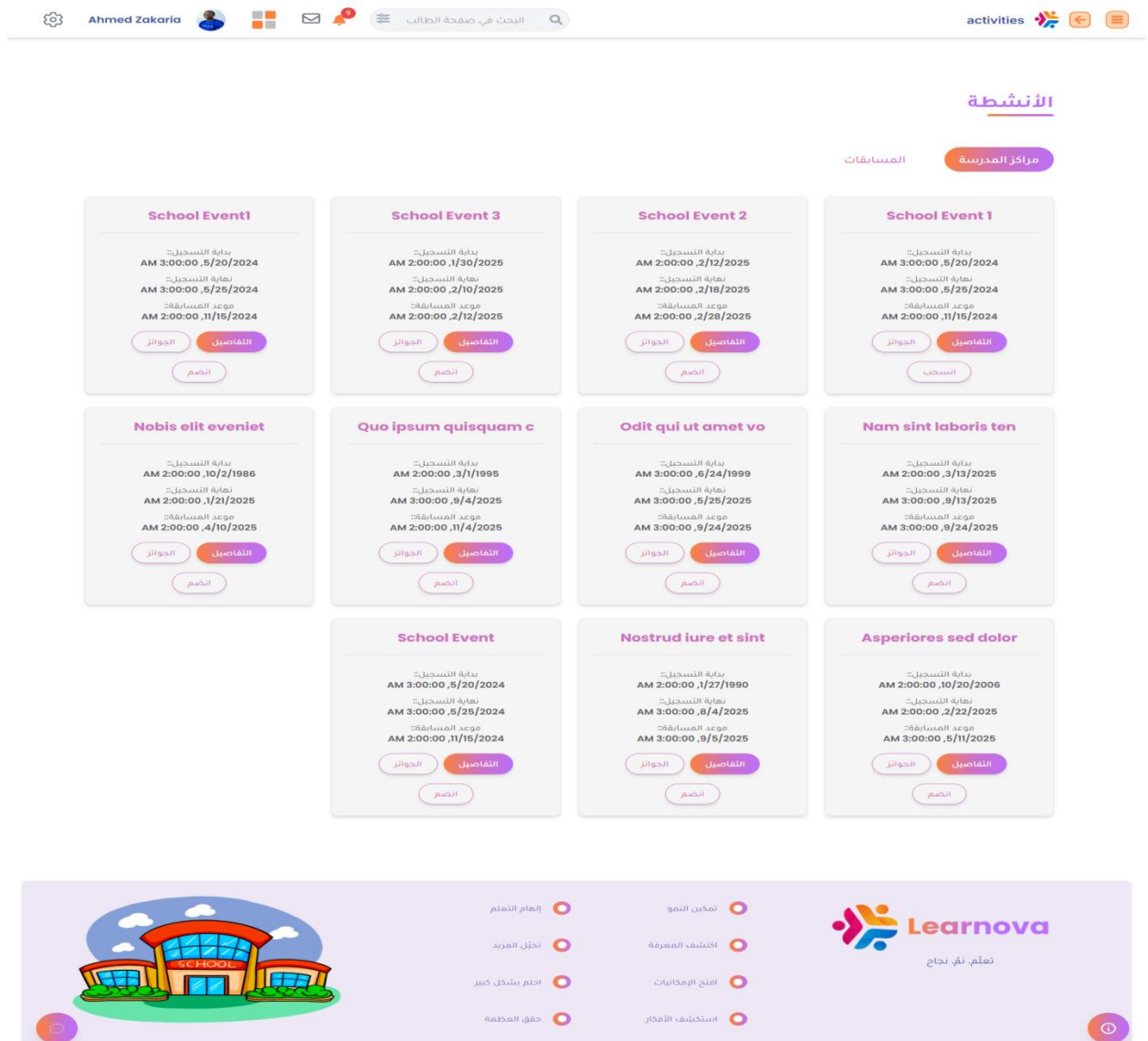
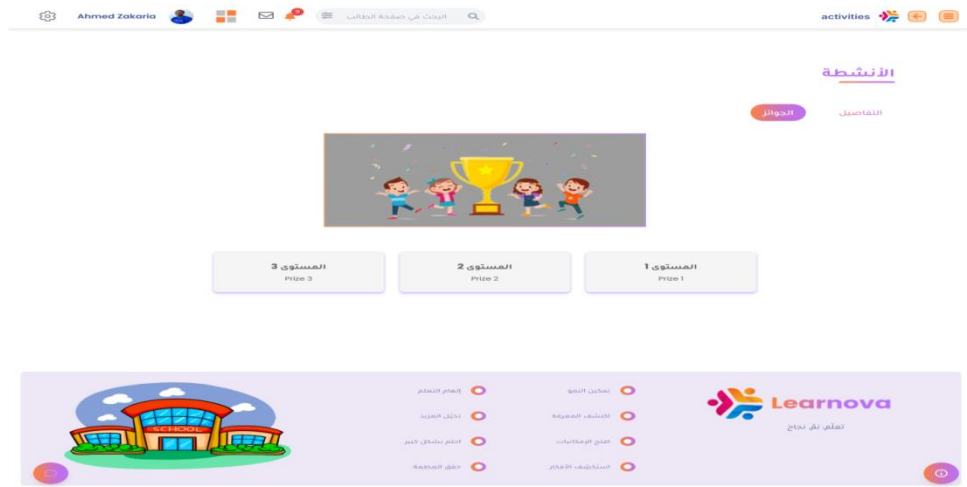
- Activity title and description
- Location and time
- Level or category
- Prize or rewards (if applicable)
- Participation Status:
 - View if they are registered or selected as participants
 - See other participants if the list is visible
 - Track participation history and performance in past events
- View School Hubs:
 - Students can explore school hubs they are linked to (e.g., clubs, academic groups).
 - Hubs are created by school staff and may host multiple activities.

Note:

All data shown in this section is added by managers or teachers. Students do not create or edit activities but can engage by viewing, joining (if permitted), and tracking participation.

Result:

This feature fosters student involvement beyond the classroom. It promotes a stronger connection to the school community and encourages participation in academic and non-academic events.



Grades

Description:

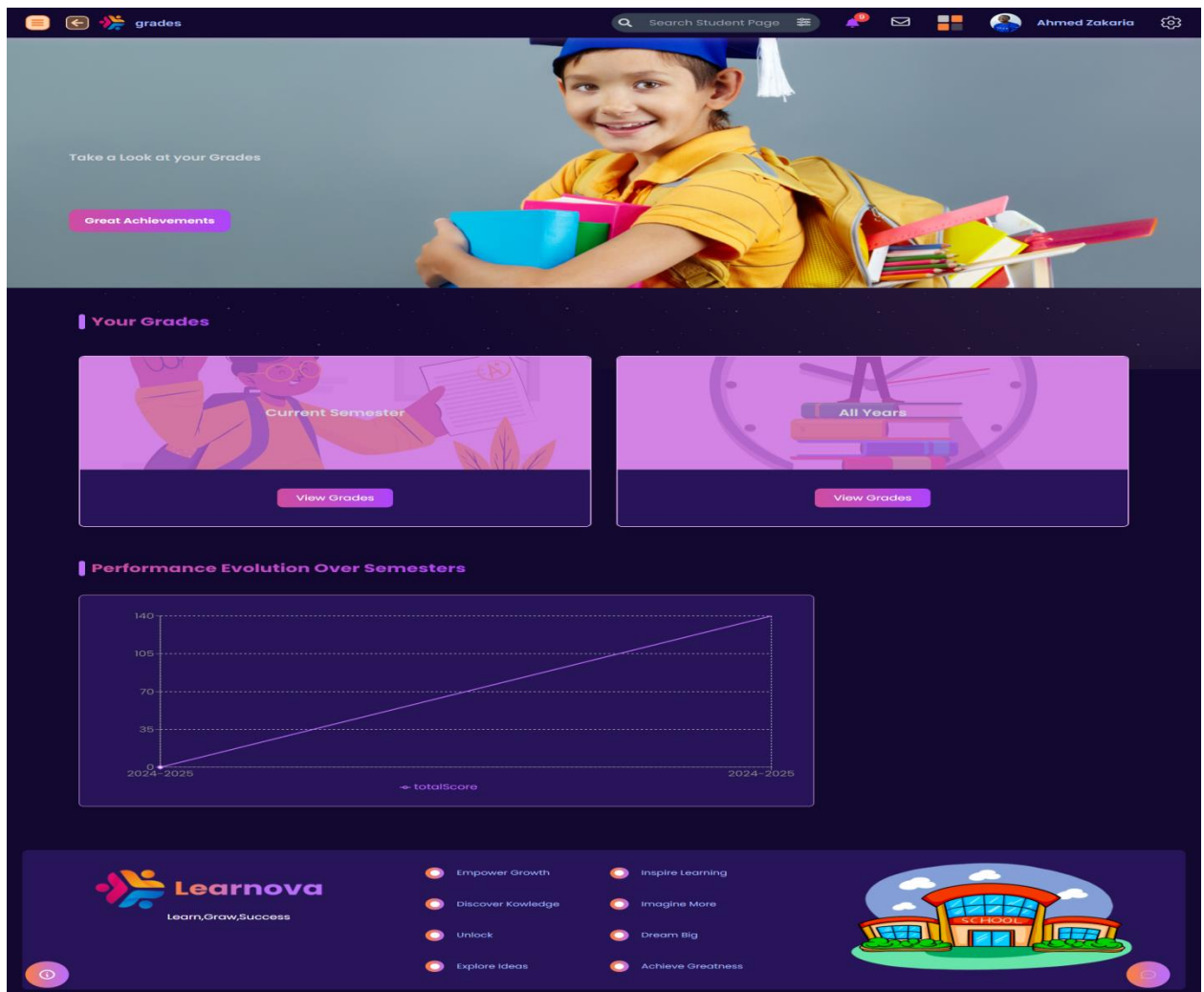
Students can view their academic performance per term and across the year.

Key Features:

- View grades by subject, class, and semester
- Check cumulative scores across all terms
- Access past exam and assignment grades

Note:

Grades are assigned by teachers and are view-only for students.



grades-for-semester

Search Student Page

Ahmed Zakaria

Grades for Semester

Back

Subject Name	Midterm Degree	Max Midterm Degree	Final Degree	Max Final Degree	Subject Score	Max Subject Score
English	10	100	10	100	20	200
Science	40	40	80	100	120	140

Learn, Grow, Success

Empower Growth

Discover Knowledge

Unlock

Explore Ideas

Inspire Learning

Imagine More

Dream Big

Achieve Greatness

grades-for-allyears

Search Student Page

Ahmed Zakaria

Grades for All Years

Back

Academic Year 2024 - 2025

Semester 1

Subject Name	Midterm Degree	Max Midterm Degree	Final Degree	Max Final Degree	Subject Score	Max Subject Score
No degrees available for this term						

Academic Year 2024 - 2025

Semester 2

Subject Name	Midterm Degree	Max Midterm Degree	Final Degree	Max Final Degree	Subject Score	Max Subject Score
Science	40	40	80	100	120	140
English	10	100	10	100	20	200

Learn, Grow, Success

Empower Growth

Discover Knowledge

Unlock

Explore Ideas

Inspire Learning

Imagine More

Dream Big

Achieve Greatness

Schedule

schedule

8
✉
☰
Ahmed Zakaria
⚙️

Upcoming Smart Classes

Weekly Schedule
Exam Schedule

	Semester 2, 2024-2025						
🕒	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
08:00	--	--	--	--	--	--	--
09:00	--	Science - Nabil Hanoh (45 minutes)	--	--	--	--	--
10:00	--	--	Science - yasseri (1 hour)	--	--	--	--
11:00	--	--	--	--	--	--	--
12:00	--	--	--	--	--	--	--
13:00	--	--	--	--	--	--	--
14:00	--	--	--	--	--	--	--

Learn, Grow, Success

🔥 Empower Growth
🔥 Inspire Learning

🔥 Discover Knowledge
🔥 Imagine More

🔥 Unlock
🔥 Dream Big

🔥 Explore Ideas
🔥 Achieve Greatness

get-exam-schedule

8
✉
☰
Ahmed Zakaria
⚙️

Upcoming Smart Classes

Weekly Schedule
Exam Schedule

Subject	Exam Date	Start Time	End Time
Math	7/1/2025	10:00	11:00
Science	7/2/2025	10:00	11:00
French	7/2/2025	10:00	11:00
English	7/12/2025	09:00	11:00

Learn, Grow, Success

🔥 Empower Growth
🔥 Inspire Learning

🔥 Discover Knowledge
🔥 Imagine More

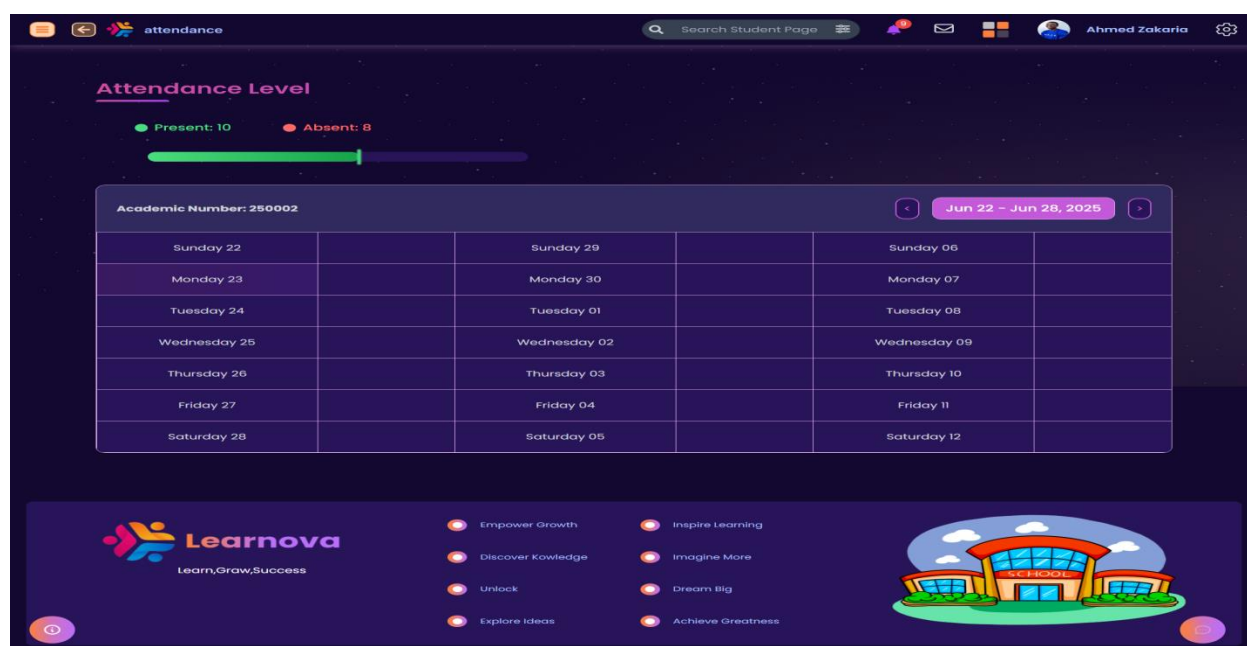
🔥 Unlock
🔥 Dream Big

🔥 Explore Ideas
🔥 Achieve Greatness

Absence Tracking

Description:

Students can view their daily attendance records .whether the student was present or absent.



Library

Description:

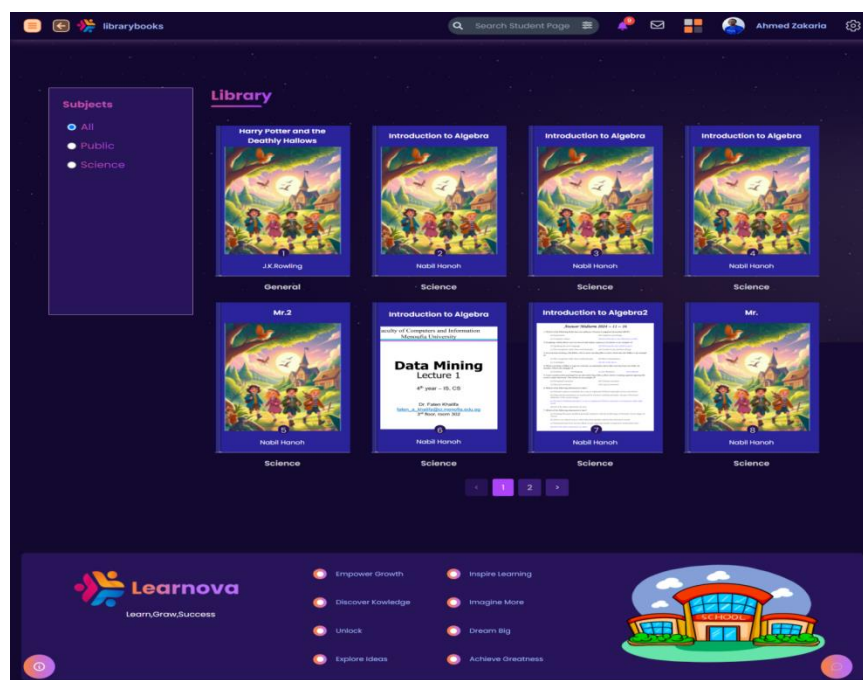
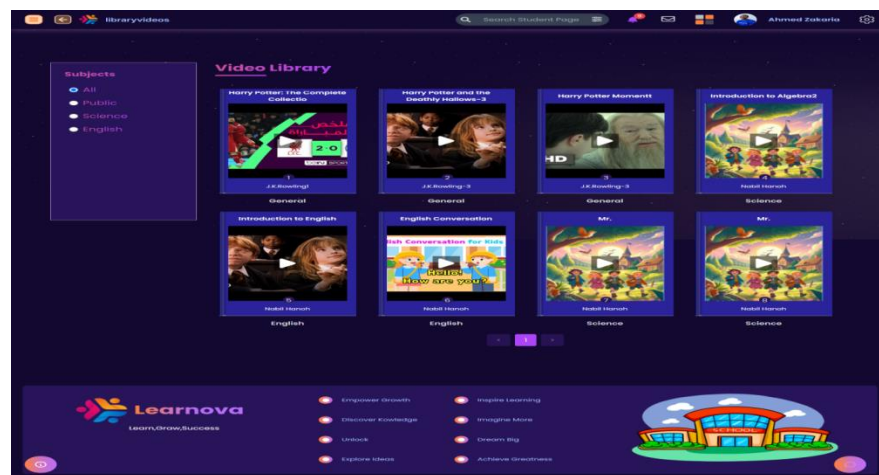
Students can browse and download learning resources shared by teachers or the school.

Key Features:

- Access materials by subject or grade
- View videos, PDFs, assignments, or documents
- Organized into categories like general or course-specific libraries

Note:

Content is uploaded by teachers; students have view/download access only.

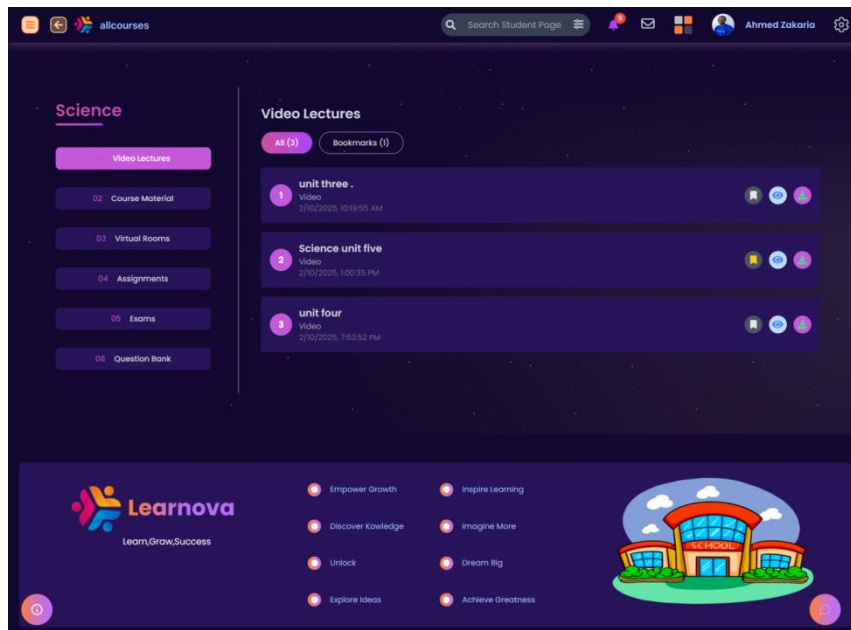
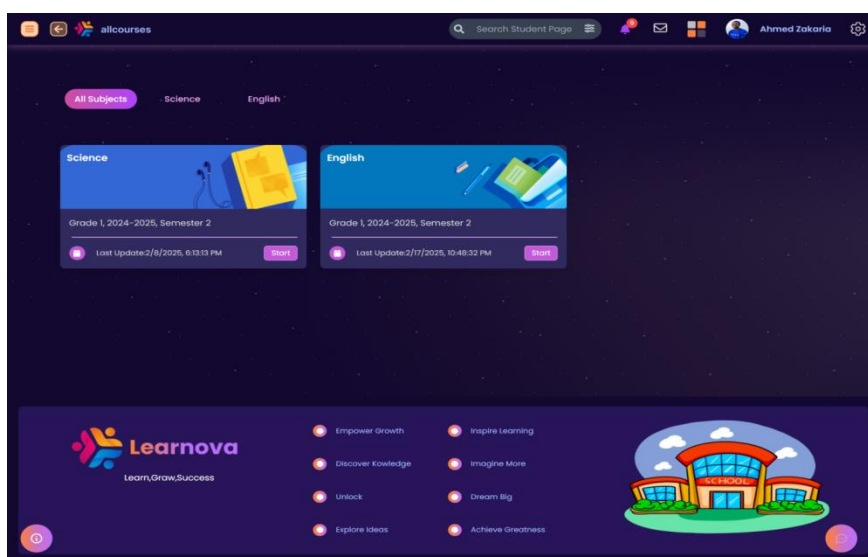


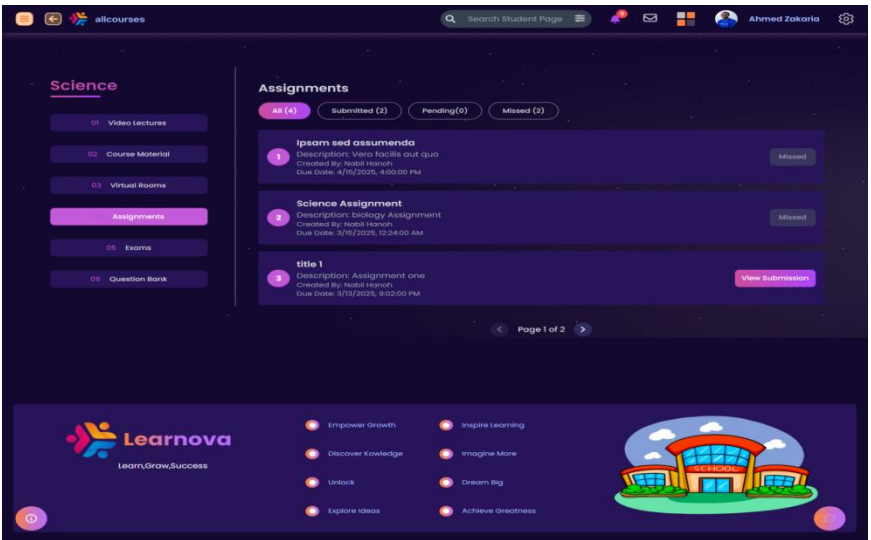
Courses

Description:

Students can access all academic content shared by their teachers within each enrolled course.

This includes videos, materials, virtual classrooms, question banks, exams, and assignments.





Expected Performance Model

Description:

The system can display an estimated or expected academic performance based on historical grades, attendance, and assignment submission trends.

Key Features:

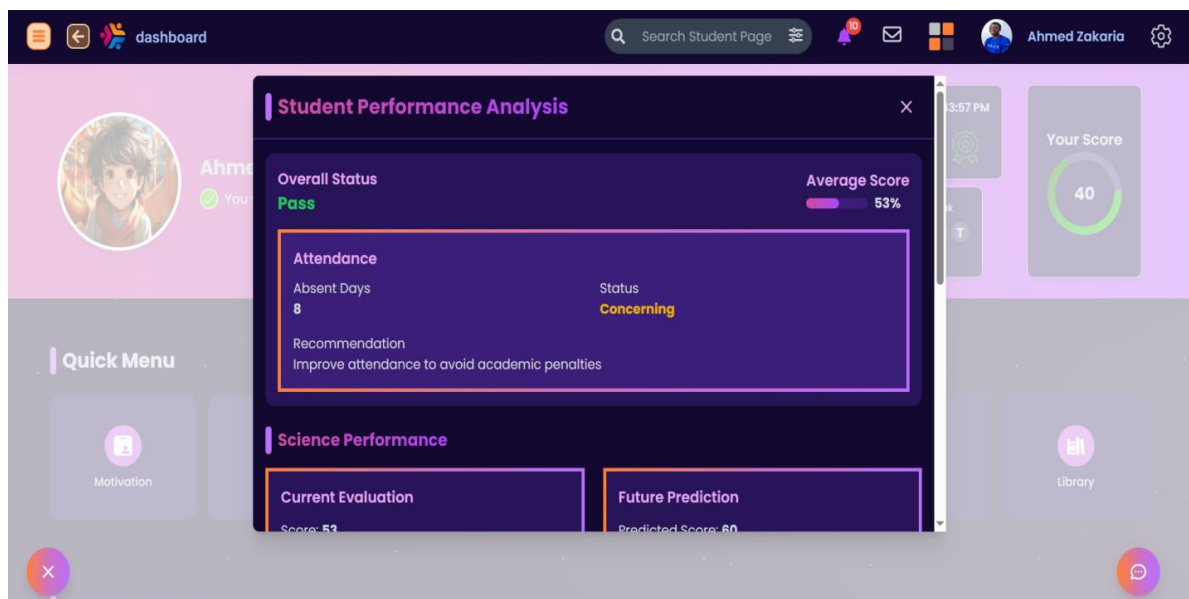
- Predict final grade trends based on past terms
- Visual progress indicators (e.g., color bars, graphs)
- Performance warnings if the student is at risk of underperforming

Data Sources Used:

- Previous grades
- Absences
- Late or missing assignments
- Exam scores

Result:

Supports self-monitoring and encourages students to improve their performance before the end of the term.



Chatbot

Description:

The platform includes an AI-powered chatbot designed to support students by answering questions and guiding them through the system.

Key Features:

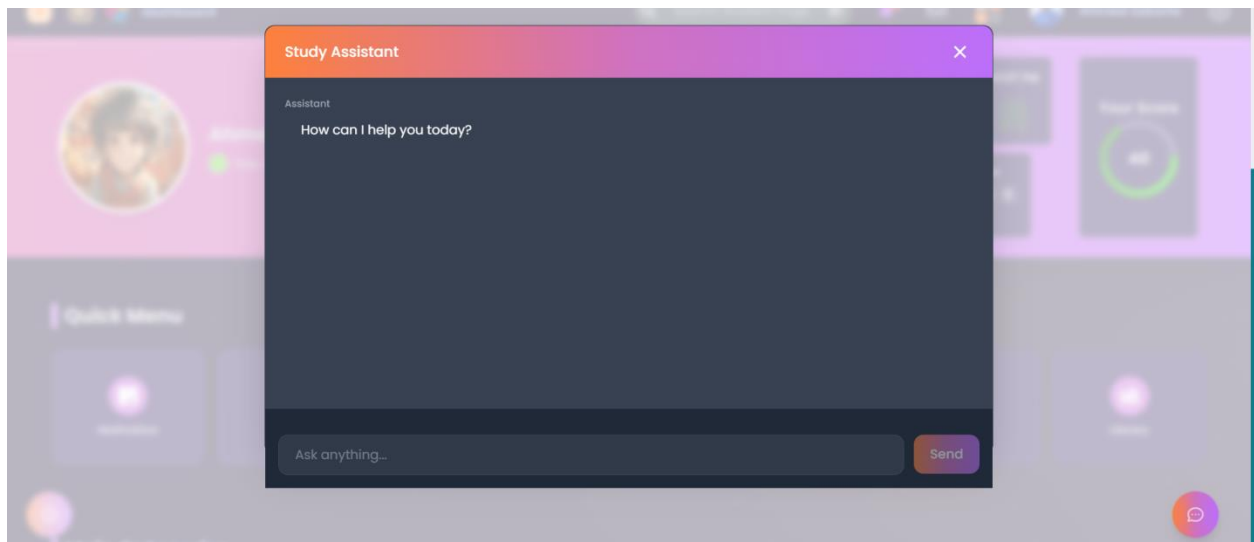
- Ask questions like “Where can I find my grades?” or “When is my next exam?”
- Get quick answers on system features, deadlines, or navigation
- 24/7 availability within the platform

Use Cases:

- Help students understand how to use the system
- Reduce dependency on admin or teacher support for simple queries
- Improve engagement and satisfaction

Result:

Enhances the student experience with instant support, improving access to information and reducing confusion.



Notification System

Description:

The notification system alerts students about important updates, deadlines, and academic activity.

Types of Notifications:

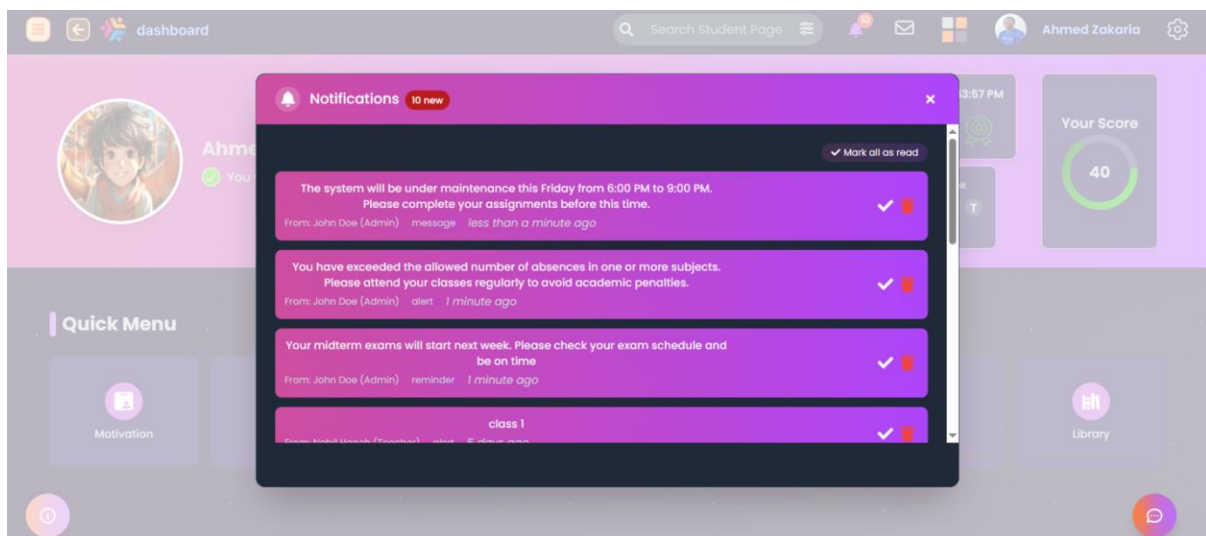
- Exam and assignment deadlines
- Grade release alerts
- Activity participation or approval
- Absence warnings
- Admin messages or system announcements

Features:

- Visual badge on the notification icon
- Clickable message previews
- Read/unread status
- Delivered in real-time

Result:

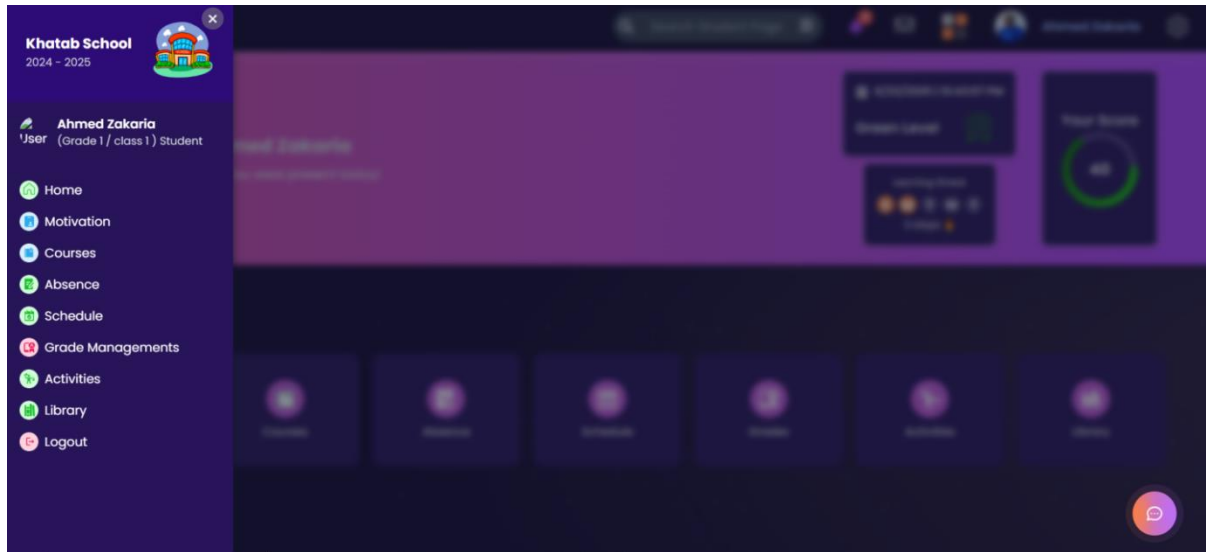
Ensures students stay informed about their responsibilities and receive timely reminders to remain engaged in their academic journey



Student Sidebar

Description:

The sidebar provides students with quick and organized access to all main features of the platform. It remains visible throughout the student interface for consistent navigation.



➤ Parent Role Implementation :

Description

The parent interface is designed to mirror the student dashboard, providing parents with a **read-only view** of their child's academic progress, schedule, performance, and participation. The goal is to keep parents informed and engaged without allowing data modification.

Access Overview

Parents have their own login credentials and account, but all data is automatically synced from their linked student(s). The design layout is similar to the student portal for familiarity and ease of use.

My Children

 Full Name	 Academic Number	 Email	 Gender
 zeko	250003	zeko@gmail.com	 Male
 Mahmoud Elframawi	250001	mahmoud@gmail.com	 Male

Features Available to Parents (Read-Only)

1. Dashboard Overview

- See a summary of the student's active courses, performance, alerts, and updates

2. Grades

- View subject-wise grades by term or semester
- See exam and assignment marks with breakdowns

3. Attendance Records

- Track daily attendance (Present/Absent)
- View attendance history with filters by date
- 4. Exam Schedule**
 - See upcoming exams, their subjects, and timing
 - Help students prepare ahead of time
- 5. Weekly Class Schedule**
 - View daily class timings and subject sessions
 - Understand what the student is studying each day
- 6. Assignments**
 - View assigned tasks, deadlines, and submission status
 - Know which assignments are pending or completed
- 7. Virtual Classrooms**
 - View scheduled virtual sessions with title, duration, and links
 - Confirm the student attends live classes
- 8. Motivation & Performance Score**
 - Track the student's motivational badge (Green, Golden, Diamond)
 - View term-based score summaries and progress
- 9. Activities & School Hubs**
 - View all activities or events the student has joined
 - See hub names, descriptions, levels, prizes, and participation status
- 10. Notifications**
 - Receive system messages from the school (e.g., exam reminders, attendance alerts)
 - Stay updated with school-wide announcements
- 11. Library Resources**
 - View academic materials (videos, files, documents) uploaded for the student
 - Understand the type of resources teachers are using
- 12. Profile & Settings**
 - View their own profile information and linked student(s)
 - No ability to update data, grades, or attendance

Data Control Note

Parents do not enter, edit, or manage data. All information comes from:

- **Teachers** (grades, attendance, course content)
- **Managers/Admins** (exam schedules, events)
- **System logic** (motivation score, deadlines)

Result

The parent role enhances transparency between school and home. It keeps guardians informed in real time, while ensuring student data remains accurate and secure.

Chapter 7

Conclusion And Future Work

Learnova - Smart School System

Learnova is an innovative platform designed to overcome the limitations of existing systems, offering a robust solution for school management and learning. Key features include:

- **Five User Roles:** Tailored dashboards for Admin, Manager, Teacher, Parent, and Student, ensuring seamless interaction and role-specific functionalities.
- **Enhanced Communication:** Real-time tools for effective student-teacher collaboration, including notifications and discussion boards.
- **AI-Powered Chatbot:** Supports Arabic and English, assisting users with navigation and queries.
- **Motivation System:** Gamified scoring with badges (Green, Golden, Diamond) to boost engagement.
- **Resource Marketplace:** A platform for sharing and accessing educational materials.
- **AI-Driven Performance Analytics:** Advanced insights for tracking and improving student performance.
- **Multi language Support:** Fully optimized Arabic and English interfaces with light/dark mode.
- **Event and Contest Management:** Tools for organizing school activities and competitions.

Comparison Between Educational Platforms

Feature / Tool	Classera	Google Classroom	Learnova
<i>Five User Roles (Admin, Manager, etc.)</i>			
<i>Enhanced Student-Teacher Communication</i>		Limited	
<i>AI-Powered Chatbot</i>			
<i>Motivation System (Gamification)</i>	Limited		

<i>Resource Marketplace</i>			
<i>AI-Driven Performance Analytics</i>			
<i>Event and Contest Management</i>	Limited		
<i>Virtual Classroom Integration</i>			
<i>Ease of Use</i>			
<i>Scalability for Small Schools</i>			

Conclusion

The analysis of **Classera** and **Google Classroom** revealed critical gaps in communication, engagement, and advanced functionality. **Learnova** surpasses these platforms by offering a user-centric system with five distinct user roles, enhanced student-teacher communication, AI-driven tools, and a comprehensive motivation system. By addressing the limitations identified through discussions with Salah Khattab Secondary School, **Learnova** delivers an innovative solution that streamlines educational processes and fosters academic excellence.

Future Work

To advance the Learnova - Smart School System, the following improvements are proposed:

1. **Emerging Technologies:**

- Add AR/VR for immersive learning (e.g., virtual labs, historical reenactments).

2. **AI/ML Enhancements:**

- Upgrade chatbot with advanced NLP for more languages and personalized responses.
- Develop predictive analytics to identify at-risk students and suggest interventions.

3. **Mobile Apps:**

- Create iOS/Android apps with offline access and push notifications.
- Add location-based attendance tracking via GPS/QR codes.

4. **Accessibility:**

- Improve screen reader support, voice commands, and UI customization (e.g., font size, high-contrast modes).

5. **External Integration:**

- Integrate with Zoom, Microsoft Teams, Khan Academy, and others.

6. **Feedback & Improvement:**

- Implement a user feedback system for feature suggestions.
- Run pilot programs with schools to refine functionality.